Common AI Information and Capital Allocation *

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Abstract

This paper studies how AI progress affects capital allocation through financial information aggregation. Using institutional portfolio holdings, we document that manager pairs with greater common alignment with AI-extractable information subsequently hold more similar portfolios. Motivated by this evidence, we develop a model in which investors choose the quantity and composition of private and AI information, and firms learn from equilibrium asset prices before investing. A shared AI model error survives aggregation and enters prices, and investors may underestimate the overlap between AI information and prices. AI progress therefore changes both investors' information precision and prices' exposure to common errors. Cheaper access initially raises aggregate productivity when high costs keep AI use negligible, but further cost reductions can reverse the gain as AI research displaces private research. Improvements from very low model accuracy can instead lower productivity when increased reliance raises common-error exposure more than informed trading reduces noise-trader contamination. Numerical equilibria trace a J-shaped relationship between model accuracy and TFP. These results show how AI progress can expand investors' information acquisition while reducing the information prices convey to firms, worsening capital allocation even when firms use prices optimally.

Keywords: artificial intelligence, information acquisition, portfolio similarity, price discovery, capital allocation.
JEL Classification: D82, G11, G23, O33.

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1 Introduction

Artificial intelligence is increasingly used in financial services, including the research that informs investment decisions. Mercer's 2026 survey finds that 68% of responding asset managers use AI to support investment analysis, and the Bank of England shows that over 75% of interviewed UK financial firms were implementing AI in 2024.[1] AI offers investors a way to process large amounts of information more quickly and at lower cost. Evidence from Italy's temporary ban on ChatGPT supports these benefits: restricting access reduced analysts' forecast production and accuracy (Bertomeu et al., 2025). These individual benefits, however, may accompany greater dependence on common models and data, exposing investors' decisions to shared signals and model errors. In 2023, then SEC Chair Gary Gensler warned that market participants could make similar decisions after receiving "the same signal from a base model or data aggregator".[2] The survey conducted by the Bank of England also shows that around one thirds of the AI usages are provided by a third-party supplier, and that the three most frequently named model providers account for 44% of model-provider mentions. This implies that a small number of models could affect the trading decisions of a non-negligible fraction of market participants. The Financial Stability Board likewise identifies reliance on common data and models as a potential source of correlated predictions and financial decisions (Financial Stability Board, 2024). Understanding the economic consequences of AI therefore requires examining both how it improves individual information processing and how shared reliance on the technology changes the correlation of investment decisions across market participants.

The consequences of common information extend beyond investors' portfolios. Asset prices aggregate information about firms' prospects, and firms can use this information when making real investment decisions. Empirical studies show that firm investment responds to stock prices that contain more information (Chen et al., 2007), and survey evidence in China also finds that listed firms extract demand information from stock prices, which in turn guides their production decisions (Goldstein et al., 2026). Bond et al. (2012) reviews broader implications of this learning channel. If a common model error enters investors' demands, market clearing can transmit it into the prices firms learn from. Greater AI reliance can then make these prices less informative even as investors acquire more precise signals. The question is whether the in-

[1] Another survey done by Mercer in 2024 reports that 54% of responding managers already used AI in investment strategies or asset-class research; see *AI Integration in Investment Management: 2024 Global Manager Survey*, https://insightcommunity.mercer.com/api/v1/uploads/Mercer_AI_integration_in_investment_management_2024_global_manager_survey_579b17aa96.pdf?public=true. Its 2026 survey covers 131 asset managers; see https://www.mercer.com/about/newsroom/how-artificial-intelligence-is-shaping-asset-management/. The Bank of England and Financial Conduct Authority's 2024 survey reports that 75% of responding financial firms use AI; see https://www.bankofengland.co.uk/report/2024/artificial-intelligence-in-uk-financial-services-2024.

[2] Gary Gensler, "Isaac Newton to AI," remarks before the National Press Club, July 17, 2023: https://www.sec.gov/newsroom/speeches-statements/gensler-isaac-newton-ai-remarks-07-17-2023.

formation gains from AI use translate into better capital allocation when the same technology also changes the dependence across investors' information.

Whether investors recognize the overlap in their information is central to this mechanism. Shared models and training data need not be transparent to their users: the Financial Stability Board notes that market participants can unknowingly rely on the same underlying data (Financial Stability Board, 2024), and the survey by the Bank of England finds that over 44% of respondents reported only a partial understanding of the AI technologies they used. Evidence from professional AI use also suggests that information overlap may be difficult to assess. In an experiment with radiologists, Agarwal et al. (2026) find that participants fail to account fully for the dependence between their own assessments and AI predictions. Although this evidence comes from medical decision-making, it motivates a specific question for financial markets: might an investor treat a price partly driven by a common AI signal as independent confirmation of that signal? How would such beliefs affect trading, price informativeness, and real investment?

This paper studies how AI progress changes the information structure of financial investors, the information that financial markets convey to firms, and aggregate productivity. Improvements in AI information need not improve aggregate allocative efficiency, which is determined by both the aggregate precision of the investors' signals and the exposure to common AI error. Cheaper access initially raises TFP when high costs keep AI use negligible, but further cost reductions can reverse this gain as AI research displaces private research. Higher model accuracy also changes both information acquisition and exposure to common errors. Near the low-accuracy boundary, the increase in common-error exposure can dominate the information gain, reducing price informativeness and TFP. Common reliance on AI can therefore generate a nonmonotonic relationship between AI progress and aggregate productivity through its effects on asset prices and capital allocation.

We first document a pattern consistent with common reliance on AI-related information. Following the revealed-reliance approach of Sheng et al. (2026), we measure managers' alignment with signals extracted from earnings calls by a large language model. Manager pairs with greater common alignment subsequently hold more similar portfolios, controlling for current similarity. This relationship is stronger after ChatGPT's public release and in settings where language models appear to have a clearer information advantage. These predictive associations motivate a shared information source.

Motivated by this evidence, we develop a production economy with endogenous information acquisition. Investors choose the quantity and composition of information between private research and AI before trading claims on firms' revenues. Private research errors are investor-specific, whereas AI model errors are shared across investors researching a given firm. Investors also learn from prices, but may underestimate the covariance between their AI infor-

mation and the price signal. With financial claims in fixed net supply, common shifts in desired holdings enter equilibrium prices. Firms observe these prices, update their beliefs, and choose capital. The model thus connects a common source of investment information to real allocation through endogenous price formation.

We characterize the local productivity effects analytically for a general degree of recognition of common errors, near two boundaries at which AI use vanishes. At fixed model accuracy, a marginal reduction in a sufficiently high acquisition cost expands total information even though it displaces some private research. The resulting increase in informed trading reduces noise-trader contamination at first order. Exposure to the common error contributes only at second order because its price loading starts near zero. Information gains therefore dominate, and cheaper AI raises price informativeness and aggregate TFP sufficiently close to this boundary, regardless of investors' recognition of common errors.

The balance differs when negligible AI use reflects low model accuracy. When the model accuracy is low enough, investors acquire nearly all the effective precision available from a very inaccurate model, even though the AI share of their information remains close to zero. A marginal accuracy improvement then expands effective AI information almost one-for-one. The induced common-error exposure and the reduction in noise-trader contamination both affect price informativeness at first order. TFP initially falls when the former effect dominates. This comparison is affected by beliefs about model-error correlation. Recognition changes acquisition incentives and reduces the common-error price loading near the low-accuracy boundary. Numerical analysis traces these responses beyond the local boundaries. Improvements in accuracy can generate a J-shaped TFP curve, while reductions in acquisition costs can generate a hump. Extensions to more general acquisition costs identify the conditions on marginal costs and interactions between research sources that preserve the local results. A further extension allows market outcomes to affect subsequent model training and preserves the local comparison under a smooth, locally stable training feedback.

In general, this paper implies that the productivity consequences of AI depend on how investors use its information and how prices convey that information to firms. Investors can acquire more precise signals while firms receive less informative prices. Distinguishing cheaper access from greater model accuracy, and individual precision from the information conveyed by prices, helps explain why AI progress need not translate monotonically into better capital allocation.

Related literature. This paper relates to research on AI and data technologies in financial decision-making. Coleman et al. (2022) compare robo-analyst and traditional analyst recommendations, while Cao et al. (2024) study the gains from combining machine and human stock analysis. Sheng et al. (2026) examine generative AI in asset management using a revealed-

reliance measure, which we adapt to study common information exposure across managers.[3] Bertomeu et al. (2025) use Italy's ChatGPT ban to study the benefits of generative AI for analysts' information processing. Huang and You (2026) find that AI adoption by mutual funds shifts information acquisition toward shorter horizons, with corresponding changes in price informativeness and investment–price sensitivity. Their study examines the allocation of information across forecast horizons; ours examines the dependence across investors who use a common source. Data technologies can also change the allocation of financing. Bonelli (2026) finds that venture capitalists adopting these technologies improve their screening of startups resembling historical observations but shift investment toward those firms and become less likely to finance innovative startups that achieve rare, major successes. Our paper studies how shared information affects dependence across investors and the prices that guide firms' investment. It connects individual information acquisition to real allocation through the aggregation of common AI signals in financial markets.

The portfolio evidence also relates to institutional herding and common holdings. Lakonishok et al. (1992), Wermers (1999), and Sias (2004) study correlated institutional trading, and Pool et al. (2015) connect portfolio similarity to social interaction among fund managers. Common ownership is associated with return comovement (Antón and Polk, 2014) and liquidity commonality (Koch et al., 2016); Girardi et al. (2021) examine the relationship between portfolio similarity and asset liquidation.[4] We document that common alignment with AI-extractable information predicts subsequent portfolio convergence, and use a model that is aligned with this phenomenon to study the consequences for price discovery and capital allocation.

Theoretically, the paper builds on endogenous information acquisition in financial markets (Grossman and Stiglitz, 1980; Verrecchia, 1982; Van Nieuwerburgh and Veldkamp, 2010). Veldkamp (2006) shows how information-production costs encourage investors to acquire common signals, generating comovement in asset prices. Hellwig and Veldkamp (2009) explain how strategic motives in decisions carry over to information choices, while Myatt and Wallace (2012) study the allocation of attention across information sources. Goldstein and Yang (2015) examine the role of information diversity in trading and acquisition incentives, and Colombo et al. (2014) analyze the welfare consequences of endogenous information acquisition. Particularly close to our question, Farboodi and Veldkamp (2020) study how advances in data processing change investors' learning about asset fundamentals and other traders' demands. We consider a choice between information sources about the same fundamental that differ in the extent to which their errors are shared. AI progress changes both total effective precision and its compo-

[3] Related work studies machine learning and language models in return prediction and portfolio construction (Gu et al., 2020; Lopez-Lira and Tang, 2026; Cong et al., 2026), as well as the portfolio effects of robo-advice (D'Acunto et al., 2019; Rossi and Utkus, 2024).

[4] Related work studies crowded mutual fund portfolios (Gonzalez et al., 2024) and the implications of common exposures for systemic liquidation risk (Wagner, 2011; Greenwood et al., 2015).

sition between private research and AI. This distinction allows us to compare cheaper access with greater model accuracy and to characterize when the induced common-error exposure outweighs the reduction in noise-trader contamination, worsening capital allocation.

In our model, investors could have biased beliefs about model error commonality, which is related to the literature on correlation neglect and repeated information. DeMarzo et al. (2003) and Eyster and Rabin (2010) study how failures to account for the origins of information distort learning, and Enke and Zimmermann (2019) provide experimental evidence on the treatment of correlated signals. In financial settings, Fedyk and Hodson (2023) document difficulties in recognizing recombined old news, while Guo and Wachter (2026) relate stock-market return patterns to investors' incomplete recognition of repetitive earnings information. These studies motivate imperfect recognition of information overlap in our setting. Investors learn from prices and correctly assess the marginal precision of their own AI signals, but may underestimate their interdependence. We study how this belief affects the information they choose to acquire, the common-error component of equilibrium prices, and the investment decisions firms make using those prices.

The macroeconomic analysis contributes to research on the real effects of financial information and information-driven capital misallocation. Chen et al. (2007) provide evidence consistent with firms learning from stock prices when investing, and Bond et al. (2012) review the channels through which financial markets affect real decisions. Goldstein et al. (2026) provide direct survey evidence on firms' use of market information in investment decisions. David et al. (2016) quantify the role of imperfect information in capital misallocation, and Benhabib et al. (2016) and Benhabib et al. (2019) study the feedback between information acquisition, uncertainty, and economic activity.[5] We introduce endogenous reliance on an AI signal with a shared error into this environment. Firms learn from the price signal and invest optimally, but the information available to them depends on investors' acquisition choices and beliefs. The resulting allocation effects distinguish the precision an individual investor acquires from the information financial markets convey to firms.

Finally, our findings relate to the productivity paradox highlighted by Solow (1987), in which rapid technological progress coexists with weak aggregate productivity growth. Research on general purpose technologies emphasizes the complementary innovations needed to realize their productivity gains (Bresnahan and Trajtenberg, 1995). David (1990) traces the delayed gains from electrification to slow diffusion and the reorganization of production, while Greenwood and Yorukoglu (1997) show how learning to use new technologies can temporarily slow productivity growth. In the context of AI, Brynjolfsson et al. (2019) emphasize implementation lags and complementary investment. Brynjolfsson et al. (2021) formalize a productivity

[5]More broadly, imperfect information shapes expectations and macroeconomic dynamics (Angeletos and Huo, 2021; Schaal and Taschereau-Dumouchel, 2025).

J-curve in which measured productivity growth is initially depressed by unmeasured intangible investment and rises as its returns materialize. We provide an additional explanation through financial information and capital allocation. Improvements in model accuracy can increase reliance on AI, raising the common-error component of prices and reducing their informativeness. Firms then base investment on less informative prices, so aggregate TFP can fall even as total information acquisition rises. This effect is especially prominent when the investors underestimate the correlation between AI signals. Further accuracy gains can restore price informativeness and TFP, generating a J-shaped relationship between AI model accuracy and equilibrium TFP. This comparison across equilibria isolates an informational channel through capital allocation that complements explanations based on implementation lags and intangible investment.

The remainder of the paper is organized as follows. Section 2 presents the portfolio evidence and the economic motivation for the model. Section 3 introduces the environment, and Section 4 characterizes information acquisition, price formation, and capital allocation. Section 5 studies the effects of AI progress analytically and numerically. Section 6 considers general information costs, and Section 7 concludes.

2 Evidence and Economic Motivation

This section examines whether common alignment with AI-extractable information predicts subsequent portfolio convergence and whether this relationship is stronger when language models have an information advantage. Because institutional adoption is not directly observed, we interpret the estimates as predictive associations. We then discuss the external evidence and economic mechanisms motivating investors' beliefs and firms' use of financial prices.

2.1 Data and Measurement

Institutional holdings and sample coverage. We obtain quarterly institutional holdings from Form 13F filings and construct a manager–stock–quarter panel. Position values are converted into portfolio weights, and consecutive reports are used to measure portfolio adjustments. In each quarter, we retain the 300 managers with the largest reported portfolios and form every unordered pair of managers. The resulting universe contains up to $300 \times 299/2 = 44{,}850$ manager pairs per quarter. The top 300 managers collectively account for more than 85% of portfolio value in the source Form 13F panel.

The earnings-call sample consists primarily of S&P 500 constituents and related large-cap U.S. firms for which transcript text, firm identifiers, and the required controls can be matched. These scored firms account for more than 65% of the portfolio value held by the top 300 man-

agers. Combining the manager and firm coverage, the manager–firm observations underlying the AI-reliance measure represent more than 55% of total portfolio value in the full Form 13F holdings panel. The transcript sample is concentrated among large public firms, but it covers an economically important segment of institutional portfolios in which correlated adjustments could matter at the aggregate level.

LLM-based firm signals. We follow Sheng et al. (2026) and use earnings-call transcripts to construct fourteen directional firm-quarter signals. The prompt asks GPT-4o-mini to evaluate the implications of each call for U.S. and global economic conditions, industry conditions, and eleven firm-level outcomes covering financial prospects, earnings, revenue, investment, labor costs, employment, product demand, production, output prices, input prices, and the cost of capital. We do not supply firm names, tickers, industry labels, or call dates as separate prompt fields, and explicitly instruct the model to use only the excerpt and not to infer from company identity, industry, dates, or later events.[6] Each score lies between -1 and 1; zero is assigned when the transcript does not support a directional assessment. Long transcripts are divided sequentially into chunks of approximately 12,000 words. GPT-4o-mini has a 128,000-token context window, and this conservative chunk size leaves room for the complete instructions and structured JSON response while avoiding context truncation. We aggregate chunk-level outputs to the transcript level and then average across transcripts for the same firm-quarter. The fourteen scores enter the portfolio-adjustment regressions separately rather than through a composite index.

Manager-level revealed AI reliance. Following the empirical logic of Sheng et al. (2026), we construct a revealed measure of alignment with AI-extractable information from the incremental explanatory power of LLM-extracted information for a manager's portfolio decisions. For every manager i and quarter t, we estimate two cross-sectional regressions across securities s:

$$\Delta w_{i,s,t} = \alpha_{i,t} + \gamma'_{i,t} X_{s,t-1} + \varepsilon_{i,s,t}, \tag{2.1}$$

$$\Delta w_{i,s,t} = \alpha_{i,t} + \gamma'_{i,t} X_{s,t-1} + \sum_{m=1}^{14} \beta^m_{i,t} GPT^m_{s,t-1} + \varepsilon_{i,s,t}, \tag{2.2}$$

where $\Delta w_{i,s,t} = w_{i,s,t} - w_{i,s,t-1}$ is the baseline portfolio-weight change. The public-information vector $X_{s,t-1}$ contains log market capitalization, book-to-market, return on assets, the stock return, and the change in the I/B/E/S recommendation consensus. All five controls and all

[6] The transcript itself can contain identifying information. In a masked-identity exercise on a random 10% sample stratified by signal quarter, scores from the original and masked texts have a pooled correlation of 0.87, with 98.6% signals having the same direction and 90.7% of firm-quarter signals being identical. Appendix D.9 reports the procedure and results.

GPT signals are dated $t-1$, so the information set precedes the portfolio adjustment.

Based on the two regressions, we define manager i's revealed AI-reliance measure as:

$$Reliance_{i,t} = R^2_{full,i,t} - R^2_{base,i,t}. \tag{2.3}$$

This construction follows the revealed-public-information approach of Kacperczyk and Seru (2007) and its recent application to generative AI by Sheng et al. (2026). Sheng et al. (2026) further show that a closely related holdings-based measure aligns with survey evidence on managers' reported use of generative AI. In the present setting, the measure captures a manager's revealed behavioral alignment with information that a language model extracts from corporate disclosures. It is therefore an information-alignment measure rather than a provider-specific indicator of model adoption. Because the full regression nests the baseline specification, the raw increase in in-sample R^2 is mechanically nonnegative. Appendix D considers partial-R^2, adjusted-R^2, and adjusted partial-R^2 measures that either normalize the incremental explanatory power of the GPT scores or penalize the additional regressors.

At the pair level, we define common AI reliance as:

$$Common_Reliance_{i,k,t} = Reliance_{i,t} \times Reliance_{k,t}. \tag{2.4}$$

We also define $AverageReliance_{i,k,t} = (Reliance_{i,t} + Reliance_{k,t})/2$. Conditional on average reliance, the product has a particularly useful interpretation:

$$Reliance_{i,t} \times Reliance_{k,t} = AverageReliance^2_{i,k,t} - \frac{(Reliance_{i,t} - Reliance_{k,t})^2}{4}. \tag{2.5}$$

Thus, holding the average fixed, a larger product means a smaller gap between the two managers' reliance and captures greater common alignment with the same AI-extractable information. Finally, we define a pair-level indicator, $Joint_GPT_Alignment_{i,k,t}$, which equals one when the joint F-tests on the fourteen GPT coefficients are significant at the 5% level for both managers. This indicator provides an extensive-margin measure of joint GPT alignment.

Portfolio similarity. We use two methods to measure the similarity between two asset managers' portfolio structures. Following Wahal and Wang (2011), Cici et al. (2017), and Gonzalez et al. (2024), we use portfolio overlap as our main measure:

$$Overlap_{i,k,t} = \sum_s \min\{w_{i,s,t}, w_{k,s,t}\}, \tag{2.6}$$

where $w_{i,s,t}$ denotes the portfolio share of asset s by manager i at time t. The portfolio overlap measure captures the fraction of portfolio holdings that can be directly matched between two

asset managers. This measure ranges from zero to one. A value of zero indicates that the two managers have no common holdings, whereas a value of one indicates identical portfolios. Economically, this measure represents the share of portfolio mass that can be matched security by security and therefore provides a direct measure of the magnitude of their common asset exposure.

We also use cosine similarity as an alternative measure (Hoberg et al., 2018; Girardi et al., 2021):

$$Cosine_{i,k,t} = \frac{\sum_s w_{i,s,t} w_{k,s,t}}{\sqrt{\sum_s w_{i,s,t}^2} \sqrt{\sum_s w_{k,s,t}^2}}. \tag{2.7}$$

Cosine similarity measures the extent to which the two portfolios are oriented in the same *direction* in the asset-weight space. Relative to portfolio overlap, it places greater weight on securities that constitute large positions in both portfolios and is therefore more sensitive to similarity in managers' dominant exposures. By contrast, the overlap measure has a more direct interpretation as the total amount of common portfolio exposure. Thus, while both measures increase when two managers hold similar portfolios, portfolio overlap primarily captures the magnitude of shared holdings, whereas cosine similarity emphasizes relative allocation patterns across securities. The main empirical patterns are similar across the two measures.

2.2 Common AI Reliance and Future Portfolio Similarity

We first examine whether common AI reliance predicts incremental portfolio convergence. Period-t reliance is estimated from holdings in $t-1$ and t, while the outcome is portfolio similarity in $t+1$. Controlling for period-t similarity absorbs the substantial persistence in portfolio composition. The construction of the reliance measure also imposes no directional restriction: given the same GPT-generated information, managers may increase or decrease their holdings of a stock, and neither the direction nor the magnitude of their adjustments needs to coincide. The specification therefore asks whether pairs with greater common reliance converge further from their current level of similarity. We estimate:

$$\begin{aligned} Similarity_{i,k,t+1} = {} & \beta \times Common_Reliance_{i,k,t} + \lambda \times Average_Reliance_{i,k,t} \\ & + \rho \times Similarity_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}. \end{aligned} \tag{2.8}$$

where $\varphi_{i,k}$ and ψ_t are manager-pair and quarter fixed effects. The vector $Z_{i,k,t}$ contains the control variables, including pair average and absolute difference in log portfolio value and number of holdings (Wahal and Wang, 2011). Standard errors are clustered by manager pair.[7]

[7] Appendix D reports a two-manager clustering sensitivity exercise. The continuous baseline coefficients remain statistically significant, although some indicator specifications are estimated less precisely.

The timing and controls serve distinct purposes. Using $t+1$ similarity places the outcome after the portfolio adjustment used to estimate reliance. Current similarity absorbs the considerable persistence in institutional portfolios, so β measures whether common reliance predicts additional convergence from the pair's existing position. Average reliance separates commonality from the overall level of AI exposure: conditional on the average, the product is larger when the two managers have more similar reliance estimates. Pair fixed effects remove time-invariant matching patterns between institutions, and quarter fixed effects absorb aggregate shifts in portfolio similarity and the information environment.

Table 1 presents four specifications for each similarity measure. Panel A is the baseline and controls for the pair's average AI reliance. Conditional on this average, pairs with greater common AI reliance have significantly higher next-quarter overlap and cosine similarity. The same pattern appears when Panel B controls separately for the two managers' reliance and when Panel C uses the joint-alignment indicator. Panel D adds squared average reliance to allow for a nonlinear relationship between the overall level of reliance and future similarity. The common-reliance coefficient remains positive for both measures and statistically significant for overlap. The current-similarity coefficients, approximately 0.80 for overlap and 0.68 for cosine similarity, show that portfolio composition is highly persistent. The reliance coefficients therefore capture differences in next-quarter similarity around this persistent baseline.

The continuous estimates describe modest one-quarter changes in portfolio similarity. Appendix Table D.1 reports a standard deviation of approximately 0.011 for common AI reliance. A one-standard-deviation increase in common reliance predicts increases of 0.0003 in future portfolio overlap and 0.0007 in future cosine similarity. These estimates correspond to approximately 0.2% and 0.3% of a standard deviation in the respective similarity measures. The indicator estimates imply larger differences. In a pair-quarter in which both managers exhibit significant GPT alignment, future overlap is 0.0020 higher and future cosine similarity is 0.0105 higher, equivalent to approximately 1.2% and 4.1% of a standard deviation, respectively. Because the regressions condition on current similarity and pair fixed effects, these magnitudes measure incremental within-pair convergence over the next quarter.

Validation and heterogeneity. We next examine whether the common-reliance measure behaves as expected in settings where language models are plausibly more useful for processing corporate disclosures. These exercises provide construct validation beyond the baseline predictive relationship. The first exercise uses the broader accessibility and salience of generative AI after the public release of ChatGPT. If the measure captures economically meaningful reliance on AI-extractable information, its association with subsequent portfolio convergence should be stronger in the post-release period. We define $PostGPT_t$ to equal one from 2023Q1 onward. Because ChatGPT was released on November 30, 2022, 2022Q4 is treated as the last pre-period.

Table 1: Common AI Reliance and Future Portfolio Similarity

	(1) Future overlap	(2) Future cosine similarity
Panel A: Controlling for average AI reliance		
Common AI reliance	0.0271***	0.0627***
	(6.27)	(6.80)
Average AI reliance	-0.0097***	0.0052**
	(-11.29)	(2.23)
Current similarity	0.7957***	0.6760***
	(371.02)	(223.62)
Panel B: Controlling for individual AI reliance		
Common AI reliance	0.0272***	0.0628***
	(6.29)	(6.81)
AI reliance, manager i	-0.0057***	0.0018
	(-10.46)	(1.18)
AI reliance, manager k	-0.0041***	0.0033**
	(-7.62)	(2.30)
Current similarity	0.7956***	0.6760***
	(371.03)	(223.62)
Panel C: Joint GPT alignment		
Joint GPT alignment	0.0020***	0.0105***
	(9.86)	(13.47)
Average AI reliance	-0.0076***	0.0059***
	(-10.86)	(3.19)
Current similarity	0.7956***	0.6760***
	(371.06)	(223.84)
Panel D: Controlling for nonlinear average AI reliance		
Common AI reliance	0.0118**	0.0180
	(2.34)	(1.51)
Average AI reliance	-0.0173***	-0.0170***
	(-9.70)	(-3.41)
Squared average AI reliance	0.0291***	0.0850***
	(5.29)	(5.56)
Current similarity	0.7956***	0.6760***
	(370.80)	(223.57)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair
Observations	890,394	890,394

Notes: Each panel represents a separate regression. Robust t-statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Quarter fixed effects absorb the level of the post indicator, while its interaction with common reliance measures the change in the predictive relationship. We estimate:

$$\begin{aligned} Similarity_{i,k,t+1} = \beta \times Common_Reliance_{i,k,t} + \lambda \times Common_Reliance_{i,k,t} \times PostGPT_t \\ + \mathcal{L}_{i,k,t} + \rho \times Similarity_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}, \end{aligned} \tag{2.9}$$

where $\mathcal{L}_{i,k,t}$ contains average reliance and its interaction with $PostGPT_t$. A positive λ means that the relationship between common AI reliance and future portfolio similarity is stronger in the post-GPT period, consistent with the broader availability of general-purpose AI tools. We also interact average reliance with the post-GPT indicator so that the interaction of interest continues to capture commonality conditional on the pair's average reliance.

The second exercise studies two dimensions of the firm information environment in which language models may have a greater comparative advantage:

First, earnings-call length. We define firm-quarter earnings-call length as the mean transcript word count, which captures the amount of unstructured information that investors must process. Language models are plausibly more useful when disclosures are longer, so we expect a stronger post-GPT association for manager pairs jointly exposed to firms with longer calls.

Second, human analyst coverage. We define analyst coverage as the number of I/B/E/S forecasts, which proxies for the supply of human-produced information. AI-extractable information may carry greater incremental value for firms with fewer analyst forecasts.

Each firm characteristic is standardized within GICS sector and quarter, then aggregated to the manager-pair level using historical portfolio-overlap weights. For manager pair (i, k), the constructed moderator variable for firm characteristic X is defined as:

$$M^X_{i,k,t} = \frac{\sum_s \min\{\bar{w}^{hist}_{i,s,t}, \bar{w}^{hist}_{k,s,t}\} Z^X_{s,t}}{\sum_s \min\{\bar{w}^{hist}_{i,s,t}, \bar{w}^{hist}_{k,s,t}\}}, \tag{2.10}$$

where $\bar{w}^{hist}_{i,s,t}$ is manager i's average portfolio weight on asset s over $t-1$ through $t-4$, and $Z^X_{s,t}$ is the sector-quarter standardized z-score for firm characteristic X. Historical overlap weights make the moderator reflect the information environment of firms to which the pair was jointly exposed before the current portfolio adjustment. A higher $M^X_{i,k,t}$ implies that the asset manager pair (i, k) is jointly more exposed to firms with higher levels of X.

For each moderator, we estimate a saturated triple-interaction model:

$$\begin{aligned} Similarity_{i,k,t+1} = \beta_1 \times Common_Reliance_{i,k,t} + \beta_2 \times Common_Reliance_{i,k,t} \times PostGPT_t \\ + \beta_3 \times Common_Reliance_{i,k,t} \times M^X_{i,k,t} \\ + \lambda \times Common_Reliance_{i,k,t} \times M^X_{i,k,t} \times PostGPT_t \end{aligned}$$

$$+ \mathcal{L}_{i,k,t} + \rho Similarity_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}, \tag{2.11}$$

where $\mathcal{L}_{i,k,t}$ contains average reliance and its corresponding interactions, the moderator, $M^X \times PostGPT$, and all remaining lower-order terms. In this regression, β_1 is the pre-GPT common-reliance slope at $M^X = 0$, β_2 is the post-GPT shift at that reference value, and λ measures how the shift varies with the information environment. We expect λ to be positive for earnings-call length and negative for analyst coverage. These signs would indicate that the post-GPT association is stronger when manager pairs focus on firms for which machine processing is plausibly more useful.

Table 2 reports the results. Column (1) shows a positive and statistically significant post-GPT interaction for both similarity measures. Columns (2) and (3) relate this shift to the information environment. The triple interaction with call length is positive for both outcomes, consistent with a stronger post-GPT association when the underlying text is more costly to process. For analyst coverage, the triple interaction is negative, consistent with AI-extractable information having greater incremental value when human information production is limited. Taken together, these patterns provide construct validation for the revealed-reliance measure. The evidence is strongest for the post-GPT shift and call length. The analyst-coverage result has the predicted sign for both outcomes but is less precisely estimated for portfolio overlap. Two-manager clustering widens the confidence intervals around the post-GPT and moderator interactions, as Appendix D reports.

Appendix D examines alternative fit measures, definitions of portfolio adjustment, manager-time controls, joint-alignment thresholds, and covariance estimators. The common-reliance coefficient for portfolio overlap remains positive across these specifications, although penalizing the additional GPT regressors reduces its precision. The cosine result remains positive under the alternative fit and portfolio-adjustment measures but is more sensitive to manager-by-quarter fixed effects. Under those fixed effects, the joint-alignment indicator remains positive and significant for both outcomes. Two-manager clustering increases standard errors but leaves the continuous baseline coefficients statistically significant. The leave-one-quarter-out checks in Appendix D.8 further show that the post-GPT pattern is not driven by any single predictor quarter.

Portfolio similarity also predicts common selling and downside comovement, as reported in Appendix D.1. These additional associations describe the consequences of common positions without identifying whether the information behind those positions is accurate.

Table 2: Validation and Heterogeneity

	(1) Post-GPT	(2) Call length	(3) Analyst coverage
Panel A: Future portfolio overlap			
Common reliance	0.0192***	0.0148***	0.0040
	(4.10)	(2.95)	(0.52)
Common reliance × Post-GPT	0.0202***	0.0207***	0.0574***
	(2.72)	(2.77)	(4.04)
Common reliance $\times M^X \times$ Post-GPT	–	0.0773***	-0.0135
		(3.44)	(-1.29)
Current similarity	0.7948***	0.7907***	0.7872***
	(370.75)	(360.76)	(349.73)
Panel B: Future cosine similarity			
Common reliance	0.0180*	0.0121	-0.0745***
	(1.81)	(1.16)	(-4.42)
Common reliance × Post-GPT	0.1575***	0.1704***	0.4174***
	(8.81)	(9.57)	(11.36)
Common reliance $\times M^X \times$ Post-GPT	–	0.2386***	-0.1610***
		(4.47)	(-6.13)
Current similarity	0.6736***	0.6705***	0.6644***
	(222.89)	(220.68)	(213.14)
All lower-order interactions	Yes	Yes	Yes
Pair controls	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
SE cluster	Pair	Pair	Pair
Observations	890,394	859,357	859,357

Notes: Each panel and column is a separate regression. Column (1) has no firm moderator. Columns (2)–(3) include all lower-order terms implied by the triple interaction and the corresponding interactions with average reliance. Robust t-statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

2.3 Implications for the Model

The evidence above identifies a predictive relationship between common alignment with AI-extractable information and subsequent portfolio similarity. The relationship is consistent with investors responding to a shared information source. In a general equilibrium model, more similar portfolio decisions can come from more accurate fundamental information and higher exposure to a common signal error; we therefore use the evidence to motivate common information in the model while keeping fundamental information and model error separate. Appendix A shows how information changes can generate similar desired holdings at given prices. In the baseline model, market clearing determines how a common shift in demand enters prices when the supply of financial claims is fixed.

Recognition of shared information. The literature and survey evidence suggest that investors may not fully recognize shared information sources. The Bank of England and Financial Conduct Authority survey reports that 46% of respondents using or planning to use AI had only a partial understanding of the technologies they employed (Bank of England and Financial Conduct Authority, 2024). About one-third of reported AI use cases relied on third-party implementations, which may themselves use a common underlying model. The survey report also raises concerns about model concentration and correlated outcomes. The Financial Stability Board points more specifically to limited transparency about training data and the possibility of unknowingly using common information sources (Financial Stability Board, 2024). These observations motivate allowing investors to underestimate the common component of AI model error, which we introduce in our baseline model.

Experimental evidence helps distinguish this hypothesis from a general tendency to overtrust AI. Agarwal et al. (2026) find that radiologists both underweight AI advice and fail to account fully for its dependence on their own information. Thus, placing too little weight on AI on average can coexist with giving too much weight to overlapping evidence. The financial analog in our model is an investor who treats an AI signal and a price partly driven by that signal as more independent than they are. Evidence on correlation neglect (Enke and Zimmermann, 2019) and on financial markets' response to recombined old news (Fedyk and Hodson, 2023) provides related support for difficulties in tracing information overlap. We represent this difficulty through the correlation-belief parameter introduced below.

Prices and real investment. The final link is firms' use of financial information. Chen et al. (2007) find that investment is more sensitive to stock prices when those prices contain more private information, consistent with managers learning from markets. Goldstein et al. (2026) provide direct evidence from surveys of Chinese listed firms. In their 2022 survey, 80.4% of

firms that monitored their own stock price reported doing so to learn information relevant to real investment. Among firms reporting such learning, 59.5% agreed that stock prices help them learn about customer demand. This finding motivates our focus on demand information: managers may understand their own production conditions while still benefiting from information about their customers and market environment. The broader literature reviewed by Bond et al. (2012) explains how this learning connects financial prices to real decisions. We adopt the learning channel, with firms updating their beliefs about demand and choosing capital optimally. Financial trading has real effects through the information in prices even when it does not directly finance firms' investment. Recent evidence also connects investor AI use to the information firms obtain from prices. Huang and You (2026) report that stocks held by AI-adopting mutual funds become more informative about short-term earnings and less informative about longer-term earnings, while corporate investment becomes less sensitive to prices. This implies that AI usage could indeed affect the informativeness of the market price, which in turn affects the firm's investment decisions.

The above observations motivate a model with common AI information, potentially imperfect recognition of its overlap with prices, and firms that learn from those prices. The portfolio evidence bears directly on the first element; the other two draw on external evidence and established economic mechanisms. Portfolio similarity alone does not determine whether prices become more informative or capital allocation improves. The model studies these equilibrium consequences, and its aggregate productivity effects remain theoretical implications to be assessed with further evidence.

3 Model

The model studies how investors' information choices affect firms through financial prices. Its production and price-learning environment builds on Benhabib et al. (2019). The main departure is the choice between private research and AI information whose model error is shared across investors. Investors may underestimate this common component, while firms understand the true information content of prices.

3.1 Economic Environment and Timing

There are N intermediate-good firms, indexed by $j = 1, \ldots, N$. Define

$$\delta \equiv \frac{1}{\theta}, \qquad \chi \equiv 1 - \frac{1}{\theta}, \qquad \theta > 1. \tag{3.1}$$

A competitive final-good producer combines intermediate goods according to:

$$Y_N = \left[\frac{1}{N} \sum_{j=1}^{N} \mathcal{E}_j^{\delta} Y_j^{\chi} \right]^{1/\chi}, \tag{3.2}$$

where $\mathcal{E}_j$ is a firm-specific product-demand shifter. The associated inverse demand curve is:

$$P_j = Y_N^{\delta} \mathcal{E}_j^{\delta} Y_j^{-\delta}. \tag{3.3}$$

Firm j produces with:

$$Y_j = Z A_j K_j^{\eta}, \qquad 0 < \eta < 1, \tag{3.4}$$

where Z is an aggregate productivity component known before trading. Capital fully depreciates, and its price is normalized to one. Firm fundamental shocks $\{A_j, \mathcal{E}_j\}$ follow independent log-normal distributions:

$$\log A_j = \mu_a + a_j, \qquad \log \mathcal{E}_j = \mu_d + d_j, \tag{3.5}$$

where:

$$a_j \sim \mathcal{N}(0, \tau_a^{-1}), \qquad d_j \sim \mathcal{N}(0, \tau_d^{-1}). \tag{3.6}$$

The fundamental shocks are independent across firms. Firms and investors share these priors. Signal errors and noise-trader demands are independent of fundamentals and independent across firms; the AI error introduced below is common across investors within a firm.

Firms and investors observe a public signal of firm productivity:

$$s_j = a_j + e_j, \qquad e_j \sim \mathcal{N}(0, \tau_s^{-1}), \tag{3.7}$$

Define

$$X_j \equiv \mathcal{E}_j^{\delta} A_j^{\chi}, \qquad g \equiv \eta\chi \in (0,1), \qquad \mu_L \equiv \chi\mu_a + \delta\mu_d. \tag{3.8}$$

Then the firm's revenue is

$$V_j = P_j Y_j = Y_N^{\delta} Z^{\chi} X_j K_j^{g}. \tag{3.9}$$

Following Benhabib et al. (2019), the financial claim pays log revenue. Lowercase letters denote the logarithms of the corresponding capital, output, and aggregate productivity variables:

$$v_j \equiv \log V_j = \delta y_N + \chi z + \mu_L + \chi a_j + \delta d_j + g k_j. \tag{3.10}$$

Trading a log-revenue claim is a tractability device: its payoff is conditionally Gaussian, so portfolio demand and the equilibrium price are linear. The claim can be interpreted as a firm-

value derivative rather than literal equity. The claim is in zero net supply; real capital remains an endogenous choice of the firm.

There is a unit mass of atomistic asset managers, indexed by $i \in [0,1]$. We solve the economy in the limit as the number of firms tends to infinity, conditional on known Z. Aggregate output is then known before trading and investment. Finite-N expressions below provide allocation accounting under the limiting investment rule.

At date 0, managers choose how much private and AI information to acquire. At date 1, firm fundamentals, signal errors, and noise-trader demands are realized. Managers observe their signals and trade firm-value claims. Firms then observe their public productivity signals and the resulting prices and choose capital. Managers anticipate the firms' investment rule when trading. At date 2, firms produce and financial claims pay off.

3.2 Information Acquisition and Correlation Beliefs

Managers observe the firm's productivity signal s_j and can acquire private and AI signals about its demand shock d_j:

$$\begin{aligned} s_{ij}^P &= d_j + u_{ij} + \epsilon_{ij}^P, & u_{ij} &\sim \mathcal{N}(0, \tau_u^{-1}), \\ s_{ij}^A &= d_j + b_j + \epsilon_{ij}^A, & b_j &\sim \mathcal{N}(0, \tau_b^{-1}). \end{aligned} \tag{3.11}$$

The investor-specific errors satisfy $\epsilon_{ij}^P \sim \mathcal{N}(0, (x_{ij}^P)^{-1})$ and $\epsilon_{ij}^A \sim \mathcal{N}(0, (x_{ij}^A)^{-1})$. Managers pay to increase the raw precisions x_{ij}^P and x_{ij}^A. Private and acquisition errors are mutually independent and independent across managers. They are also independent of b_j and noise-trader demand, while b_j is shared by all investors researching firm j. The effective precisions and their quantity–composition representation are:

$$\begin{aligned} h_{ij}^P &= \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P}, & h_{ij}^A &= \frac{\tau_b x_{ij}^A}{\tau_b + x_{ij}^A}, \\ h_{ij} &= h_{ij}^P + h_{ij}^A, & \alpha_{ij} &= \frac{h_{ij}^A}{h_{ij}}. \end{aligned} \tag{3.12}$$

Thus h measures total effective precision and α the share supplied by AI. More acquisition reduces investor-specific noise, but effective precision remains bounded by the source's underlying error: $h_{ij}^P < \tau_u$ and $h_{ij}^A < \tau_b$. When suppressing the firm index, we also write $h_{Pi} = h_i^P$ and $h_{Ai} = h_i^A$.

The AI signal represents costly processing of a shared model forecast. Access to the same technology permits different signal realizations because acquisition noise remains investor-specific; the underlying forecast is not directly observed by all investors. This interpretation

follows the distinction between a common information source and the signals obtained by processing it (Myatt and Wallace, 2012). We take the model's precision as given in the baseline.[8]

We assume that investors pay separable power costs for the signals, with positive cost parameters and exponent $\nu > 1$, as in the precision-cost approach of Colombo et al. (2014). In effective-precision coordinates, the per-firm cost is:

$$\mathcal{C}^I(h_i, \alpha_i) = \frac{c_A}{\nu}\left(\frac{\tau_b \alpha_i h_i}{\tau_b - \alpha_i h_i}\right)^{\nu} + \frac{c_P}{\nu}\left(\frac{\tau_u(1-\alpha_i)h_i}{\tau_u - (1-\alpha_i)h_i}\right)^{\nu}, \qquad \nu > 1. \tag{3.13}$$

Section 6 relaxes this specification. The distinction between quantity and composition will also be useful there.

Managers also extract information from the market price q_j. Since the production function is log-linear and the firm-value claim traded is the log of firm revenue, we conjecture that:

$$q_j = \beta_0 + \beta_s s_j + \beta_q \tilde{q}_j, \qquad \tilde{q}_j = d_j + \psi b_j + \zeta n_j, \qquad n_j \sim \mathcal{N}(0, \tau_n^{-1}), \tag{3.14}$$

where $\tilde{q}_j$ is the standardized price signal and n_j is noise-trader demand. Observing s_j and the equilibrium price coefficients allows investors to recover $\tilde{q}_j$ from q_j. They therefore combine three signals about demand, $(s_{ij}^P, s_{ij}^A, \tilde{q}_j)$. Accounting for a shared error helps distinguish demand information from the component repeated in AI and prices.

Correlation beliefs. Although the AI model error b_j is fully common, managers may perceive only a fraction $\kappa \in [0,1]$ of its variance as common. Their subjective Gaussian representations of the AI signal and the normalized price signal are:

$$s_{ij}^A = d_j + \sqrt{\kappa} b_j^c + \sqrt{1-\kappa} b_{ij}^I + \epsilon_{ij}^A, \tag{3.15}$$

$$\tilde{q}_j = d_j + \psi\sqrt{\kappa} b_j^c + \zeta n_j, \tag{3.16}$$

where b_j^c is the perceived AI common error, and b_{ij}^I is the perceived idiosyncratic shock. They are subjectively independent, and both have precision τ_b. This representation preserves the manager's perceived marginal variance of the AI signal, while changing only the perceived separation between the common and idiosyncratic parts. Investors use the equilibrium price coefficients but misspecify the cross-sectional structure of AI errors: they perceive a fraction κ of model-error variance as common. Consequently, they also understate the variance of the price error when $\kappa < 1$. At $\kappa = 1$, beliefs are correct. At $\kappa = 0$, a manager believes that the

[8] Financial models can use market outcomes as training inputs. For example, Liu et al. (2023) use stock-price changes to construct labels for fine-tuning. Appendix B.10 allows the information quality of one period's market outcomes to determine the next period's model precision. Under a smooth, locally stable training feedback, the local productivity comparisons in Proposition 5 continue to hold.

fixed AI error is idiosyncratic and therefore washes out of the price. The manager still learns from the price, but incorrectly treats it as an independent confirmation of the AI signal.

The parameter κ formalizes the difficulty of recognizing overlapping information discussed in Section 2.3. It does not change the perceived marginal accuracy of the investor's own AI signal, and it is not a general measure of trust in AI. A low value captures the possibility that shared data or models make seemingly distinct information sources less independent than their users believe. We treat it as a sensitivity parameter, rather than estimate it from the portfolio evidence.

3.3 Decisions and Equilibrium

Managers have mean-variance preferences with risk-aversion coefficient $\rho > 0$. Suppressing the firm index where there is no ambiguity, they anticipate a log-linear investment rule:

$$k = k_0 + k_s s + k_q \tilde{q}, \tag{3.17}$$

where (k_0, k_s, k_q) are coefficients common to all firms. We will verify this log-linear investment decision later. Although investment occurs immediately after price formation, managers know this rule when trading. Hence, conditional on (s, q), subsequent investment is known and affects expected payoff but not conditional payoff variance.

Managers allocate wealth between a risk-free asset and firm-value claims. The risk-free return is normalized to zero, and m_i denotes manager i's position in a given claim.[9] Let $\mathcal{I}_i$ contain the public productivity signal, the two acquired signals, and the price. Conditional on this information, manager i evaluates position m_i according to:

$$\mathcal{V}_i(m_i \mid \mathcal{I}_i) = m_i \left(\mathbb{E}_i^{\kappa}[v \mid \mathcal{I}_i] - q\right) - \frac{\rho}{2} m_i^2 \operatorname{Var}_i^{\kappa}(v \mid \mathcal{I}_i). \tag{3.18}$$

Before signals are realized, the information choice solves:

$$\max_{h_i, \alpha_i} \left\{ \mathbb{E}_{0,i}^{\kappa} \left[\max_{m_i} \mathcal{V}_i(m_i \mid \mathcal{I}_i) \right] - \mathcal{C}^I(h_i, \alpha_i) \right\}. \tag{3.19}$$

This criterion first evaluates the mean-variance trading problem conditional on each information realization and then averages its optimized value.

Firms take aggregate output as given and maximize expected revenue net of capital cost, $\mathbb{E}[V_j \mid s_j, q_j] - K_j$. They use the true distribution of the price signal. This separates the effect

[9] Following Van Nieuwerburgh and Veldkamp (2010) and Benhabib et al. (2019), we assume that asset managers can hold unrestricted positions in the risk-free asset, so the claim position is not restricted by a binding portfolio budget constraint.

of investors' beliefs on information aggregation from the firms' optimal use of the information they receive.

The firm-value claim is in zero net supply, and noise traders hold n_j. Market clearing for each firm implies:

$$\int_0^1 m_{ij}^* \, di + n_j = 0. \tag{3.20}$$

A symmetric equilibrium consists of common information choices, individual portfolio rules, price coefficients, and firm investment rules. Managers' information and portfolio choices are optimal under their subjective beliefs, firms' investment choices are optimal under the true signal distribution, and financial markets clear under realized signals. Each manager takes the aggregate price and investment rules as given when choosing information. We focus on interior, regular equilibria satisfying the individual second-order conditions.

4 Information Aggregation and Capital Allocation

We solve backward from portfolio choice to information acquisition and price formation, then characterize the investment decisions supported by those prices. The key distinction is between investors' perceived information and the true precision of the signal available to firms.

4.1 Investor Learning and Portfolio Choice

Since all the shocks are independent across firms, we suppress the subscript j for simplicity. Define:

$$v_{Ai} \equiv (\alpha_i h_i)^{-1}, \qquad v_{Pi} = ((1-\alpha_i)h_i)^{-1}, \qquad v_q^\kappa \equiv \frac{\kappa\psi^2}{\tau_b} + \frac{\zeta^2}{\tau_n}, \qquad c_{Aq}^\kappa \equiv \frac{\kappa\psi}{\tau_b}. \tag{4.1}$$

Conditional on d, the perceived error covariance matrix of $(s_i^P, s_i^A, \tilde{q})'$ is:

$$R_i^\kappa = \begin{pmatrix} h_{Pi}^{-1} & 0 & 0 \\ 0 & v_{Ai} & c_{Aq}^\kappa \\ 0 & c_{Aq}^\kappa & v_q^\kappa \end{pmatrix}. \tag{4.2}$$

Define $D_i^\kappa = v_{Ai} v_q^\kappa - (c_{Aq}^\kappa)^2 > 0$. The manager's subjective posterior distribution of demand d satisfies:

Lemma 1 (Managers' subjective posterior) *The subjective posterior mean of d is*

$$\widehat{d}_i^\kappa = \ell_{Pi}^\kappa s_i^P + \ell_{Ai}^\kappa s_i^A + \ell_{qi}^\kappa \tilde{q}, \tag{4.3}$$

where:

$$V_{di}^{\kappa} \equiv \mathrm{Var}_i^{\kappa}(d \mid s_i^P, s_i^A, \tilde{q}) = \left[\tau_d + h_{Pi} + \frac{v_{Ai} + v_q^{\kappa} - 2c_{Aq}^{\kappa}}{D_i^{\kappa}}\right]^{-1}, \tag{4.4}$$

and

$$\ell_{Pi}^{\kappa} = V_{di}^{\kappa} h_{Pi}, \qquad \ell_{Ai}^{\kappa} = V_{di}^{\kappa} \frac{v_q^{\kappa} - c_{Aq}^{\kappa}}{D_i^{\kappa}}, \qquad \ell_{qi}^{\kappa} = V_{di}^{\kappa} \frac{v_{Ai} - c_{Aq}^{\kappa}}{D_i^{\kappa}} \tag{4.5}$$

Proof. See Appendix B.1. ■

The public signal separately determines the posterior of productivity:

$$\widehat{a}_j = \omega_s s_j, \qquad \omega_s \equiv \frac{\tau_s}{\tau_a + \tau_s}, \qquad \mathrm{Var}(a_j \mid s_j) = \frac{1}{\tau_a + \tau_s}. \tag{4.6}$$

Thus, the conditional payoff variance satisfies:

$$\Omega_i^{\kappa} \equiv \mathrm{Var}_i^{\kappa}(v \mid \mathcal{I}_i) = \frac{\chi^2}{\tau_a + \tau_s} + \delta^2 V_{di}^{\kappa}. \tag{4.7}$$

Investment does not contribute to conditional payoff variance because managers know the rule $k(s, \tilde{q})$. The following lemma characterizes each asset manager's optimal portfolio choice:

Lemma 2 *The optimal holding is*

$$m_i^* = \frac{\widehat{v}_i^{\kappa} - q}{\rho \Omega_i^{\kappa}}, \qquad \widehat{v}_i^{\kappa} \equiv \mathbb{E}_i^{\kappa}[v \mid \mathcal{I}_i]. \tag{4.8}$$

The optimized conditional value is

$$\max_{m_i} \mathcal{V}_i(m_i \mid \mathcal{I}_i) = \frac{(\widehat{v}_i^{\kappa} - q)^2}{2\rho \Omega_i^{\kappa}}. \tag{4.9}$$

Proof. See Appendix B.2. ■

4.2 Information Acquisition and Price Formation

Before signals arrive, managers choose (h_i, α_i) to maximize the expected optimized trading value net of acquisition costs. An atomistic deviator changes its own posterior coefficients $(\ell_{Pi}^{\kappa}, \ell_{Ai}^{\kappa}, \ell_{qi}^{\kappa})$ when changing (h_i, α_i), but takes the aggregate price rule, the firm's public investment rule, and aggregate demand as given. In equilibrium, the expected optimized conditional value is determined by $\mathbb{E}_{0,i}^{\kappa}(\widehat{v}_i^{\kappa} - q)^2$, which is characterized by the following proposition:

Proposition 1 *Let $\pi_{Ai} \equiv x_i^A = \tau_b h_{Ai}/(\tau_b - h_{Ai})$ denote raw AI precision, and define*

$$H_i^\kappa = \frac{(\gamma_{Pi} + \gamma_{Ai} + \gamma_{qi})^2}{\tau_d} + \frac{\kappa(\gamma_{Ai} + \psi\gamma_{qi})^2}{\tau_b} + \frac{(1-\kappa)\gamma_{Ai}^2}{\tau_b} + \frac{\gamma_{Pi}^2}{h_{Pi}} + \frac{\gamma_{Ai}^2}{\pi_{Ai}} + \frac{(\zeta\gamma_{qi})^2}{\tau_n}, \tag{4.10}$$

where:

$$\gamma_{Pi} = \delta\ell_{Pi}^\kappa, \qquad \gamma_{Ai} = \delta\ell_{Ai}^\kappa, \qquad \gamma_{qi} = \delta(\ell_{qi}^\kappa - \bar{\ell}_T^\kappa), \qquad \bar{\ell}_T^\kappa \equiv \int \ell_{Pi}^\kappa di + \int \ell_{Ai}^\kappa di + \int \ell_{qi}^\kappa di, \tag{4.11}$$

Then the manager's subjective gross ex ante information value is

$$\mathcal{G}_i^\kappa(h_i, \alpha_i) = \mathbb{E}_{0,i}^\kappa\left[\frac{(\widehat{v}_i^\kappa - q)^2}{2\rho\Omega_i^\kappa}\right] = \frac{H_i^\kappa}{2\rho\Omega_i^\kappa}. \tag{4.12}$$

Proof. See Appendix B.3. ■

The numerator H_i^κ measures the ex ante variation in perceived expected excess returns, or the trading opportunities generated by information. The denominator Ω_i^κ measures residual payoff risk after the signals are observed. Information can be valuable both because it changes the state-contingent expected return and because it lowers residual risk.

For an interior information choice, the first-order conditions are:

$$\frac{\partial\mathcal{G}_i^\kappa}{\partial h_i} = \mathcal{C}_h^I, \qquad \frac{\partial\mathcal{G}_i^\kappa}{\partial\alpha_i} = \mathcal{C}_\alpha^I. \tag{4.13}$$

When differentiating (4.12), the deviator allows its own V_{di}^κ, posterior coefficients, H_i^κ, and Ω_i^κ to change, while holding $(\psi, \zeta, \bar{\ell}_T^\kappa)$ fixed.

In equilibrium, idiosyncratic signal errors wash out, whereas the common AI error does not:

$$\int_0^1 s_{ij}^P\, di = d_j, \qquad \int_0^1 s_{ij}^A\, di = d_j + b_j. \tag{4.14}$$

Considering a symmetric equilibrium, where all investors make the same information acquisition decisions, the true signal realizations give:

$$\int_0^1 \widehat{d}_{ij}^\kappa\, di = (\ell_P^\kappa + \ell_A^\kappa + \ell_q^\kappa)d_j + (\ell_A^\kappa + \psi\ell_q^\kappa)b_j + \zeta\ell_q^\kappa n_j. \tag{4.15}$$

Combining the conjectured investment decisions, managers' portfolio holding decisions, and the market clearing condition, we can solve for the equilibrium price q_j:

Proposition 2 (Financial-information equilibrium) *Suppose the symmetric information choices*

are interior, the covariance matrices are positive definite, and the individual information objective is locally concave. In equilibrium, the price coefficients are:

$$\beta_q = \delta(\ell_P^\kappa + \ell_A^\kappa + \ell_q^\kappa) + gk_q, \tag{4.16}$$

$$\beta_s = \chi\omega_s + gk_s, \tag{4.17}$$

$$\beta_0 = \delta y_N + \chi z + \mu_L + gk_0. \tag{4.18}$$

Proof. See Appendix B.4. ■

The symmetric equilibrium solves the following four equations:

$$\begin{aligned} \psi &= \frac{\ell_A^\kappa}{\ell_P^\kappa + \ell_A^\kappa}, \qquad \zeta = \frac{\rho\Omega^\kappa}{\delta(\ell_P^\kappa + \ell_A^\kappa)}, \\ \mathcal{G}_{h_P}^\kappa &= \mathcal{C}_{h_P}^I, \qquad \mathcal{G}_{h_A}^\kappa = \mathcal{C}_{h_A}^I. \end{aligned} \tag{4.19}$$

The equilibrium is a subjective-beliefs equilibrium. Managers optimize using their perceived covariance matrix, but prices clear under the true realization of the common error. When managers underestimate commonness, they treat overlapping information in AI and prices as more independent than it is. Two limiting cases clarify the mechanism. At $\kappa = 1$, the covariance matrix in (4.2) is the correct covariance matrix and investors correctly account for the dependence among all three demand signals. At $\kappa = 0$,

$$\ell_P^0 = V_d^0 h_P, \qquad \ell_A^0 = V_d^0 h_A, \qquad \psi = \frac{h_A}{h_P + h_A} = \alpha. \tag{4.20}$$

Thus complete correlation neglect delivers the reduced-form benchmark in which the true common-error loading of the price equals the AI information share. Price learning itself is not shut down: ℓ_q^0 generally remains positive. What disappears is the covariance correction connecting the price error to the manager's own AI error.

4.3 Firm Investment and Aggregate Productivity

Firm j observes (s_j, q_j) and, using the equilibrium price coefficients, recovers the normalized price:

$$\widetilde{q}_j = \frac{q_j - \beta_0 - \beta_s s_j}{\beta_q} = d_j + \psi b_j + \zeta n_j. \tag{4.21}$$

Firms know the true distribution of the price signal. This assumption isolates the effect of managers' correlation neglect on price formation; it does not require firms to make the same

mistake. Define the true price-error variance and precision by:

$$S_q^T \equiv \frac{\psi^2}{\tau_b} + \frac{\zeta^2}{\tau_n}, \qquad \tau_q^T \equiv (S_q^T)^{-1}. \tag{4.22}$$

Given the productivity and price signals, the firm's posterior means of its productivity and demand shocks are, respectively:

$$\widehat{a}_j^F = \omega_s s_j, \qquad \omega_s = \frac{\tau_s}{\tau_a + \tau_s}, \tag{4.23}$$

$$\widehat{d}_j^F = \omega_q^T \widetilde{q}_j, \qquad \omega_q^T = \frac{\tau_q^T}{\tau_d + \tau_q^T}. \tag{4.24}$$

Because $\log X_j = \mu_L + \chi a_j + \delta d_j$, the conditional expectation of revenue productivity is:

$$M_j \equiv \mathbb{E}[X_j \mid s_j, \widetilde{q}_j] = \exp\left\{\mu_L + \chi\omega_s s_j + \delta\omega_q^T \widetilde{q}_j + \frac{1}{2}\Sigma^F\right\}, \tag{4.25}$$

where

$$\Sigma^F \equiv \frac{\chi^2}{\tau_a + \tau_s} + \frac{\delta^2}{\tau_d + \tau_q^T}, \tag{4.26}$$

denotes the conditional variance of the log revenue productivity $\log X_j$. Given these posterior beliefs, each firm takes aggregate output as given and chooses capital. The optimal investment rule is log-linear, verifying our conjecture:

Proposition 3 (Firm investment rules) *Let:*

$$\Theta \equiv \frac{1}{1-g} > 1. \tag{4.27}$$

Conditional on known aggregate output, the atomistic investment rule is:

$$K_j = \left[g Y_N^{\delta} Z^{\chi} M_j\right]^{\Theta}. \tag{4.28}$$

Thus, the coefficients in the log-linear investment rule (3.17) are given by:

$$k_s = \Theta\chi\omega_s, \tag{4.29}$$

$$k_q = \Theta\delta\omega_q^T, \tag{4.30}$$

$$k_0 = \Theta\left[\log g + \delta y_N + \chi z + \mu_L + \frac{1}{2}\Sigma^F\right]. \tag{4.31}$$

Proof. See Appendix B.5. ■

A realized common AI error affects investment according to:

$$\frac{\partial \log K_j}{\partial b_j} = k_q \psi > 0. \tag{4.32}$$

The firm understands that the price is noisy and optimally attenuates its response through ω_q^T. It nevertheless cannot distinguish d_j from b_j in a single price realization. Managers' correlation neglect therefore matters because it changes the true equilibrium coefficients (ψ, ζ) and hence the precision of the demand signal observed by the firm.

Aggregate TFP. Let $K_N = N^{-1} \sum_j K_j$ denote aggregate investment. The individual investment rule implies $K_j = K_N M_j^\Theta / [N^{-1} \sum_\ell M_\ell^\Theta]$, so aggregate output can be written as $Y_N = Z \mathcal{A}_N^R K_N^\eta$, where $\mathcal{A}_N^R$ can be regarded as the aggregate TFP. Appendix B.6 derives the realized finite-sample index $\mathcal{A}_N^R$, which depends on the realized firm-level fundamentals. As in Benhabib et al. (2019), the equilibrium interpretation applies as $N \to \infty$, when each firm is atomistic. Conditional on known Z, the law of large numbers replaces cross-sectional averages by their expectations, so limiting aggregate TFP is nonstochastic.

Proposition 4 (Limiting aggregate TFP) *Suppose the primitive firm-level shocks are independent and identically distributed across j and have finite relevant moments. Then as the number of firms $N \to \infty$, the aggregate TFP becomes:*

$$\mathcal{A}_N^R \xrightarrow{a.s.} \mathcal{A} = \left\{ \mathbb{E}[M^\Theta] \right\}^{1/\chi - \eta}. \tag{4.33}$$

Define:

$$\Sigma_0 \equiv \frac{\chi^2}{\tau_a} + \frac{\delta^2}{\tau_d}, \tag{4.34}$$

Then aggregate TFP can be expressed as:

$$\log \mathcal{A} = \left(\frac{1}{\chi} - \eta \right) \left[\Theta \mu_L + \frac{\Theta^2}{2} \Sigma_0 - \frac{\Theta(\Theta - 1)}{2} \Sigma^F \right]. \tag{4.35}$$

Proof. See Appendix B.7. ■

Although b_j is independent across firms and therefore washes out of the cross-sectional average in levels, its variance remains in each firm's price signal. It consequently affects the deterministic limiting TFP through Σ^F. For aggregate capital, the unit capital cost and firms' first-order conditions imply $K = gY$ in the limit with $N \to \infty$. Combining this identity with $Y = Z \mathcal{A} K^\eta$ gives:

$$K = (gZ\mathcal{A})^{1/(1-\eta)}, \qquad Y = g^{\eta/(1-\eta)} (Z\mathcal{A})^{1/(1-\eta)}. \tag{4.36}$$

The allocation-efficiency object $\mathcal{A}$ separates changes in the distribution of investment from changes in aggregate capital. Holding production parameters fixed, output and TFP move in the same direction, with the output response amplified by endogenous aggregate investment.

Price informativeness and aggregate TFP. The effect of the financial equilibrium on real allocation is summarized by the true error variance of the standardized price signal,

$$S_q^T(\kappa; c_A, \tau_b) = \underbrace{\frac{\psi(\kappa; c_A, \tau_b)^2}{\tau_b}}_{\text{common-AI component}} + \underbrace{\frac{\zeta(\kappa; c_A, \tau_b)^2}{\tau_n}}_{\text{noise-trader component}}. \tag{4.37}$$

The variance S_q^T measures noise in the demand signal conveyed by prices. The model error is common across investors researching a given firm and independent across firms, so it can distort capital allocation without creating an aggregate shock in the economy.

AI reliance $\alpha = h_A/(h_P + h_A)$ matters because it changes the equilibrium price loadings, but it is not by itself a sufficient statistic for allocation. A rise in AI reliance can increase ψ while a simultaneous increase in total information reduces ζ. Firms care about the sum of the two true error components. Since:

$$\frac{1}{\tau_d + \tau_q^T} = \frac{S_q^T}{1 + \tau_d S_q^T}, \tag{4.38}$$

the price error variance determines the equilibrium aggregate TFP. Only the second term of Σ^F depends on S_q^T. Equations (4.26) and (4.38) imply:

$$\frac{\partial \Sigma^F}{\partial S_q^T} = \frac{\delta^2}{(1 + \tau_d S_q^T)^2}.$$

Differentiating (4.35) then gives:

$$\frac{\partial \log \mathcal{A}}{\partial S_q^T} = -\frac{(1/\chi - \eta)\Theta(\Theta - 1)\delta^2}{2(1 + \tau_d S_q^T)^2} < 0. \tag{4.39}$$

Thus a parameter change improves allocation if and only if it reduces the true error variance of the price signal, holding production parameters fixed. AI progress affects both components through information acquisition. More informed trading can reduce the price response to noise demand, while greater reliance on a shared signal can increase exposure to its common error. The equilibrium determines which effect dominates. Firms optimally discount a less precise price, but the resulting increase in posterior uncertainty still lowers allocation efficiency.

At fixed model accuracy, a change in AI acquisition costs affects price noise only through

the equilibrium loadings:

$$\frac{dS_q^T}{dc_A} = \frac{2\psi}{\tau_b}\frac{d\psi}{dc_A} + \frac{2\zeta}{\tau_n}\frac{d\zeta}{dc_A}. \tag{4.40}$$

A cost reduction raises TFP if its effects on information acquisition reduce the total price-error variance. There is no direct change in the variance of the underlying AI error.

Higher model accuracy also changes the underlying variance of the common AI error, adding a direct information benefit:

$$\frac{dS_q^T}{d\tau_b} = \frac{2\psi}{\tau_b}\frac{d\psi}{d\tau_b} - \frac{\psi^2}{\tau_b^2} + \frac{2\zeta}{\tau_n}\frac{d\zeta}{d\tau_b}. \tag{4.41}$$

The middle term is the direct benefit of a less volatile common error. The other two terms capture endogenous changes in price exposure and noise. The next section uses the equilibrium system to compare these responses near the no-AI boundary and then studies their global shape numerically.

5 AI Progress and Aggregate Productivity

The previous analysis shows that investors' information-acquisition decisions affect the precision of firms' price signals, which in turn changes the equilibrium capital allocation. This section examines how AI progress affects information acquisition and capital allocation. As we have discussed earlier, an advancement in AI technology could be characterized by either a decrease in access cost c_A or an increase in model accuracy τ_b. We first characterize their local productivity effects for general beliefs about the common error. Complete correlation neglect then provides a simple account of the roles of information quantity and composition, while correct recognition clarifies how accounting for overlapping information changes the comparison. Numerical analysis traces the equilibrium effects beyond the no-AI boundary.

5.1 Local Productivity Effects of AI Progress

We consider two routes to negligible AI use: prohibitively high acquisition costs at fixed accuracy, and vanishing model accuracy at fixed acquisition costs. Both approach the same economy without AI, in which beliefs about the common AI error have no effect. They differ in how additional AI use changes the true price-error variance. Throughout this subsection, $\kappa \in [0, 1]$ is fixed along each technology comparison.

To describe the common reference economy, let $r(h)$ be the precision of the price signal when investors acquire private information of effective precision h and use no AI information. Given h, the price equations determine $r(h)$; the corresponding individual marginal gross return to

private information, evaluated at the symmetric price rule, is $m(h)$. Write $s(h) = 1/r(h)$ and $\Omega_a = \chi^2/(\tau_a + \tau_s)$. Appendix B.8 derives these functions and establishes $r'(h) > 0$, $s'(h) < 0$, and $m'(h) < 0$. The no-AI information choice h_0 then satisfies $m(h_0) = \mathrm{MC}_P(h_0)$.

Under the acquisition cost in (3.13), the marginal costs of effective precision are

$$\mathrm{MC}_P(h_P) = c_P \frac{\tau_u^{\nu+1} h_P^{\nu-1}}{(\tau_u - h_P)^{\nu+1}}, \tag{5.1}$$

$$\mathrm{MC}_A(h_A; c_A, \tau_b) = c_A \frac{\tau_b^{\nu+1} h_A^{\nu-1}}{(\tau_b - h_A)^{\nu+1}}. \tag{5.2}$$

At the no-AI equilibrium, write $r_0 = r(h_0)$ and $s_0' = s'(h_0)$, and define

$$\begin{aligned} m_0 = m(h_0), \qquad \lambda_0 = \frac{\mathrm{MC}_P'(h_0)}{\mathrm{MC}_P'(h_0) - m'(h_0)} \in (0,1), \qquad S_0 = s(h_0), \\ \Gamma_0 = \frac{(1/\chi - \eta)\Theta(\Theta - 1)\delta^2}{2(1 + \tau_d S_0)^2} > 0. \end{aligned} \tag{5.3}$$

These objects are independent of κ. The coefficient Γ_0 converts a marginal increase in the true price-error variance into a loss of log TFP at the common reference economy.

Proposition 5 (Local TFP effects of AI progress) *Suppose $\nu > 1$ and fix $\kappa \in [0,1]$. Assume an interior, regular no-AI equilibrium $h_0 \in (0, \tau_u)$ and symmetric equilibrium branches approaching it that satisfy the individual local second-order conditions. For positive AI use, consider the branch with $0 < \psi < 1$ and a positive definite perceived signal covariance matrix. Then:*

(i) Fix $\tau_b > 0$ and let $c_A \to \infty$. Effective AI information vanishes, and

$$h - h_0 = \lambda_0 h_A + o(h_A), \qquad \frac{d \log \mathcal{A}}{d c_A} < 0 \quad \text{for sufficiently large } c_A. \tag{5.4}$$

Thus a marginal reduction in a sufficiently high acquisition cost raises aggregate TFP for every fixed correlation belief.

(ii) Fix $c_A \in (0, \infty)$ and let $\tau_b \downarrow 0$. Then

$$\frac{h_A}{\tau_b} \to 1, \qquad \frac{\psi}{\tau_b} \to \frac{1}{h_0 + \kappa r(h_0)}, \qquad \frac{\alpha}{\tau_b} \to \frac{1}{h_0}. \tag{5.5}$$

The right-hand derivative of log TFP at the zero-accuracy boundary is

$$\left.\frac{d \log \mathcal{A}}{d \tau_b}\right|_{0^+} = \frac{\Gamma_0 h_0^2}{(h_0 + \kappa r_0)^2} \left[-s_0' \lambda_0 - \frac{1}{h_0^2} + \frac{\kappa B_0}{h_0^2} \right], \tag{5.6}$$

where B_0 depends only on the no-AI equilibrium. Its expression is given in (B.39) *in Appendix B.9.*

Proof. See Appendix B.9. ■

The proposition distinguishes two reasons why AI use may initially be low. At a fixed positive model accuracy, the common-error price loading starts from zero and grows at first order in effective AI information. Its contribution to price-error variance is therefore second order. Additional information reduces noise-trader contamination at first order, so the initial productivity gain from cheaper access holds even when investors recognize the common error.

The accuracy boundary differs because the variance of the underlying model error diverges as AI use vanishes. Along this path, $h_A/\tau_b \to 1$: investors acquire nearly all the effective precision available from the AI model, even though both h_A and the AI share approach zero. With $\nu > 1$, the marginal cost of raw precision tends to zero as acquisition vanishes, allowing investors to reduce acquisition noise relative to the highly volatile model error. A marginal accuracy improvement therefore expands effective AI information almost one-for-one. Common-error exposure and the reduction in noise-trader contamination both affect price informativeness at first order, so TFP falls if the former effect dominates. Recognition changes this comparison through both information acquisition and inference. For $\kappa > 0$, investors account for the overlap between AI and prices, reducing the common-error loading relative to the AI share: $\psi/\alpha \to h_0/(h_0 + \kappa r_0) < 1$. The coefficient B_0 in (5.6) summarizes how recognition changes the expression that determines the sign of the initial accuracy effect. When $B_0 > 0$, a greater recognition shifts this expression toward a productivity gain. If the accuracy effect is negative at $\kappa = 0$ and positive at $\kappa = 1$, there exists a unique threshold $\bar{\kappa}$ such that a marginal increase in model accuracy reduces TFP when $\kappa < \bar{\kappa}$ and increases TFP when $\kappa > \bar{\kappa}$.

5.2 Information Acquisition and Correlation Beliefs

This subsection considers two limiting cases: complete neglect with $\kappa = 0$ and correct recognition with $\kappa = 1$. Complete correlation neglect separates the roles of information quantity and composition in Proposition 5. Setting $\kappa = 0$ leaves price learning in place but removes the perceived covariance between AI and price errors. The resulting equilibrium reduction makes the acquisition and aggregation mechanisms explicit. Let $r \equiv \tau_n/\zeta^2$ denote the inverse variance of the noise-trader component of the price signal. The following lemma shows that the functions $r(h)$ and $m(h)$ used to describe the no-AI economy also characterize an economy with positive AI use when $\kappa = 0$.

Lemma 3 (Information equilibrium under complete correlation neglect) *Suppose $\kappa = 0$ and maintain the conditions for an interior symmetric financial-information equilibrium. The equilibrium*

conditions reduce to

$$\psi = \alpha, \tag{5.7}$$

$$r = \frac{\tau_n \delta^2 h^2}{\rho^2 [\Omega_a(\tau_d + h + r) + \delta^2]^2}, \tag{5.8}$$

$$\mathcal{C}_h^I(h, \alpha) = m(h), \tag{5.9}$$

$$\mathcal{C}_\alpha^I(h, \alpha) = 0. \tag{5.10}$$

For each $h > 0$, the price equation has a unique positive solution $r(h)$, with $r'(h) > 0$. The marginal gross return $m(h)$ is given in (B.14), with $m(h) > 0$ and $m'(h) < 0$.

Proof. See Appendix B.8. ■

Lemma 3 shows that under complete neglect, the common-error price loading ψ equals the AI information share α, and the information acquisition problem separates into a choice of total information and a cost-minimizing composition at a given quantity. With $\kappa = 0$, the private and AI information have the same marginal gross value per unit of effective precision for the investors. Investors therefore choose the least costly mix of the two sources and equate its marginal cost to marginal benefit $m(h)$:

$$m(h) = \mathrm{MC}_P(h_P) = \mathrm{MC}_A(h_A; c_A, \tau_b), \qquad h = h_P + h_A. \tag{5.11}$$

Composition nevertheless changes the information that prices convey to firms. Since $\psi = \alpha$, the variance of the price-signal error simplifies to:

$$S_q^T = \frac{\alpha^2}{\tau_b} + s(h), \qquad s(h) = \frac{1}{r(h)}. \tag{5.12}$$

More total information acquisition reduces the noise-trader component. A larger AI share instead raises the common-error component at a given model accuracy. AI progress can increase total information while raising prices' exposure to the common model error, leaving its effect on price informativeness ambiguous.

Near the no-AI equilibrium, differentiating the private-information condition in (5.11) gives

$$h - h_0 = \lambda_0 h_A + o(h_A), \qquad \alpha = \frac{h_A}{h_0} + o(h_A). \tag{5.13}$$

Each additional unit of effective AI information displaces $1 - \lambda_0$ units of private research and

increases total precision by λ_0 units. As a result, the price signal error satisfies:

$$S_q^T - S_0 = h_A \left[s'(h_0)\lambda_0 + \frac{1}{h_0^2}\frac{h_A}{\tau_b} \right] + o(h_A). \tag{5.14}$$

The negative first term captures the reduction in noise-trader contamination. The positive second term captures the common AI error transmitted through prices. The ratio h_A/τ_b is the share of an AI signal's error variance attributable to the common model error.

When high acquisition costs constrain AI use and accuracy is fixed, $h_A/\tau_b \to 0$. The acquired signal remains far below the precision ceiling of the model, and investor-specific acquisition noise dominates its error. Cheaper access expands total information, while common-error exposure is negligible at first order. When low model accuracy constrains AI use, the information-choice condition instead implies $h_A/\tau_b \to 1$. With a fixed acquisition-cost parameter and $\nu > 1$, the marginal cost of raw precision approaches zero as acquisition vanishes. Even a small research input can then reduce acquisition noise relative to the highly volatile model error. Investors obtain nearly all the effective precision available from the model, although h_A and α both approach zero.

Greater accuracy therefore raises effective AI information almost one-for-one at this boundary. The common error exposure thus rises even though the underlying common model error variance falls. In the complete-neglect case, the derivative of endogenous aggregate TFP is

$$\left.\frac{d\log \mathcal{A}}{d\tau_b}\right|_{0^+,\,\kappa=0} = \Gamma_0 \left[-s'(h_0)\lambda_0 - \frac{1}{h_0^2} \right]. \tag{5.15}$$

TFP initially falls when $h_0^{-2} > -s'(h_0)\lambda_0$. In this case, investors acquire more precise information, but the resulting price reveals less about demand when the increase in common-error exposure dominates. Firms discount the noisier price, yet their remaining uncertainty rises and capital allocation deteriorates.

Recognition of common errors. Accounting for overlapping information changes both acquisition incentives and the weight placed on AI in prices. At an interior symmetric equilibrium with $0 < \psi < 1$, the posterior formulas and the information first-order conditions imply

$$\frac{\mathcal{G}_{h_A}^{\kappa}}{\mathcal{G}_{h_P}^{\kappa}} = \frac{\mathrm{MC}_A(h_A)}{\mathrm{MC}_P(h_P)} = \left(\frac{v_q^{\kappa} - c_{Aq}^{\kappa}}{v_q^{\kappa} - (c_{Aq}^{\kappa})^2 h_A} \right)^2 < 1, \qquad 0 < \kappa \leq 1. \tag{5.16}$$

AI information has lower marginal value because some of its content is already present in the price. With strictly convex acquisition costs, investors choose a smaller AI share than would

minimize acquisition costs at the same total precision. Meanwhile, when using AI signals, investors also account for the common model error and use the signal more prudently, reducing the equilibrium loading of common model error to $\psi < \alpha$.

Under correct recognition, $\kappa = 1$, investors account fully for this overlap. At the low-accuracy boundary, h_A/τ_b still tends to one, but ψ/τ_b tends to $1/[h_0 + r(h_0)] < 1/h_0$. Correct recognition thus reduces the common-error loading without eliminating it. Indeed, the common component of price-error variance initially rises at rate $1/[h_0 + r(h_0)]^2$. Whether TFP improves depends on whether the reduction in noise-trader contamination is larger. A sufficient condition for a strictly positive initial accuracy effect under correct recognition is

$$\nu \geq 2 - \frac{3h_0}{\tau_u}. \tag{5.17}$$

Appendix B.9 proves this sufficient condition and gives the exact sign criterion for general $\nu > 1$. Our baseline parameterization uses quadratic costs, $\nu = 2$, and therefore satisfies the condition. Under this parameterization, the initial accuracy effect is positive under correct recognition and negative under complete neglect. Proposition 5 then implies a unique threshold $\bar{\kappa} \in (0, 1)$: a marginal accuracy improvement raises TFP when $\kappa > \bar{\kappa}$ and lowers it when $\kappa < \bar{\kappa}$. The positive effect under correct recognition need not extend to all cost curvatures.[10]

5.3 Numerical Analysis

Proposition 5 characterizes the effects of AI progress near the no-AI equilibrium. We now examine how these effects change as AI use expands. The exercises vary model accuracy and acquisition costs separately, and trace their effects on the quantity and composition of information, price informativeness, and aggregate productivity and output. We solve the symmetric equilibrium in the large-N limit and compare different beliefs about the common AI error.

Parameterization. We first parameterize an economy without AI as a common reference for the exercises. Following Benhabib et al. (2019), we set the elasticity of substitution between intermediate goods to $\theta = 6$, the production exponent to $\eta = 0.8$, and the risk-aversion coefficient to $\rho = 8$.[11] To discipline prior and signal uncertainty, we use the U.S. estimates in David et al. (2016) where the standard deviations of fundamental innovations and firms' private-signal errors are 0.45 and 0.39, respectively. We use these magnitudes as a guide to uncertainty in our

[10] Figure B.1 gives an example with nearly linear acquisition costs, $\nu = 1.01$, in which a marginal accuracy improvement initially lowers TFP even under correct recognition of the common model error.

[11] Benhabib et al. (2019)'s choice of η draws on Gopinath et al. (2017). The latter estimate revenue elasticities whose sum ranges from 0.75 to 0.91, providing evidence on curvature rather than a direct estimate of our production exponent.

static environment, rather than as direct estimates of its priors. Taking inverse variances gives $\tau_a = 0.45^{-2} = 4.9383$ and $\tau_s = 0.39^{-2} = 6.5746$. The resulting posterior standard deviation of log productivity is approximately 0.295. We hold τ_s fixed as financial information changes. We set the prior precision of demand shocks to $\tau_d = 0.04$, using the value in Benhabib et al. (2019) as a literature benchmark.

For noise-trader demand, we set $\tau_n = 1000$, corresponding to a noise-demand standard deviation of approximately 0.0316, using the quarterly noise-trading intensity in Peress and Schmidt (2021) as an order-of-magnitude reference.[12] For the information acquisition cost, we use quadratic costs, $\nu = 2$, and set the private-signal precision ceiling to $\tau_u = 0.02$, as illustrative choices: private research remains noisy even as acquisition effort increases. Conditional on these values, we choose $c_P = 1.12605$ so that the equilibrium price-signal precision without AI, $1/S_q^T$, matches the price-signal precision of 0.0655 in the higher-information equilibrium reported by Benhabib et al. (2019). The resulting private-information precision is $h_P = h = 0.01358 < \tau_u$, which implies that the private research reaches around 68% of the maximum private signal precision, τ_u.

In the numerical exercise, we compare four cases with $\kappa = 0$, $\kappa = 0.25$, $\kappa = 0.75$, and $\kappa = 1$, representing complete correlation neglect, partial recognition, and correct beliefs about the common error. These values illustrate how beliefs affect the transmission of AI progress; they are not estimates of investors' recognition of shared errors. The AI cost c_A and accuracy τ_b are specified within each exercise. We report TFP as an index with its value in the economy without AI normalized to 100; that reference economy is the same across beliefs because it contains no AI signal.

Holding production parameters fixed, output and TFP move in the same direction, with $\Delta \log Y = \Delta \log \mathcal{A}/(1-\eta)$. We therefore show four panels for each exercise: TFP, the common-error component of price noise, total information, and the AI share. The no-AI economy provides a common normalization, not a calibration of current AI adoption.

Changes in AI accuracy. We first increase model accuracy τ_b while holding the acquisition-cost parameters equal, $c_A = c_P$. Figure 1 shows a J-shaped TFP response under complete correlation neglect: productivity initially declines, before recovering and rising above the no-AI benchmark.

Higher model accuracy improves the precision of a given amount of AI research and raises the return to additional acquisition. Investors therefore acquire more AI information and partly substitute away from private research, increasing both total information h and the AI share α

[12] At the quarterly frequency, Peress and Schmidt (2021) reports bounds of 2.2% to 4.3% for noise-trading intensity using price-improved TAQ trades from 2010 to 2015. These bounds concern market-level trading intensity constructed from turnover. They are not a direct estimate of the variance of our normalized claim demand; the mapping is illustrative and depends on the demand normalization.

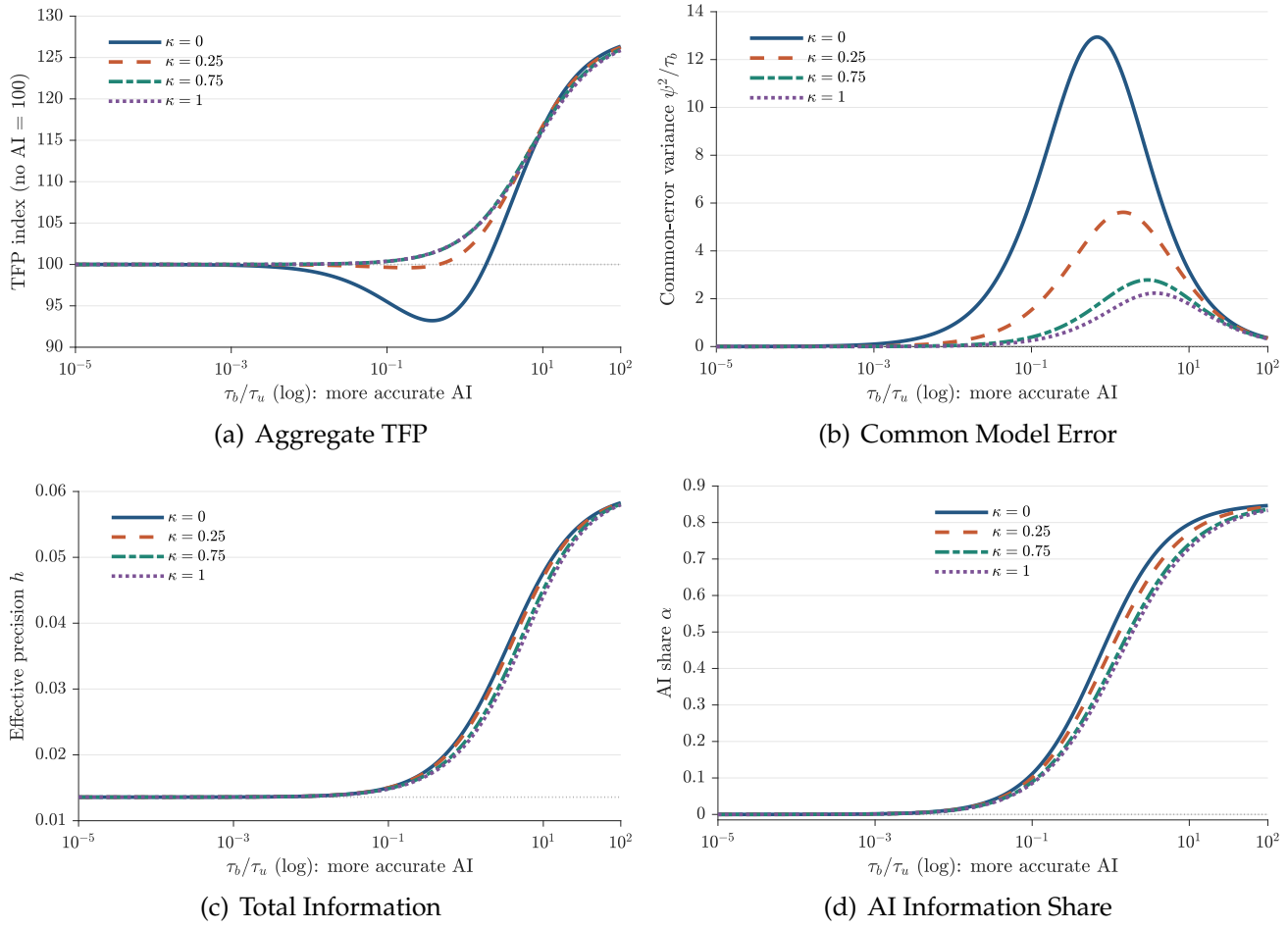
in Panels C and D. The expansion strengthens informed trading relative to noise demand and reduces the noise-trader component of price error. However, greater AI reliance also raises the equilibrium loading ψ. Starting from low accuracy, this loading increases sufficiently rapidly that the common-error component ψ^2/τ_b rises in Panel B, despite the decline in the underlying error variance $1/\tau_b$. A more accurate model thus exposes prices to more common error because investors use its information more extensively. Under complete correlation neglect, this increase initially outweighs the reduction in noise-trader contamination, worsening firms' information about demand and lowering TFP.

As accuracy improves further, the reduction in noise-trader contamination overtakes the increase in the common component, and TFP starts to recover. The recovery therefore begins before the common-error component itself falls. At higher accuracy, acquisition costs and the remaining investor-specific noise increasingly limit the gains from further improvements in the underlying model. The response of AI reliance moderates, and the decline in model-error variance eventually dominates the further increase in ψ. The common component then falls as well, reinforcing the recovery in capital allocation, TFP, and output. The J-shaped response reflects this changing balance between the expansion of AI reliance and the improvement in the information on which investors rely.

Away from the boundary, recognition changes both the information mix and its weight in prices along these equilibrium paths. At $\kappa = 0.25$, TFP exhibits a shallow initial decline, whereas it increases throughout the plotted range at $\kappa = 0.75$ and $\kappa = 1$. Even under correct beliefs, the common-error component initially rises, but the reduction in noise-trader contamination is sufficient to improve price informativeness. The positive initial effect at $\kappa = 1$ is consistent with the sufficient condition in (5.17), since the exercise uses quadratic costs. At higher accuracy, the TFP ordering across beliefs can reverse: greater reliance on AI then conveys more demand information through prices. These comparisons illustrate why recognition of common errors has no uniform implication for allocative productivity.

These results also connect to the productivity J-curve literature. Brynjolfsson et al. (2021) show how complementary intangible investment and its incomplete measurement can depress measured productivity growth during the early adoption of a general-purpose technology, before the returns to these investments become visible. Our model identifies a complementary mechanism operating through information aggregation and capital allocation. When investors underestimate the AI model's common errors, improvements from low AI accuracy induce greater reliance on the shared signal. Although investors acquire more precise information, the resulting increase in common-error exposure initially outweighs the reduction in noise-trader contamination. Prices become less informative about demand, and firms allocate capital less efficiently. The productivity loss therefore reflects a deterioration in actual allocation, even as investors actively use the improving technology. Further gains in accuracy eventually improve

price informativeness and restore the productivity gains from AI. The horizontal axis here is model accuracy, not time. The exercise therefore describes a comparative-static J-shaped response in actual allocative productivity, rather than a transition path or a measurement shortfall. It shows how the information generated by a new technology can affect productivity through the prices guiding investment.



(a) Aggregate TFP

(b) Common Model Error

(c) Total Information

(d) AI Information Share

Figure 1: Equilibrium Effect of AI Accuracy Changes

Notes: This figure shows the equilibrium effect as AI accuracy τ_b changes. The acquisition cost is held at $c_A = c_P$. Moving right corresponds to an increase in τ_b; the horizontal axis shows τ_b/τ_u on a logarithmic scale. The four curves correspond to $\kappa = 0, 0.25, 0.75, 1$. Gray dotted lines show the economy without AI. All other structural parameters are fixed across curves.

Changes in AI acquisition costs. We next reduce AI acquisition costs c_A while holding model accuracy at $\tau_b = \tau_u$. Figure 2 shows the equilibrium TFP index $\mathcal{A}$, the common-error component of price-signal noise ψ^2/τ_b, and the information structure (h, α). Panel A shows that the initial productivity gain in Proposition 5 can reverse as AI becomes cheaper. Under complete correlation neglect, TFP first rises and then falls below its level in the economy without AI.

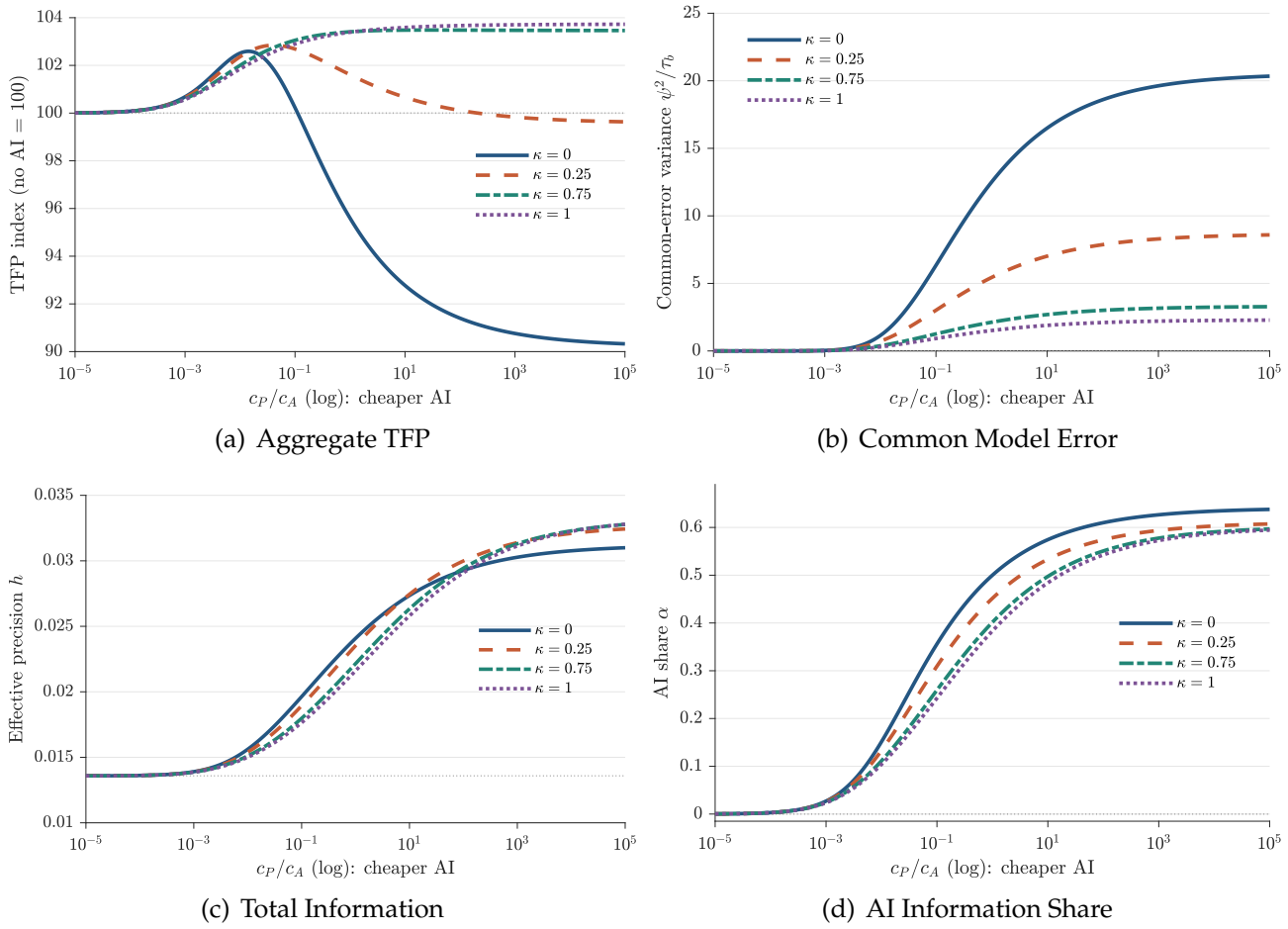
Lower acquisition costs make additional AI information cheaper at the margin, inducing investors to expand AI research. Since AI and private signals convey information about the same demand fundamental, this expansion partly substitutes for private research. Along the equilibrium path, investors acquire less private information, while both total information h and the AI share α increase, as shown in Panels C and D. The distinction between the two sources matters for information aggregation: additional AI acquisition reduces investor-specific signal noise but leaves the underlying common error unchanged. Whereas private signal errors average out across investors, the common AI error shifts their portfolio demands in the same direction. As investors place greater weight on AI information, market clearing therefore gives this error a larger loading in the price signal. Under complete correlation neglect, this connection is particularly direct: $\psi = \alpha$. With τ_b fixed, the rising AI share thus increases ψ^2/τ_b, as shown in Panel B.

At the same time, greater information acquisition strengthens informed trading relative to noise-trader demand, reducing the noise-trader component ζ^2/τ_n along all four paths. Initially, this reduction outweighs the increase in the common-error component, so prices become more informative and firms allocate capital more closely to demand fundamentals. As AI reliance grows, however, the balance between these effects changes. Under complete correlation neglect, $\psi = \alpha$, so the common-error component is quadratic in the AI share. A given increase in α therefore adds more common price error when investors already rely heavily on AI.

As acquisition costs decrease, further gains in total information also shrink. With model accuracy fixed, effective AI precision is bounded by τ_b: additional acquisition removes investor-specific noise but cannot eliminate the common model error. The accompanying decline in private research further limits the net expansion in h. Consequently, further cost reductions yield progressively smaller reductions in noise-trader contamination, while the continued shift toward AI raises the common-error loading. Under complete correlation neglect, the latter effect eventually dominates along the equilibrium path. Prices then become less informative even though investors continue to acquire more information. Firms optimally discount the less precise price signal, but their remaining uncertainty about demand rises, worsening capital allocation and lowering TFP. Output follows TFP, with the response amplified by the adjustment in aggregate investment.

Recognizing the common error moderates the shift toward AI along these paths and preserves a greater role for private research. The hump remains at $\kappa = 0.25$. At $\kappa = 0.75$, the later decline is small and TFP remains above the no-AI benchmark, while at $\kappa = 1$ it increases throughout the plotted range. The ordering across beliefs also changes with costs. When AI use is limited, the larger information expansion under greater neglect can improve price discovery. When AI becomes cheap, the associated common-error exposure can instead reduce TFP. Cheaper access therefore illustrates how expanding investors' information can eventually

impair the information that prices convey to firms.



(a) Aggregate TFP

(b) Common Model Error

(c) Total Information

(d) AI Information Share

Figure 2: Equilibrium Effect of AI Cost Changes

Notes: This figure shows the equilibrium effect as AI information cost c_A changes. AI accuracy is held at $\tau_b = \tau_u$. Moving right corresponds to a decline in c_A; the horizontal axis shows c_P/c_A on a logarithmic scale. The four curves correspond to $\kappa = 0, 0.25, 0.75, 1$.

6 General Information Acquisition Costs

This section considers the robustness of our main results under general information acquisition costs. Conditional on information choices, the portfolio and price equations do not require a particular acquisition cost. Costs determine how those choices respond to technological progress. We first extend the local results to general separable cost functions, then consider the entropy-based cost structure under certain conditions. To isolate the role of costs, we use the complete-neglect reduction in Section 5.2 throughout this section. The relevant equilibrium branches satisfy individual second-order conditions and the stated regularity conditions.

6.1 General Separable Acquisition Costs

We first retain the precision technology in (3.12) and allow general separable acquisition costs:

$$C = c_A f(x^A) + g(x^P, c_P), \qquad x^A = \frac{\tau_b h_A}{\tau_b - h_A}, \qquad x^P = \frac{\tau_u h_P}{\tau_u - h_P}. \tag{6.1}$$

This formulation follows the literature that specifies costs of attention or signal precision. Colombo et al. (2014) assume an increasing, convex private-precision cost with zero initial marginal cost. Models with several information sources also specify costs directly over the vector of attention inputs.[13] Equation (6.1) makes the source costs separable; the next subsection allows composition and total information to interact in costs.

Define the private cost in effective-precision coordinates by

$$G(p) = g\left(\frac{\tau_u p}{\tau_u - p}, c_P\right). \tag{6.2}$$

Let $h_0 \in (0, \tau_u)$ satisfy $m(h_0) = G'(h_0)$, where m is the marginal information value in Lemma 3. For this cost specification, let $m_0 = m(h_0)$ and define:

$$\lambda_0 = \frac{G''(h_0)}{G''(h_0) - m'(h_0)}, \qquad \Gamma_0 = -\left.\frac{\partial \log \mathcal{A}}{\partial S_q^T}\right|_{S_q^T = s(h_0)} > 0. \tag{6.3}$$

Proposition 6 *Suppose $f(0) = 0$, $f'(x) > 0$ for $x > 0$, and f is convex, continuously differentiable at zero, and twice continuously differentiable at positive inputs. When $f'_+(0) > 0$, additionally assume $\lim_{\pi \downarrow 0} \pi f''(\pi) = 0$. Suppose G is twice continuously differentiable near h_0, with $G''(h_0) > 0$. Then:*

(i) Fix $\tau_b > 0$. On the interior equilibrium branch, a marginal reduction in c_A raises TFP whenever h_A is sufficiently close to zero:

$$\frac{d \log \mathcal{A}}{d c_A} < 0 \qquad \text{for sufficiently small } h_A > 0. \tag{6.4}$$

The no-AI boundary is approached as $c_A \to \infty$ if $f'_+(0) = 0$, and as $c_A \uparrow m_0 / f'_+(0)$ from below if $f'_+(0) > 0$.

(ii) Fix $c_A > 0$ satisfying $c_A f'_+(0) < m_0$. As $\tau_b \downarrow 0$:

$$\frac{d \log \mathcal{A}}{d \tau_b}\Big|_{\tau_b \to 0} = \Gamma_0 \left(1 - \sqrt{\frac{c_A f'_+(0)}{m_0}}\right) \times \left[[-s'(h_0)]\lambda_0 - \frac{1}{h_0^2}\left(1 - \sqrt{\frac{c_A f'_+(0)}{m_0}}\right)\right]. \tag{6.5}$$

[13] For example, Pavan (2025) uses a linear cost of total attention in the main text and allows a general increasing convex cost in the appendix. The information-source structure follows Myatt and Wallace (2012).

Proof. See Appendix C.1. ■

Proposition 6 shows which features of information costs matter for the two forms of AI progress. Under correlation neglect, the AI acquisition condition can be written as:

$$\left(1 - \frac{h_A}{\tau_b}\right)^2 = \frac{c_A f'(\pi_A)}{m(h)}. \tag{6.6}$$

The fraction of available AI precision that investors acquire therefore depends on the marginal cost of acquisition relative to the marginal value of effective information. With accuracy fixed, high costs keep raw acquisition small, so $h_A/\tau_b \to 0$. The resulting gain in total information reduces noise-trader contamination at first order, while exposure to the common error increases only at second order, leading to a marginal TFP gain. A positive initial marginal cost changes the location of the adoption boundary: above the finite threshold $m_0/f'_+(0)$, investors do not use AI, and cost reductions have no effect until they cross that threshold.

When accuracy approaches zero at a fixed acquisition cost, raw acquisition also vanishes. If $f'_+(0) = 0$, the marginal cost of acquisition reaches zero and investors choose to obtain nearly all the precision the model can provide. Thus $h_A/\tau_b \to 1$, even though the AI share α approaches zero. Equation (6.5) then recovers the $\kappa = 0$ boundary formula in Proposition 5. The power exponent determines how quickly acquisition approaches this limit, not the limit itself. If, instead, the marginal cost at the adoption margin is positive, $f'_+(0) > 0$, the investors would acquire information such that the aggregate AI information precision is a fraction of its frontier:

$$\frac{h_A}{\tau_b}|_{\tau_b \to 0} = 1 - \sqrt{\frac{c_A f'_+(0)}{m_0}}, \tag{6.7}$$

which remains positive if $c_A f'_+(0) < m_0$. This weakens both the gain in total information and the increase in common-error exposure. As a result, the marginal TFP effect of an increase in τ_b changes to (6.5). However, as long as $c_A f'_+(0) < m_0$, the sign of this marginal effect could be either positive or negative, as the expansion in common error exposure dominates when:

$$1 - \sqrt{\frac{c_A f'_+(0)}{m_0}} > -h_0^2 s'(h_0)\lambda_0. \tag{6.8}$$

6.2 Entropy-Based Non-Separable Information Costs

Information costs can also be expressed in terms of the entropy reduction from acquired signals (Van Nieuwerburgh and Veldkamp, 2010). For the two direct Gaussian signals, the entropy

reduction is:

$$\mathcal{K} = I(d; s^A, s^P) = \frac{1}{2}\log\left(1 + \frac{h}{\tau_d}\right), \qquad h = H(\mathcal{K}) = \tau_d(e^{2\mathcal{K}} - 1). \tag{6.9}$$

The composition variable α retains its definition as the share of effective precision supplied by AI. At a given $\mathcal{K}$, different values of α may require different research inputs and generate different exposure to the common error. Inspired by this observation, we set the information acquisition cost as:

$$\mathcal{C}(\mathcal{K}, \alpha; c_A, c_P, \tau_b). \tag{6.10}$$

This representation accommodates entropy-based costs with source-dependent acquisition technology. It also includes (6.1): substitute $h_A = \alpha H(\mathcal{K})$ and $h_P = (1-\alpha)H(\mathcal{K})$ into the raw-precision formulas. We retain the original feasibility bounds and consider interior equilibria. Assume that $\mathcal{C}$ is twice continuously differentiable at interior choices and that $\mathcal{C}(\mathcal{K}, 0; c_A, c_P, \tau_b) = \mathcal{C}_0(\mathcal{K}; c_P)$, so AI technology does not change private-research costs when AI is unused. For this cost, let $G(p) = \mathcal{C}_0(\frac{1}{2}\log(1 + p/\tau_d); c_P)$ and let $h_0 \in (0, \tau_u)$ solve $G'(h_0) = m(h_0)$. Assume G is twice continuously differentiable near h_0, with $G''(h_0) - m'(h_0) > 0$. The constants $m_0 = m(h_0)$, λ_0, and Γ_0 are evaluated at this no-AI equilibrium.

In the complete neglect benchmark, $\kappa = 0$, the information acquisition first-order conditions in a symmetric equilibrium are given by:

$$\mathcal{C}_\alpha = 0, \qquad \mathcal{C}_\mathcal{K} = H'(\mathcal{K})m(H(\mathcal{K})). \tag{6.11}$$

This specification permits cost interactions across sources: $\mathcal{C}_{h_A h_P}$ can be nonzero. An additional unit of AI precision may raise or lower the marginal cost of private research. In quantity–composition coordinates, the response also depends on how total information changes the preferred mix. For $\vartheta \in \{c_A, \tau_b\}$, differentiating the composition condition gives:

$$\alpha_\vartheta = -\frac{\mathcal{C}_{\alpha\vartheta}}{\mathcal{C}_{\alpha\alpha}} - \frac{\mathcal{C}_{\mathcal{K}\alpha}}{\mathcal{C}_{\alpha\alpha}}\mathcal{K}_\vartheta. \tag{6.12}$$

The first term is the change in the preferred AI share at a fixed information quantity. The second is the further adjustment in composition as total information changes. The quantity response accounts for this adjustment as well as the equilibrium change in the marginal value of information.[14]

We first consider the equilibrium response to a change in AI information cost c_A. A cost

[14] Let $B(\mathcal{K}) = H'(\mathcal{K})m(H(\mathcal{K}))$. We restrict attention to equilibrium branches on which $\mathcal{C}_{\alpha\alpha} > 0$ and the denominator below is positive. These are equilibrium regularity conditions in addition to individual optimality. The

reduction can make AI more attractive relative to private research and reduce the cost of acquiring additional information. The interaction between quantity and composition determines how much of the response is an expansion in information rather than a replacement of private research. Define:

$$U_c = \mathcal{C}_{\alpha\alpha}\mathcal{C}_{\mathcal{K}c_A} - \mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\alpha c_A},$$
$$V_c = [\mathcal{C}_{\mathcal{K}\mathcal{K}} - B'(\mathcal{K})]\mathcal{C}_{\alpha c_A} - \mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\mathcal{K}c_A}.$$

Then (6.12) implies that $\alpha_{c_A} < 0$, or a marginal decrease in AI information cost increases the share of AI information, if and only if $V_c > 0$. Given $V_c > 0$, a cost reduction also expands total information near the no-AI boundary if:

$$\liminf_{h_A \downarrow 0} \frac{h_{c_A}}{\alpha_{c_A}} = \liminf_{h_A \downarrow 0} \frac{H'(\mathcal{K})U_c}{V_c} > 0, \tag{6.13}$$

where the limit follows this cost path.[15] Under these two conditions, the increase in total information does not vanish relative to the increase in the AI share. At fixed τ_b, the marginal common-error contribution is proportional to α and vanishes at the no-AI boundary. The positive lower bound in (6.13) ensures that the reduction in noise-trader contamination remains first order relative to the change in composition. It therefore dominates sufficiently close to that boundary, and a marginal cost reduction raises TFP.

We next consider higher AI accuracy. The following lemma links the fraction of available precision acquired and the expansion of total information to the behavior of marginal costs near the boundary:

Lemma 4 (Information acquisition at low AI accuracy) *Fix c_A, c_P and consider an interior equilibrium branch with $h \to h_0$ as $\tau_b \downarrow 0$. Suppose $\mathcal{C}_{\alpha\alpha} > 0$ on the feasible composition interval for total information near h_0. Assume the following conditions:*

(i) For each $q \in (0,1)$, the following limit exists:

$$F(q) = \lim_{\substack{\tau_b \downarrow 0 \\ H(\mathcal{K}) \to h_0}} \mathcal{C}_\alpha\left(\mathcal{K}, \frac{q\tau_b}{H(\mathcal{K})}; c_A, c_P, \tau_b\right) \tag{6.14}$$

There is a $q_0 \in (0,1]$ such that $F(q) < 0$ for $q < q_0$ and $F(q) > 0$ for $q > q_0$.

response of aggregate information acquisition intensity is:

$$\mathcal{K}_\vartheta = -\frac{\mathcal{C}_{\mathcal{K}\vartheta} - \mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\alpha\vartheta}/\mathcal{C}_{\alpha\alpha}}{\mathcal{C}_{\mathcal{K}\mathcal{K}} - B'(\mathcal{K}) - \mathcal{C}_{\mathcal{K}\alpha}^2/\mathcal{C}_{\alpha\alpha}}.$$

[15] The responses of $\mathcal{K}$ and α are given in Appendix C.2.

(ii) In effective-precision coordinates, $\mathcal{C}_{h_P}$ is continuous at $h_A = 0$ for each $\tau_b > 0$, and:

$$\lim_{\substack{\tau_b \downarrow 0,\ h_P \to h_0 \\ 0 < h_A < \tau_b}} \mathcal{C}_{h_A h_P}(h_A, h_P; c_A, c_P, \tau_b) = \gamma_0 \in \mathbb{R}. \tag{6.15}$$

Then:

$$\frac{h_A}{\tau_b} \longrightarrow q_0, \qquad h = h_0 + L_0 \tau_b + o(\tau_b), \qquad L_0 = q_0 \frac{G''(h_0) - \gamma_0}{G''(h_0) - m'(h_0)}. \tag{6.16}$$

Proof. See Appendix C.3. ■

Condition (i) in Lemma 4 compares feasible allocations that use a fixed fraction $q \in (0,1)$ of available AI precision as accuracy vanishes. Its sign conditions imply that equilibrium utilization converges to q_0. They also cover $q_0 = 1$: when $F(q) < 0$ for every $q < 1$, investors approach the precision frontier. This recovers (6.7) under separable costs. Condition (ii) measures how AI information acquisition changes the marginal cost of private research. A positive γ_0 implies that higher AI information acquisition increases the marginal cost of private research, while a negative γ_0 implies that AI information acquisition complements private research. The finite joint limit rules out an arbitrarily large change in private marginal cost per unit of AI precision as the AI frontier shrinks. It does not require the interaction to vanish. Total information rises at first order with accuracy if and only if $\gamma_0 < G''(h_0)$: the increase in private marginal cost induced by AI must be smaller than the curvature of private research cost.

These acquisition responses determine the marginal TFP effects:

Proposition 7 *Under the maintained equilibrium regularity conditions, the local TFP effects are:*

(i) Fix $\tau_b > 0$. On an interior cost path approaching $(h, \alpha) = (h_0, 0)$ and satisfying $V_c > 0$ and (6.13), a marginal reduction in c_A raises TFP sufficiently close to the boundary:

$$\frac{d \log \mathcal{A}}{d c_A} < 0. \tag{6.17}$$

(ii) Under the conditions of Lemma 4, the right derivative of TFP at zero accuracy is:

$$\left.\frac{d \log \mathcal{A}}{d \tau_b}\right|_{0^+} = \Gamma_0 \left\{ [-s'(h_0)] L_0 - \frac{q_0^2}{h_0^2} \right\}, \tag{6.18}$$

Proof. See Appendix C.4. ■

Condition (6.13), Lemma 4, and Proposition 7 characterize the marginal effect of AI progress on the aggregate TFP index. Under general cost functions, the marginal response of information acquisition depends on how acquisition from one source affects the marginal cost of the

other. Under the stated conditions, a marginal cost reduction raises TFP sufficiently close to the no-AI boundary, while a marginal accuracy increase can reduce allocation efficiency at the adoption margin.

Relation to separable costs. For separable costs, $\mathcal{C}_{h_A h_P} = 0$ and hence $L_0 = \lambda_0 q_0$. The relative marginal-cost limit in (6.14), divided by h_0, is $c_A f'_+(0)/(1-q)^2 - m_0$. Under $c_A f'_+(0) < m_0$, it gives $q_0 = 1 - \sqrt{c_A f'_+(0)/m_0}$. When $f'_+(0) = 0$, this limit is negative at every $q < 1$, so $q_0 = 1$ and $L_0 = \lambda_0$. Equation (6.18) therefore recovers the boundary results in Proposition 6.

These extensions identify the acquisition incentives behind the local productivity effects of AI progress. Marginal acquisition costs determine how much of the available AI precision is obtained, while cost interactions between information sources govern the accompanying adjustment in private research. Under correlation neglect and the stated cost conditions, cheaper access at fixed accuracy raises TFP near the no-AI boundary because the expansion in total information reduces noise-trader contamination more than it increases common-error exposure. Improvements from very low accuracy can instead increase common-error exposure enough to outweigh the information gain. This comparison depends on how private research adjusts. An increase in its marginal cost induced by AI acquisition strengthens displacement and limits the expansion in total information, whereas cost complementarities support private research and reinforce that expansion. The local comparison continues to depend on how investors' acquisition choices balance the reduction in noise-trader contamination against the common error transmitted through prices to firms' investment decisions.

7 Conclusion

This paper studies how common reliance on artificial intelligence affects the information that financial markets convey to firms and, through that information, the allocation of capital. The portfolio evidence shows that manager pairs with greater common alignment with AI-extractable information subsequently hold more similar portfolios. Motivated by this association, this paper then builds a model that connects investors' choices over the quantity and composition of information to price formation and firms' investment. Shared AI information helps investors learn about fundamentals, but also exposes their decisions to common model errors that survive aggregation and enter financial prices.

The analysis distinguishes two forms of AI progress: cheaper access to information and greater model accuracy. For any degree of recognition of common errors, a marginal cost reduction raises TFP when high acquisition costs initially keep AI use close to zero. Additional information strengthens informed trading, and the reduction in noise-trader contamination

dominates the increase in common-error exposure. Improvements from very low model accuracy can instead reduce TFP, with the sign depending on correlation beliefs and acquisition incentives. Investors acquire more precise AI information, but their increased reliance on the shared signal can make prices less informative about fundamentals. Thus, economies with similarly limited AI use can have different marginal productivity effects, depending on whether costs or accuracy constrain acquisition. Numerical exercises illustrate how these local effects can extend to a hump-shaped relationship between cheaper access and TFP, and a J-shaped relationship between model accuracy and TFP.

These results distinguish the information an investor acquires from the information that markets aggregate. Improvements in individual signals need not translate into more informative prices when investors adjust their reliance on shared sources. Firms learn from market prices and invest optimally, yet their allocation decisions deteriorate when those prices reveal less about demand. With production technology held fixed, the resulting productivity changes reflect actual capital allocation. Recognition of common errors changes both information acquisition and price formation, and its effect on aggregate TFP depends on which force dominates the changes in TFP. The general-cost extensions identify the acquisition incentives under which the local comparison between cheaper access and greater accuracy continues to hold. Evaluating the productivity gains from financial AI therefore requires attention to both individual signal quality and the information that prices convey.

Further empirical work could connect these mechanisms more directly to institutional AI use. Observations of the models investors employ, the overlap in their information, and their beliefs about shared errors would help assess the magnitude of the effects on prices and real investment. The J-shaped relationship in this paper compares equilibria at different levels of model accuracy. A dynamic extension could examine how this information channel interacts with capital adjustment and investment in complementary technologies. These questions extend the paper's central finding: the productivity consequences of AI depend on how investors use its information and how financial markets transmit that information to firms.

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A A Fixed-Price Portfolio Benchmark

This appendix isolates the portfolio implications of the information mechanism at given prices. Investors first acquire information and then choose budget-constrained portfolios. The exercise illustrates how common information can produce similar desired holdings; it does not impose financial-market clearing or provide an additional step needed to solve the macroeconomic model. We characterize portfolio choice, information acquisition, and aggregate desired positions, and then provide proofs and extensions.

A.1 Environment and Information Acquisition

The portfolio benchmark follows the information-acquisition tradition in treating initial asset prices and the stochastic payoff process as exogenous (Van Nieuwerburgh and Veldkamp, 2010). This setting isolates how the quantity and composition of information affect portfolios. Prices convey no additional information in this benchmark.

Timing and assets. The economy has three dates, $t = 0, 1, 2$. There are $N \geq 2$ risky assets indexed by $j \in \{1, \ldots, N\}$. Asset j has a stochastic date-2 payoff:

$$f_j \sim \mathcal{N}(\mu_j, \sigma_j^2), \tag{A.1}$$

and payoffs are independent across assets. There is a continuum of investors indexed by $i \in [0, 1]$. Each investor has one unit of wealth at date 1 and takes the vector of strictly positive asset prices $\boldsymbol{p} = (p_1, \ldots, p_N)'$ and the gross risk-free payoff r as given.

At date 0, investor i chooses how much information to acquire about each asset and how to divide this information between private research and an AI source. At date 1, the signals are observed, and the investor chooses a portfolio. Asset payoffs are realized at date 2.

Prior beliefs and signals. Investors share a common prior that coincides with the unconditional payoff distribution:

$$f_j \mid \mathcal{I}_{i0} \sim \mathcal{N}\left(\mu_j, (\lambda_j^0)^{-1}\right), \qquad \lambda_j^0 = \sigma_j^{-2}. \tag{A.2}$$

Investor i can acquire a private signal s_{ij}^P and an AI signal s_{ij}^A about asset j:

$$s_{ij}^P = f_j + u_{ij} + \epsilon_{ij}^P, \tag{A.3}$$

$$s_{ij}^A = f_j + b_j + \epsilon_{ij}^A. \tag{A.4}$$

The private component $u_{ij} \sim \mathcal{N}(0, \tau_u^{-1})$ is specific to investor i, whereas the AI model error $b_j \sim \mathcal{N}(0, \tau_b^{-1})$ is shared by all investors who use the AI signal. The remaining errors satisfy:

$$\epsilon_{ij}^P \sim \mathcal{N}\left(0, (x_{ij}^P)^{-1}\right), \qquad \epsilon_{ij}^A \sim \mathcal{N}\left(0, (x_{ij}^A)^{-1}\right).$$

Fundamentals and signal errors are mutually independent across assets. The investor-specific errors are also independent across investors. Investors pay to increase x_{ij}^P and x_{ij}^A. The precisions τ_u and τ_b describe the irreducible errors in the two information technologies.

We define the effective precisions contributed by the private and AI signals as:

$$h_{ij}^P = \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P}, \qquad h_{ij}^A = \frac{\tau_b x_{ij}^A}{\tau_b + x_{ij}^A}, \tag{A.5}$$

and we can summarize the quantity and composition of information by:

$$h_{ij} = h_{ij}^P + h_{ij}^A, \qquad \alpha_{ij} = \frac{h_{ij}^A}{h_{ij}}. \tag{A.6}$$

Effective AI precision is $a_{ij} = \alpha_{ij} h_{ij}$. The feasible set satisfies:

$$h_{ij} > 0, \qquad 0 < \alpha_{ij} < 1, \qquad \alpha_{ij} h_{ij} < \tau_b, \qquad (1 - \alpha_{ij}) h_{ij} < \tau_u. \tag{A.7}$$

The first two inequalities describe an interior information structure. The last two follow from the upper bounds on effective precision generated by the irreducible signal errors.

Bayesian updating from $\{s_{ij}^P, s_{ij}^A\}$ gives the payoff posterior mean $\widehat{\mu}_{ij}$ and precision $\widehat{\lambda}_{ij}$:

$$\widehat{\mu}_{ij} = \frac{\lambda_j^0 \mu_j + (1 - \alpha_{ij}) h_{ij} s_{ij}^P + \alpha_{ij} h_{ij} s_{ij}^A}{\lambda_j^0 + h_{ij}}, \qquad \widehat{\lambda}_{ij} = \lambda_j^0 + h_{ij}. \tag{A.8}$$

For a single investor, h_{ij} determines the total information acquired. The composition variable α_{ij} determines how much of that information is exposed to the common AI model error.

The payoff posterior already accounts for uncertainty about b_j. An investor can also infer the common error from the two signals, but this inference provides no additional observation beyond those signals. Here the posterior distribution of f_j is sufficient for portfolio choice because payoffs depend only on fundamentals. Section 3 allows managers to learn from prices, so the covariance between their own AI signal and the price also enters their inference. Initial prices in the present section are fixed and supply no information.

Information costs. Raising the raw precisions $\{x_{ij}^A, x_{ij}^P\}$ reduces investor-specific signal noise and requires costly research. Write this cost in effective-information coordinates as

$$C(h_{ij}, \alpha_{ij}; c_A, c_P, \tau_b, \tau_u).$$

We assume that the cost is increasing in total effective precision and locally convex in the AI share. A useful specification, employed in the comparative-static analysis, is

$$C(h, \alpha; c_A, c_P, \tau_b, \tau_u) = c_A f\left(\frac{\tau_b \alpha h}{\tau_b - \alpha h}\right) + g\left(\frac{\tau_u (1-\alpha) h}{\tau_u - (1-\alpha)h}, c_P\right), \tag{A.9}$$

where f and $g(\cdot, c_P)$ are nonnegative, strictly increasing, and convex functions, and $x^A = \frac{\tau_b \alpha h}{\tau_b - \alpha h}$ and $x^P = \frac{\tau_u (1-\alpha) h}{\tau_u - (1-\alpha)h}$ denote the raw signal precisions required to produce the desired effective precisions (Colombo et al., 2014). A decline in c_A lowers the price of AI information within a given technological frontier. An increase in τ_b improves the common-error precision and expands the upper bound on effective AI information.

Sequential optimization. After observing the signals, investor i chooses asset holdings $\boldsymbol{q}_i = (q_{i1}, \ldots, q_{iN})'$ to maximize mean-variance utility:

$$U_i = \mathbb{E}_i(W_i) - \frac{\rho}{2} \mathbb{V}_i(W_i),$$

where W_i denotes wealth at date $t = 2$, $\rho > 0$ measures risk aversion, and the expectations are taken conditional on the information set at date $t = 1$, after the signals are revealed. Conditional on posterior beliefs, the date-1 portfolio-choice problem can be written as:

$$\max_{\boldsymbol{q}_i} \quad \sum_{j=1}^{N} q_{ij}(\widehat{\mu}_{ij} - r p_j) - \frac{\rho}{2} \sum_{j=1}^{N} q_{ij}^2 \widehat{\lambda}_{ij}^{-1} \qquad \text{subject to} \quad \sum_{j=1}^{N} p_j q_{ij} = 1. \tag{A.10}$$

We abstract from short-sale constraints to keep the portfolio rule linear in posterior beliefs and permit a transparent characterization of portfolio similarity (Van Nieuwerburgh and Veldkamp, 2010). At date 0, the investor chooses $\{h_{ij}, \alpha_{ij}\}_{j=1}^N$ to maximize the expected value of the utility, $\mathbb{E}_0[U_i]$, net of information costs. We solve the portfolio problem first and then return to information acquisition.

A.2 Optimal Portfolios and Pairwise Similarity

For a given information structure, the portfolio problem (A.10) has the following solution.

Lemma A.1 (Optimal portfolio structure) *Suppose that asset payoffs are conditionally independent, posterior precisions are strictly positive, and short selling is unrestricted. Conditional on posterior beliefs, investor i's optimal holding of asset j is*

$$q_{ij}^* = \frac{\widehat{\lambda}_{ij} p_j}{B_i} + \frac{\widehat{\lambda}_{ij}}{\rho}\left(m_{ij} - p_j \frac{A_i}{B_i}\right), \tag{A.11}$$

where

$$m_{ij} \equiv \widehat{\mu}_{ij} - r p_j, \qquad A_i \equiv \sum_{\ell=1}^{N} \widehat{\lambda}_{i\ell} p_\ell m_{i\ell}, \qquad B_i \equiv \sum_{\ell=1}^{N} \widehat{\lambda}_{i\ell} p_\ell^2.$$

See Appendix A.5.

The first term in (A.11) exhausts the budget using positions based on posterior precision. It assigns more weight to assets with less residual uncertainty, independently of signal realizations. The second term is an active position: the investor increases holdings when an asset's expected excess payoff rises relative to the portfolio-wide benchmark. Higher precision strengthens this response, while greater risk aversion dampens it.

To characterize the portfolio similarity between different asset managers, we first define the ex ante average portfolio of investor i as:

$$\overline{\boldsymbol{q}}_i = \mathbb{E}_0[\boldsymbol{q}_i^*], \tag{A.12}$$

and the centered inner-product similarity between two distinct investors i and k can be expressed as:

$$\Omega_{ik} = (\boldsymbol{q}_i^* - \overline{\boldsymbol{q}}_i)'(\boldsymbol{q}_k^* - \overline{\boldsymbol{q}}_k). \tag{A.13}$$

This measure removes average portfolio differences generated by asset prices, prior expected payoffs, and posterior precisions. It isolates the extent to which common signal realizations move the two portfolios in the same direction. Taking expectations at date $t = 0$ gives:

$$\mathbb{E}_0[(\boldsymbol{q}_i^*)'\boldsymbol{q}_k^*] = \overline{\boldsymbol{q}}_i'\overline{\boldsymbol{q}}_k + \mathbb{E}_0[\Omega_{ik}].$$

The following proposition characterizes the expected portfolio similarity given the information structure $\{(h_{ij}, \alpha_{ij})\}$:

Proposition A.1 (Common AI exposure and portfolio similarity) *Suppose that asset fundamentals and common AI model errors are independent across assets. Investor-specific signal errors are independent across investors. For two distinct investors i and k, given the information structure* $\{(h_{ij}, \alpha_{ij})\}$,

the expected portfolio similarity is given by:

$$E_0[\Omega_{ik}] = \frac{1}{\rho^2} \sum_{\ell=1}^{N} \omega_{ik,\ell} \left[h_{i\ell} h_{k\ell} \sigma_\ell^2 + \alpha_{i\ell} h_{i\ell} \alpha_{k\ell} h_{k\ell} \tau_b^{-1} \right], \tag{A.14}$$

where $\omega_{ik,\ell} \geq 0$ depends on asset prices, posterior precisions, and the budget constraint. Its expression is given in Appendix A.6. Conditional on total information quantities and posterior precisions, expected portfolio similarity weakly increases with investors' joint exposure to each asset's common AI signal.

See Appendix A.6.

Proposition A.1 identifies two economically distinct sources of portfolio convergence. First, investors' active positions covary when their signals reveal the same underlying payoff. This component reflects common learning about fundamentals and is present even when research errors are investor-specific. Second, AI information introduces an additional source of comovement. A model error shared across users enters both investors' posterior beliefs and moves their portfolios in the same direction, although the error is unrelated to the asset's true payoff. Conditional on total effective information, the strength of this AI-specific component is proportional to the product of the investors' effective AI exposures, $\alpha_{ij} h_{ij} \alpha_{kj} h_{kj}$, and to the variance of the common model error, τ_b^{-1}. The weights $\omega_{ik,j}$ capture how asset prices, posterior precisions, and the budget constraint transmit a signal about asset j across the rest of each portfolio. Since these weights are nonnegative under the maintained independence assumptions, asset heterogeneity affects the size of the contribution from joint AI exposure.

Model robustness. Appendix A.8 studies two departures from the benchmark. With short-sale constraints, the portfolio rule is piecewise linear. For locally fixed active sets, common AI exposure continues to align portfolio responses, as shown in Appendix A.8.1. Appendix A.8.2 introduces multiple AI providers. Provider-specific errors diversify as the number of models grows, while an error component correlated across providers remains in the aggregate signal.

The symmetric benchmark. The general results above allow for heterogeneous assets and investors. We now impose symmetry across assets to obtain a clearer portfolio decomposition. The same specialization will also allow us to aggregate holdings across investors.

Assume

$$p_j = 1, \qquad \mu_j = \mu, \qquad \sigma_j^2 = \sigma^2, \qquad \lambda_j^0 = \sigma^{-2} \quad \forall j. \tag{A.15}$$

We focus on information choices that are symmetric across assets for each investor,

$$h_{ij} = h_i, \qquad \alpha_{ij} = \alpha_i, \qquad \widehat{\lambda}_i = \sigma^{-2} + h_i. \tag{A.16}$$

Let bars denote cross-asset averages for a given realization, for example:

$$\bar{f} = \frac{1}{N}\sum_{j=1}^{N} f_j, \qquad \bar{b} = \frac{1}{N}\sum_{j=1}^{N} b_j, \qquad \bar{u}_i = \frac{1}{N}\sum_{j=1}^{N} u_{ij}.$$

Applying Lemma A.1 gives:

$$q^*_{ij} = \frac{1}{N} + \frac{\widehat{\lambda}_i}{\rho}(\widehat{\mu}_{ij} - \bar{\widehat{\mu}}_i) = \frac{1}{N} + \frac{(1-\alpha_i)h_i}{\rho}(s^P_{ij} - \bar{s}^P_i) + \frac{\alpha_i h_i}{\rho}(s^A_{ij} - \bar{s}^A_i). \tag{A.17}$$

This is a decision rule in observed signals. Substituting the signal equations gives a decomposition in realized fundamentals and errors:

$$q^*_{ij} = \frac{1}{N} + \frac{h_i}{\rho}\left[(f_j - \bar{f}) + \alpha_i(b_j - \bar{b}) + \xi_{ij}\right], \tag{A.18}$$

where

$$\xi_{ij} = (1-\alpha_i)(u_{ij} - \bar{u}_i) + \alpha_i(\epsilon^A_{ij} - \bar{\epsilon}^A_i) + (1-\alpha_i)(\epsilon^P_{ij} - \bar{\epsilon}^P_i). \tag{A.19}$$

Equation (A.18) describes the sources of realized portfolio variation. The fundamental component, $(f_j - \bar{f})$, directs investment toward assets with higher realized payoffs. The AI component, $\alpha_i(b_j - \bar{b})$, loads on a model error shared across investors who acquire the AI signal, and greater reliance on AI information strengthens this channel. The final component contains investor-specific information and therefore creates dispersion across individual portfolios. A higher information precision h_i increases the response of portfolio choice to information, whereas greater risk aversion ρ dampens it.

In the symmetric-asset environment, the centered similarity measure reduces to

$$\Omega_{ik} = \sum_{j=1}^{N}\left(q^*_{ij} - \frac{1}{N}\right)\left(q^*_{kj} - \frac{1}{N}\right).$$

Proposition A.1 then implies:

$$\mathbb{E}_0[\Omega_{ik}] = \frac{N-1}{\rho^2} h_i h_k \left(\sigma^2 + \alpha_i \alpha_k \tau_b^{-1}\right). \tag{A.20}$$

The AI-specific term is proportional to $(\alpha_i h_i)(\alpha_k h_k)$. The product structure motivates measuring joint alignment with AI-extractable information at the pair level.

A.3 Ex Ante Information Choice

We now return to date 0, before the signals are observed. Let $\mathcal{G}_i(\boldsymbol{h}_i)$ denote the expected gross value of the optimized portfolio. Bayesian posterior means satisfy

$$\mathbb{E}_0[\widehat{\mu}_{ij}] = \mu_j, \qquad \mathbb{V}_0(\widehat{\mu}_{ij}) = \frac{1}{\lambda_j^0} - \frac{1}{\widehat{\lambda}_{ij}}. \tag{A.21}$$

The distribution of the investor's payoff posterior depends on total effective precision and is independent of the AI share at a fixed h_{ij}. Thus $\mathcal{G}_i$ depends on $\boldsymbol{h}_i$ alone. The investor's information problem is

$$\begin{aligned} \max_{\{h_{ij},\alpha_{ij}\}_{j=1}^N} \quad & \mathcal{G}_i(\boldsymbol{h}_i) - \sum_{j=1}^N C(h_{ij}, \alpha_{ij}; c_A, c_P, \tau_b, \tau_u) \\ \text{subject to} \quad & \alpha_{ij} h_{ij} < \tau_b, \qquad (1-\alpha_{ij}) h_{ij} < \tau_u \quad \text{for every } j. \end{aligned} \tag{A.22}$$

At an interior optimum,

$$C_\alpha(h_{ij}, \alpha_{ij}) = 0, \tag{A.23}$$

$$C_h(h_{ij}, \alpha_{ij}) = \frac{\partial \mathcal{G}_i}{\partial h_{ij}}. \tag{A.24}$$

Total information balances the value of a more precise posterior against its resource cost. Composition minimizes the cost of supplying that precision. The difference in cross-investor error correlation between the two sources affects portfolio similarity but has no separate effect on private payoffs in this benchmark. Appendix A.7 gives the gross value and its derivatives.

An interior competitive information equilibrium in this fixed-price environment is a collection

$$\left\{h_{ij}^*, \alpha_{ij}^*, q_{ij}^*\right\}_{i\in[0,1],\, j=1,\ldots,N}$$

such that beliefs satisfy (A.8), portfolios satisfy Lemma A.1, and information choices maximize (A.22), with interior conditions (A.23) and (A.24). Prices remain exogenous. Section 3 endogenizes prices through financial-market clearing.

The symmetric information conditions also simplify. Differentiating the gross value with respect to a common increase in precision across assets, we can show that the interior information choice now satisfies:

$$C_h(h_i, \alpha_i) = \frac{N-1}{2N\rho}\sigma^2 + \frac{\rho}{2N^2(\sigma^{-2}+h_i)^2}, \tag{A.25}$$

$$C_\alpha(h_i, \alpha_i) = 0. \tag{A.26}$$

The first term on the right-hand side of (A.25) is the value of learning which assets have high realized payoffs, and the second reflects the value of reducing the conditional variance of a budget-constrained portfolio.

A.4 Aggregate Holdings and the Macroeconomic Connection

Consider aggregate holdings. Integrating (A.18) across investors and applying the law of large numbers to the investor-specific terms gives

$$Q_j \equiv \int_0^1 q_{ij}^* \, di = \frac{1}{N} + \frac{1}{\rho}\left[\left(\int_0^1 h_i \, di\right)(f_j - \overline{f}) + \left(\int_0^1 \alpha_i h_i \, di\right)(b_j - \overline{b})\right]. \tag{A.27}$$

Thus, investor-specific errors disappear in the aggregate, whereas fundamentals and the common AI model error survive aggregation.

Under an ex ante symmetric information-acquisition equilibrium,

$$h_i = h, \qquad \alpha_i = \alpha \quad \forall i. \tag{A.28}$$

Individual portfolios remain heterogeneous after signals are observed because ξ_{ij} is investor-specific. Aggregate holdings become:

$$Q_j = \frac{1}{N} + \frac{h}{\rho}\left[(f_j - \overline{f}) + \alpha(b_j - \overline{b})\right]. \tag{A.29}$$

Since $\sum_j Q_j = 1$, expected aggregate portfolio concentration is

$$\mathbb{E}_0\left[\sum_{j=1}^N Q_j^2\right] = \frac{1}{N} + \frac{N-1}{\rho^2} h^2 \left(\sigma^2 + \alpha^2 \tau_b^{-1}\right). \tag{A.30}$$

Equations (A.20) and (A.30) describe the same common-information mechanism at two different levels of aggregation. Expected pairwise similarity, $\mathbb{E}_0[\Omega_{ik}]$, measures the covariance of active positions for a given investor pair. Expected aggregate concentration, $\mathbb{E}_0[\sum_{j=1}^N Q_j^2]$, measures how strongly common portfolio responses remain in aggregate holdings. Investor-specific research errors wash out from both objects in expectation, whereas variation in fundamentals and the common AI model error remain. At the pair level, the AI-specific component depends on the product of the two investors' effective AI exposures, $(\alpha_i h_i)(\alpha_k h_k)$. In the symmetric aggregate, it depends on the square of effective AI exposure, $(\alpha h)^2$. Pairwise similarity provides the model counterpart to the empirical evidence in Section 2. Aggregate concentration records the common demand response at given prices. Section 3 studies how this response enters prices when the financial market clears.

Connection to the motivating evidence. The model links common information to common portfolio adjustments. The shared-error term provides one source of the pairwise similarity documented in Section 2; common learning about fundamentals provides another. The incremental R^2 measure captures alignment with AI-extractable information, but does not directly estimate $\alpha_i h_i$, identify b_j, or separate these two sources of comovement. The evidence motivates the information channel. Sections 4 and 5 study the price and allocation consequences of the same information technology.

At fixed asset prices, common information appears in the quantities that investors demand. With a fixed net supply of financial claims, market clearing instead requires prices to adjust. The macroeconomic model uses this second setting and allows firms to learn from the resulting prices. The fixed supply of financial claims does not fix real investment, which firms choose after observing prices. Common reliance on AI can therefore affect real investment through firms' learning and lead to capital misallocation.

A.5 Proof of Lemma A.1

The first-order condition for the holding of asset j is

$$\widehat{\mu}_{ij} - r p_j - \rho q_{ij} \widehat{\lambda}_{ij}^{-1} - \zeta_i p_j = 0, \tag{A.31}$$

which gives

$$q_{ij} = \frac{\widehat{\lambda}_{ij}}{\rho} \left(m_{ij} - \zeta_i p_j \right). \tag{A.32}$$

Substituting (A.32) into the budget constraint yields

$$1 = \frac{1}{\rho} \sum_{\ell=1}^{N} \widehat{\lambda}_{i\ell} p_\ell \left(m_{i\ell} - \zeta_i p_\ell \right) = \frac{A_i - \zeta_i B_i}{\rho}.$$

Therefore,

$$\zeta_i = \frac{A_i - \rho}{B_i}. \tag{A.33}$$

Substituting (A.33) into (A.32) gives

$$\begin{aligned} q_{ij}^* &= \frac{\widehat{\lambda}_{ij}}{\rho} \left(m_{ij} - p_j \frac{A_i - \rho}{B_i} \right) \\ &= \frac{\widehat{\lambda}_{ij} p_j}{B_i} + \frac{\widehat{\lambda}_{ij}}{\rho} \left(m_{ij} - p_j \frac{A_i}{B_i} \right), \end{aligned}$$

which proves (A.11). The objective function is strictly concave because $\rho > 0$ and $\widehat{\lambda}_{ij} > 0$ for every asset. The solution is therefore unique.

A.6 Proof of Proposition A.1

From Lemma A.1, the random component of the holding of asset j is

$$q_{ij}^{*} - \overline{q}_{ij} = \frac{\widehat{\lambda}_{ij}}{\rho}\left[\widehat{\mu}_{ij} - E_0(\widehat{\mu}_{ij}) - \frac{p_j}{B_i}\sum_{\ell=1}^{N}\widehat{\lambda}_{i\ell}p_\ell\left(\widehat{\mu}_{i\ell} - E_0(\widehat{\mu}_{i\ell})\right)\right]. \tag{A.34}$$

Bayesian updating gives

$$\widehat{\mu}_{i\ell} - \mathbb{E}_0(\widehat{\mu}_{i\ell}) = \frac{h_{i\ell}(f_\ell - \mu_\ell) + \alpha_{i\ell}h_{i\ell}b_\ell + e_{i\ell}}{\widehat{\lambda}_{i\ell}}, \tag{A.35}$$

where $e_{i\ell}$ collects investor-specific signal errors. Substituting (A.35) into (A.34) yields

$$q_{ij}^{*} - \overline{q}_{ij} = \frac{1}{\rho}\sum_{\ell=1}^{N}\chi_{ij\ell}\left[h_{i\ell}(f_\ell - \mu_\ell) + \alpha_{i\ell}h_{i\ell}b_\ell + e_{i\ell}\right], \tag{A.36}$$

where

$$\chi_{ij\ell} = \begin{cases} 1 - \dfrac{\widehat{\lambda}_{i\ell}p_\ell^2}{B_i}, & j = \ell, \\ -\dfrac{\widehat{\lambda}_{ij}p_j p_\ell}{B_i}, & j \neq \ell. \end{cases} \tag{A.37}$$

Substituting (A.36) into (A.13) gives

$$\begin{aligned}\mathbb{E}_0[\Omega_{ik}] = \frac{1}{\rho^2}\sum_{j=1}^{N}\sum_{\ell=1}^{N}\sum_{n=1}^{N}\chi_{ij\ell}\chi_{kjn} \\ \times \mathbb{E}_0\left[\left(h_{i\ell}(f_\ell - \mu_\ell) + \alpha_{i\ell}h_{i\ell}b_\ell + e_{i\ell}\right)\right. \\ \left.\times\left(h_{kn}(f_n - \mu_n) + \alpha_{kn}h_{kn}b_n + e_{kn}\right)\right].\end{aligned} \tag{A.38}$$

Independence across assets implies that all terms with $\ell \neq n$ have zero expectation. Independence of investor-specific signal errors across investors implies that these errors do not contribute to the covariance between investors i and k. Therefore,

$$\mathbb{E}_0[\Omega_{ik}] = \frac{1}{\rho^2}\sum_{\ell=1}^{N}\left(\sum_{j=1}^{N}\chi_{ij\ell}\chi_{kj\ell}\right)\left[h_{i\ell}h_{k\ell}\sigma_\ell^2 + \alpha_{i\ell}h_{i\ell}\alpha_{k\ell}h_{k\ell}\tau_b^{-1}\right]. \tag{A.39}$$

Using (A.37), the portfolio-adjustment weight is:

$$\sum_{j=1}^{N} \chi_{ij\ell}\chi_{kj\ell} = \left(1 - \frac{\widehat{\lambda}_{i\ell}p_\ell^2}{B_i}\right)\left(1 - \frac{\widehat{\lambda}_{k\ell}p_\ell^2}{B_k}\right) + \frac{p_\ell^2}{B_iB_k}\sum_{j\neq\ell}\widehat{\lambda}_{ij}\widehat{\lambda}_{kj}p_j^2.$$

$$\omega_{ik,\ell} = \left(1 - \frac{\widehat{\lambda}_{i\ell}p_\ell^2}{B_i}\right)\left(1 - \frac{\widehat{\lambda}_{k\ell}p_\ell^2}{B_k}\right) + \frac{p_\ell^2}{B_iB_k}\sum_{j\neq\ell}\widehat{\lambda}_{ij}\widehat{\lambda}_{kj}p_j^2 \geq 0. \tag{A.40}$$

Each component of this expression is nonnegative because $B_i \geq \widehat{\lambda}_{i\ell}p_\ell^2$ and $B_k \geq \widehat{\lambda}_{k\ell}p_\ell^2$. Hence, $\omega_{ik,\ell} \geq 0$, and equations (A.14)–(A.40) follow.

A.7 Information Value in the Portfolio Benchmark

Define the prior expected excess return as:

$$\overline{m}_j = \mu_j - rp_j.$$

Conditional on posterior beliefs, substitution of the optimal portfolio into the objective gives the indirect utility:

$$U_i^*(\widehat{\boldsymbol{\mu}}_i, \widehat{\boldsymbol{\Lambda}}_i) = \frac{1}{2\rho}\sum_{j=1}^{N}\widehat{\lambda}_{ij}(\widehat{\mu}_{ij} - rp_j)^2 - \frac{1}{2\rho}\frac{\left[\sum_{j=1}^{N}\widehat{\lambda}_{ij}p_j(\widehat{\mu}_{ij} - rp_j) - \rho\right]^2}{B_i}. \tag{A.41}$$

To see how the budget constraint enters this expression, write $m_{ij} = \widehat{\mu}_{ij} - rp_j$ and substitute $q_{ij}^* = \widehat{\lambda}_{ij}(m_{ij} - \zeta_i p_j)/\rho$ into the objective:

$$U_i^* = \frac{1}{\rho}\sum_j \widehat{\lambda}_{ij}(m_{ij} - \zeta_i p_j)m_{ij} - \frac{1}{2\rho}\sum_j \widehat{\lambda}_{ij}(m_{ij} - \zeta_i p_j)^2 = \frac{1}{2\rho}\left[\sum_j \widehat{\lambda}_{ij}m_{ij}^2 - \zeta_i^2 B_i\right].$$

Using $\zeta_i = (A_i - \rho)/B_i$ gives (A.41). The second term imposes the budget constraint on the otherwise unconstrained portfolio value.

Using the posterior moments in (A.21) and independence across assets gives the ex ante value. Define:

$$S_i = \sum_{j=1}^{N}\widehat{\lambda}_{ij}p_j\overline{m}_j - \rho, \qquad T_i = \sum_{j=1}^{N}p_j^2\left(\frac{\widehat{\lambda}_{ij}^2}{\lambda_j^0} - \widehat{\lambda}_{ij}\right). \tag{A.42}$$

Chosen precisions, and hence B_i, are fixed before signals arrive. For the first term in (A.41),

$$\mathbb{E}_0\left[\widehat{\lambda}_{ij}(\widehat{\mu}_{ij} - rp_j)^2\right] = \widehat{\lambda}_{ij}\left[\overline{m}_j^2 + \mathrm{Var}_0(\widehat{\mu}_{ij})\right]$$

$$= \widehat{\lambda}_{ij}\overline{m}_j^2 + \frac{\widehat{\lambda}_{ij}}{\lambda_j^0} - 1.$$

For the budget term, the random variable $W_i = \sum_j \widehat{\lambda}_{ij} p_j(\widehat{\mu}_{ij} - r p_j) - \rho$ has mean S_i. Independence across assets gives

$$\mathrm{Var}_0(W_i) = \sum_j \widehat{\lambda}_{ij}^2 p_j^2 \left(\frac{1}{\lambda_j^0} - \frac{1}{\widehat{\lambda}_{ij}} \right) = T_i.$$

Thus $\mathbb{E}_0[W_i^2] = S_i^2 + T_i$. Taking expectations in (A.41) yields the ex ante gross utility function:

$$\mathbb{E}_0[U_i^*] = \mathcal{G}_i(\boldsymbol{h}_i) = \frac{1}{2\rho} \left\{ \sum_{j=1}^{N} \left[\widehat{\lambda}_{ij}\overline{m}_j^2 - 1 + \frac{\widehat{\lambda}_{ij}}{\lambda_j^0} \right] - \frac{S_i^2 + T_i}{B_i} \right\}. \tag{A.43}$$

For completeness, the first-order condition for total information can be written in closed form. Increasing h_{ij} changes only $\widehat{\lambda}_{ij}$, with $\partial\widehat{\lambda}_{ij}/\partial h_{ij} = 1$. The derivatives of the objects in the budget term are

$$\frac{\partial B_i}{\partial h_{ij}} = p_j^2, \qquad \frac{\partial S_i}{\partial h_{ij}} = p_j\overline{m}_j, \qquad \frac{\partial T_i}{\partial h_{ij}} = p_j^2 \left(\frac{2\widehat{\lambda}_{ij}}{\lambda_j^0} - 1 \right).$$

Applying the quotient rule to $(S_i^2 + T_i)/B_i$ and differentiating the remaining terms of (A.43) gives:

$$C_h(h_{ij}, \alpha_{ij}) = \frac{1}{2\rho} \left[\overline{m}_j^2 + \frac{1}{\lambda_j^0} - \frac{R'_{ij}B_i - (S_i^2 + T_i)p_j^2}{B_i^2} \right], \tag{A.44}$$

where

$$R'_{ij} = 2S_i p_j \overline{m}_j + p_j^2 \left(\frac{2\widehat{\lambda}_{ij}}{\lambda_j^0} - 1 \right). \tag{A.45}$$

The right-hand side of (A.24) is the marginal private value of increasing effective precision for asset j. The first two terms capture the direct value of a more precise posterior. The final fraction records how the budget constraint changes the value of information through the investor's remaining positions.

With symmetric assets and information, let $L_i = \sigma^{-2} + h_i$ and $\overline{m} = \mu - r$. Then

$$B_i = NL_i, \qquad S_i = NL_i\overline{m} - \rho, \qquad T_i = N(L_i^2\sigma^2 - L_i).$$

In particular,

$$\frac{S_i^2 + T_i}{B_i} = NL_i\overline{m}^2 - 2\rho\overline{m} + \frac{\rho^2}{NL_i} + L_i\sigma^2 - 1.$$

Substitution into (A.43) cancels $NL_i\overline{m}^2$. Using $L_i\sigma^2 - 1 = h_i\sigma^2$ gives

$$\mathcal{G}_i(h_i) = \mu - r - \frac{\rho}{2N(\sigma^{-2} + h_i)} + \frac{N-1}{2\rho}h_i\sigma^2. \tag{A.46}$$

Since the same precision is acquired for each of the N assets, total information cost is $NC(h_i, \alpha_i)$. The first-order conditions (A.25) and (A.26) are therefore

$$C_h(h_i, \alpha_i) = \frac{\mathcal{G}_i'(h_i)}{N} = \frac{\rho}{2N^2(\sigma^{-2} + h_i)^2} + \frac{N-1}{2\rho N}\sigma^2, \qquad C_\alpha(h_i, \alpha_i) = 0.$$

A.8 Portfolio Extensions

This subsection considers two extensions of the portfolio benchmark. The first imposes short-selling constraints and shows that common AI exposure continues to align local portfolio responses. The second introduces multiple AI providers and identifies the residual cross-investor covariance in model errors that sustains the mechanism.

A.8.1 Short-Selling Constraint and Common AI Exposure

The portfolio benchmark allows unrestricted positions in order to obtain a globally linear portfolio rule. This appendix introduces the short selling constraint

$$q_{ij} \geq 0 \qquad \text{for every investor } i \text{ and asset } j. \tag{A.47}$$

The constraint makes portfolio choice piecewise linear because the set of assets held by an investor becomes endogenous. The common-information mechanism remains present. Within any region in which investors' active sets are unchanged, a common AI shock moves their portfolios in aligned directions, and the strength of this aligned response increases with their joint reliance on AI information. The constraint also creates an extensive-margin channel because a common signal can cause several investors to enter or exit the same asset at the same time.

Portfolio Choice with Short-Selling Constraint. Conditional on posterior beliefs, investor i solves

$$\begin{aligned} \max_{\{q_{ij}\}_{j=1}^N} \quad & \sum_{j=1}^N q_{ij}m_{ij} - \frac{\rho}{2}\sum_{j=1}^N q_{ij}^2\widehat{\lambda}_{ij}^{-1} \\ \text{subject to} \quad & \sum_{j=1}^N p_j q_{ij} = 1, \qquad q_{ij} \geq 0, \end{aligned} \tag{A.48}$$

where

$$m_{ij} \equiv \widehat{\mu}_{ij} - rp_j.$$

Prices and posterior precisions are strictly positive. The objective is strictly concave, and the feasible set is compact, so the optimal portfolio exists and is unique.

Lemma A.2 (Portfolio choice with short-selling constraints) *The unique solution to* (A.48) *is*

$$q_{ij}^{*} = \frac{\widehat{\lambda}_{ij}}{\rho}\left[m_{ij} - \zeta_i p_j\right]_{+}, \qquad [x]_{+} \equiv \max\{x, 0\}, \tag{A.49}$$

where the scalar ζ_i is uniquely determined by

$$\sum_{j=1}^{N} \frac{\widehat{\lambda}_{ij} p_j}{\rho}\left[m_{ij} - \zeta_i p_j\right]_{+} = 1. \tag{A.50}$$

Define the active set

$$\mathcal{S}_i \equiv \left\{j : m_{ij} > \zeta_i p_j\right\}. \tag{A.51}$$

For any candidate active set $\mathcal{S}_i$, let

$$A_i(\mathcal{S}_i) \equiv \sum_{j\in\mathcal{S}_i} \widehat{\lambda}_{ij} p_j m_{ij}, \qquad B_i(\mathcal{S}_i) \equiv \sum_{j\in\mathcal{S}_i} \widehat{\lambda}_{ij} p_j^2. \tag{A.52}$$

Then

$$\zeta_i = \frac{A_i(\mathcal{S}_i) - \rho}{B_i(\mathcal{S}_i)}, \tag{A.53}$$

and the optimal holding can be written as

$$q_{ij}^{*} = \begin{cases} \dfrac{\widehat{\lambda}_{ij} p_j}{B_i(\mathcal{S}_i)} + \dfrac{\widehat{\lambda}_{ij}}{\rho}\left[m_{ij} - p_j \dfrac{A_i(\mathcal{S}_i)}{B_i(\mathcal{S}_i)}\right], & j \in \mathcal{S}_i, \\ 0, & j \notin \mathcal{S}_i. \end{cases} \tag{A.54}$$

Let ζ_i be the multiplier on the budget constraint and let $\nu_{ij} \geq 0$ be the multiplier on $q_{ij} \geq 0$. The Lagrangian is

$$\mathcal{L}_i = \sum_{j=1}^{N} q_{ij} m_{ij} - \frac{\rho}{2}\sum_{j=1}^{N} q_{ij}^2 \widehat{\lambda}_{ij}^{-1} + \zeta_i\left(1 - \sum_{j=1}^{N} p_j q_{ij}\right) + \sum_{j=1}^{N} \nu_{ij} q_{ij}.$$

The KKT conditions are

$$m_{ij} - \rho q_{ij}\widehat{\lambda}_{ij}^{-1} - \zeta_i p_j + \nu_{ij} = 0, \tag{A.55}$$

$$q_{ij} \geq 0, \qquad \nu_{ij} \geq 0, \qquad \nu_{ij} q_{ij} = 0. \tag{A.56}$$

If $q_{ij} > 0$, complementary slackness implies $\nu_{ij} = 0$, and (A.55) gives

$$q_{ij} = \frac{\widehat{\lambda}_{ij}}{\rho}\left(m_{ij} - \zeta_i p_j\right) > 0.$$

If $q_{ij} = 0$, then

$$\nu_{ij} = \zeta_i p_j - m_{ij} \geq 0,$$

which is equivalent to $m_{ij} \leq \zeta_i p_j$. Combining the two cases yields (A.49). Substitution into the budget constraint gives (A.50).

The left-hand side of (A.50) is continuous and weakly decreasing in ζ_i. It is strictly decreasing whenever at least one asset is active. It equals zero for sufficiently large ζ_i and diverges as $\zeta_i \to -\infty$. Hence, the budget equation has a unique solution.

For a fixed active set, the budget condition becomes

$$1 = \frac{1}{\rho}\sum_{j\in\mathcal{S}_i}\widehat{\lambda}_{ij}p_j\left(m_{ij} - \zeta_i p_j\right) = \frac{A_i(\mathcal{S}_i) - \zeta_i B_i(\mathcal{S}_i)}{\rho}.$$

Solving for ζ_i gives (A.53). Substituting this expression into the interior rule for $j \in \mathcal{S}_i$ and rearranging gives (A.54). Strict concavity of the objective establishes uniqueness.

The long-only solution has the same benchmark-plus-active-tilt structure as the unconstrained portfolio rule. The benchmark and all portfolio-wide averages are now calculated over the endogenous active set. The positive-part operator makes the global solution nonlinear in posterior beliefs, while the rule remains linear within each active-set region.

The Symmetric Case. To connect the extension to the symmetric benchmark in Appendix A.2, suppose

$$p_j = 1, \qquad \mu_j = \mu, \qquad \lambda_j^0 = \sigma^{-2}, \qquad h_{ij} = h_i, \qquad \alpha_{ij} = \alpha_i \tag{A.57}$$

for all assets j. The common posterior precision of investor i is

$$\widehat{\lambda}_i = \sigma^{-2} + h_i.$$

Let $K_i \equiv |\mathcal{S}_i|$ and define the active-set average of a variable x_{ij} as

$$\overline{x}_{i,\mathcal{S}_i} \equiv \frac{1}{K_i}\sum_{j\in\mathcal{S}_i} x_{ij}.$$

Corollary A.1 (Symmetric long-only portfolio) *Under* (A.57)*, the cutoff and optimal holdings are*

$$\zeta_i = \overline{m}_{i,\mathcal{S}_i} - \frac{\rho}{K_i \widehat{\lambda}_i}, \tag{A.58}$$

$$q_{ij}^* = \begin{cases} \dfrac{1}{K_i} + \dfrac{\widehat{\lambda}_i}{\rho}\left(m_{ij} - \overline{m}_{i,\mathcal{S}_i}\right), & j \in \mathcal{S}_i, \\ 0, & j \notin \mathcal{S}_i. \end{cases} \tag{A.59}$$

The active set satisfies

$$\begin{aligned} m_{ij} &> \overline{m}_{i,\mathcal{S}_i} - \frac{\rho}{K_i \widehat{\lambda}_i}, \quad j \in \mathcal{S}_i, \\ m_{ij} &\leq \overline{m}_{i,\mathcal{S}_i} - \frac{\rho}{K_i \widehat{\lambda}_i}, \quad j \notin \mathcal{S}_i. \end{aligned} \tag{A.60}$$

Under symmetry,

$$A_i(\mathcal{S}_i) = \widehat{\lambda}_i \sum_{j \in \mathcal{S}_i} m_{ij}, \qquad B_i(\mathcal{S}_i) = K_i \widehat{\lambda}_i.$$

Substitution into (A.53) gives (A.58). Substituting the cutoff into the positive holding rule yields

$$\begin{aligned} q_{ij}^* &= \frac{\widehat{\lambda}_i}{\rho}\left(m_{ij} - \overline{m}_{i,\mathcal{S}_i} + \frac{\rho}{K_i \widehat{\lambda}_i}\right) \\ &= \frac{1}{K_i} + \frac{\widehat{\lambda}_i}{\rho}\left(m_{ij} - \overline{m}_{i,\mathcal{S}_i}\right) \end{aligned}$$

for $j \in \mathcal{S}_i$. The inequalities in (A.60) are the KKT conditions for active and inactive assets.

Define the effective AI exposure of investor i by

$$a_i \equiv \alpha_i h_i. \tag{A.61}$$

The posterior mean satisfies

$$\widehat{\mu}_{ij} = \frac{\sigma^{-2}\mu + h_i s_{ij}}{\widehat{\lambda}_i},$$

where

$$s_{ij} = f_j + \alpha_i b_j + (1 - \alpha_i) u_{ij} + \alpha_i \epsilon_{ij}^A + (1 - \alpha_i)\epsilon_{ij}^P. \tag{A.62}$$

The common prior mean and the risk-free payoff cancel when the posterior payoff is demeaned within the active set. Consequently, for $j \in \mathcal{S}_i$,

$$q_{ij}^* = \frac{1}{K_i} + \frac{h_i}{\rho}\left(f_j - \overline{f}_{\mathcal{S}_i}\right) + \frac{\alpha_i h_i}{\rho}\left(b_j - \overline{b}_{\mathcal{S}_i}\right) + \frac{h_i}{\rho}\xi_{ij}^{\mathcal{S}_i}, \tag{A.63}$$

where

$$\xi_{ij}^{\mathcal{S}_i} = (1-\alpha_i)\left(u_{ij} - \overline{u}_{i,\mathcal{S}_i}\right) + \alpha_i\left(\epsilon_{ij}^{A} - \overline{\epsilon}_{i,\mathcal{S}_i}^{A}\right) + (1-\alpha_i)\left(\epsilon_{ij}^{P} - \overline{\epsilon}_{i,\mathcal{S}_i}^{P}\right). \tag{A.64}$$

Equation (A.63) displays the same three components as the unconstrained solution. Investors respond to asset fundamentals, to the common AI model error, and to investor-specific information. The active-set average replaces the full-market average.

Common AI Exposure and Portfolio Similarity. With a short-selling constraint, the active set varies with the realized shocks. Expected portfolio similarity therefore contains an extensive margin, through the distribution of active sets, and an intensive margin, through holdings conditional on those sets. Its unconditional expectation is:

$$\mathbb{E}_0[\Omega_{ik}] = \sum_{\mathcal{S}_i,\mathcal{S}_k} \Pr(\mathcal{S}_i,\mathcal{S}_k) \times \mathbb{E}_0\left[\Omega_{ik} \mid \mathcal{S}_i,\mathcal{S}_k\right], \tag{A.65}$$

where $\Pr(\mathcal{S}_i,\mathcal{S}_k)$ captures the extensive margin and $\mathbb{E}_0[\Omega_{ik} \mid \mathcal{S}_i,\mathcal{S}_k]$ captures similarity conditional on a pair of active sets. For a common AI error b_ℓ, (A.63) implies

$$\frac{\partial q_{ij}^*}{\partial b_\ell} = \frac{\alpha_i h_i}{\rho}\mathbf{1}\{\ell \in \mathcal{S}_i\}\left[\mathbf{1}\{j=\ell\} - \frac{\mathbf{1}\{j \in \mathcal{S}_i\}}{K_i}\right]. \tag{A.66}$$

A positive common signal about asset ℓ raises its portfolio weight and finances this increase through reductions in the other active positions.

Proposition A.2 (Common AI exposure with fixed active set) *Consider two distinct investors i and k at a state where their active sets $\mathcal{S}_i$ and $\mathcal{S}_k$ are locally stable. Let*

$$M_{ik} \equiv |\mathcal{S}_i \cap \mathcal{S}_k|.$$

For a common AI error b_ℓ, define the similarity of the two investors' local portfolio responses by

$$\mathcal{R}_{ik,\ell}^{A} \equiv \sum_{j=1}^{N} \frac{\partial q_{ij}^*}{\partial b_\ell}\frac{\partial q_{kj}^*}{\partial b_\ell}. \tag{A.67}$$

If $\ell \notin \mathcal{S}_i \cap \mathcal{S}_k$, then $\mathcal{R}_{ik,\ell}^{A} = 0$. If $\ell \in \mathcal{S}_i \cap \mathcal{S}_k$, then

$$\mathcal{R}_{ik,\ell}^{A} = \frac{(\alpha_i h_i)(\alpha_k h_k)}{\rho^2}\frac{(K_i-1)(K_k-1)+(M_{ik}-1)}{K_i K_k} \geq 0. \tag{A.68}$$

The inequality is strict whenever both investors hold at least two assets.

If ℓ is inactive for either investor, (A.66) implies that one response vector is zero, so $\mathcal{R}_{ik,\ell}^{A} = 0$.

Suppose $\ell \in \mathcal{S}_i \cap \mathcal{S}_k$. Substituting (A.66) into (A.67) gives

$$\mathcal{R}^A_{ik,\ell} = \frac{(\alpha_i h_i)(\alpha_k h_k)}{\rho^2} \sum_{j=1}^{N} \left[\mathbf{1}\{j=\ell\} - \frac{\mathbf{1}\{j \in \mathcal{S}_i\}}{K_i}\right] \times \left[\mathbf{1}\{j=\ell\} - \frac{\mathbf{1}\{j \in \mathcal{S}_k\}}{K_k}\right].$$

Expanding the product and summing over assets yields

$$\mathcal{R}^A_{ik,\ell} = \frac{(\alpha_i h_i)(\alpha_k h_k)}{\rho^2} \left[1 - \frac{1}{K_i} - \frac{1}{K_k} + \frac{M_{ik}}{K_i K_k}\right].$$

The bracket can be rewritten as

$$\frac{K_i K_k - K_i - K_k + M_{ik}}{K_i K_k} = \frac{(K_i - 1)(K_k - 1) + (M_{ik} - 1)}{K_i K_k}.$$

Because $M_{ik} \geq 1$ whenever ℓ belongs to the intersection, the expression is nonnegative. It is strictly positive when $K_i \geq 2$ and $K_k \geq 2$.

Proposition A.2 shows that, conditional on locally fixed active sets, the alignment of portfolio responses increases with the product of effective AI exposures, $(\alpha_i h_i)(\alpha_k h_k)$. Changes in active sets add a second synchronization margin: a common AI shock can lead several investors to enter or exit the same asset. This extensive margin generally has no closed-form expression because the active sets are endogenous. The unconstrained portfolio benchmark isolates the same common-error mechanism while preserving a globally linear portfolio rule.

A.8.2 Multiple AI Providers

The portfolio benchmark uses one AI signal to isolate the common-information mechanism. Multiple providers can reduce this exposure when their model errors are sufficiently independent. Correlated errors leave a residual common component even when investors diversify across many models.

Suppose there are M AI service providers, indexed by $m \in \{1, \ldots, M\}$. The signal that investor i obtains from provider m about asset j is:

$$s^A_{ij,m} = f_j + b_{j,m} + \epsilon^A_{ij,m}, \tag{A.69}$$

where $b_{j,m}$ is provider m's model error and $\epsilon^A_{ij,m}$ is an investor-specific error whose precision can be increased at a cost. Because model errors may be correlated across providers, write the signal vector as:

$$\boldsymbol{s}^A_{ij} = \mathbf{1} f_j + \boldsymbol{b}_j + \boldsymbol{\epsilon}^A_{ij}, \tag{A.70}$$

where $\boldsymbol{b}_j \sim \mathcal{N}(0, \Sigma_b^{-1})$ and $\boldsymbol{\epsilon}^A_{ij} \sim \mathcal{N}(0, (\boldsymbol{X}^A_{ij})^{-1})$. Define $\Omega_{ij} = \Sigma_b^{-1} + (\boldsymbol{X}^A_{ij})^{-1}$. Bayesian updat-

ing then gives:

$$\widehat{\lambda}_{ij} = \lambda_{ij}^0 + \mathbf{1}_M' \Omega_{ij}^{-1} \mathbf{1}_M + \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P}, \tag{A.71}$$

$$\widehat{\mu}_{ij} = \frac{\lambda_{ij}^0 \mu_{ij}^0 + \mathbf{1}_M' \Omega_{ij}^{-1} \boldsymbol{s}_{ij}^A + \frac{\tau_u x_{ij}^P}{\tau_u + x_{ij}^P} s_{ij}^P}{\widehat{\lambda}_{ij}}. \tag{A.72}$$

Define:

$$\boldsymbol{\omega}_{ij} = \frac{\Omega_{ij}^{-1} \mathbf{1}_M}{\mathbf{1}_M' \Omega_{ij}^{-1} \mathbf{1}_M}, \tag{A.73}$$

This is investor i's weight vector for asset j. Since $\boldsymbol{\omega}_{ij}' \mathbf{1}_M = 1$, the provider signals can be aggregated into the sufficient signal:

$$\tilde{s}_{ij}^A = \boldsymbol{\omega}_{ij}' \boldsymbol{s}_{ij}^A = f_j + \boldsymbol{\omega}_{ij}' \boldsymbol{b}_j + \boldsymbol{\omega}_{ij}' \boldsymbol{\epsilon}_{ij}^A. \tag{A.74}$$

The model-error component of this aggregate signal is investor specific because aggregation weights may differ:

$$\tilde{b}_{ij} = \boldsymbol{\omega}_{ij}' \boldsymbol{b}_j. \tag{A.75}$$

For two investors i and k, the relevant object for portfolio comovement is therefore the cross-investor covariance:

$$\mathrm{Cov}(\tilde{b}_{ij}, \tilde{b}_{kj}) = \boldsymbol{\omega}_{ij}' \Sigma_b^{-1} \boldsymbol{\omega}_{kj}. \tag{A.76}$$

Positive residual covariance preserves the common-error channel even when investors use several providers. For example, suppose $\mathbb{V}(b_{j,m}) = \tau_b^{-1}$ and $\mathrm{Corr}(b_{j,m}, b_{j,n}) = \rho_b \in [0, 1)$ for every $m \neq n$. Under symmetric provider weights, $\boldsymbol{\omega}_{ij} = M^{-1} \mathbf{1}_M$ for every investor, and $\Sigma_b^{-1} = \tau_b^{-1}[(1 - \rho_b) \boldsymbol{I}_M + \rho_b \mathbf{1}_M \mathbf{1}_M']$. The variance of the common aggregate error is:

$$\mathbb{V}(\tilde{b}_{ij}) = \frac{\tau_b^{-1}}{M} + \frac{M - 1}{M} \rho_b \tau_b^{-1}. \tag{A.77}$$

Thus, as $M \to \infty$, $\mathbb{V}(\tilde{b}_{ij}) \to \rho_b \tau_b^{-1}$, which is positive whenever $\rho_b > 0$. Diversification across providers eliminates independent model-specific noise but not the component shared across models, for example because providers use similar training data or architectures. Under common provider weights, the portfolio benchmark applies directly after replacing the single AI signal by $\tilde{s}_{ij}^A$. With heterogeneous weights, the exact formulas change, but the portfolio-similarity mechanism remains whenever $\boldsymbol{\omega}_{ij}' \Sigma_b^{-1} \boldsymbol{\omega}_{kj} > 0$.

B Macroeconomic Equilibrium and Comparative Statics

B.1 Proof of Lemma 1

Let $\mathbf{1} = (1,1,1)'$. Gaussian updating gives

$$V_{di}^{\kappa} = \left[\tau_d + \mathbf{1}'(R_i^{\kappa})^{-1}\mathbf{1}\right]^{-1}$$

and

$$(\ell_{Pi}^{\kappa}, \ell_{Ai}^{\kappa}, \ell_{qi}^{\kappa})' = V_{di}^{\kappa}(R_i^{\kappa})^{-1}\mathbf{1}.$$

The private-signal block contributes h_{Pi}. The inverse of the lower 2×2 block is

$$\frac{1}{D_i^{\kappa}} \begin{pmatrix} v_q^{\kappa} & -c_{Aq}^{\kappa} \\ -c_{Aq}^{\kappa} & v_{Ai} \end{pmatrix}.$$

Substitution gives (4.4)–(4.5).Suppress the firm index. Under the manager's subjective model, d and b^c are independent, with a joint prior

$$\begin{pmatrix} d \\ b^c \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \tau_d^{-1} & 0 \\ 0 & \tau_b^{-1} \end{pmatrix}\right).$$

For this proof, write $\omega_{Ai} \equiv v_{Ai} - \kappa/\tau_b = (1-\kappa)/\tau_b + 1/x_i^A > 0$ and $\omega_q \equiv \zeta^2/\tau_n > 0$ for the error variances conditional on (d, b^c). The three signals satisfy

$$\begin{pmatrix} s_i^P \\ s_i^A \\ \widetilde{q} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 1 & \sqrt{\kappa} \\ 1 & \psi\sqrt{\kappa} \end{pmatrix}}_{\mathbf{M}^{\kappa}} \begin{pmatrix} d \\ b^c \end{pmatrix} + \begin{pmatrix} u_i + \epsilon_i^P \\ \sqrt{1-\kappa}\, b_i^I + \epsilon_i^A \\ \zeta n \end{pmatrix}, \qquad \mathbf{W}_i^{\kappa} = \operatorname{diag}(h_{Pi}^{-1}, \omega_{Ai}, \omega_q).$$

The observation errors are independent of (d, b^c) and of each other. Since b^c is included in the state vector, its variance is excluded from $\mathbf{W}_i^{\kappa}$.

Standard Gaussian updating gives the joint posterior covariance and mean:

$$\begin{aligned} \mathbf{\Sigma}_{i,\text{post}}^{\kappa} &= \left[\begin{pmatrix} \tau_d & 0 \\ 0 & \tau_b \end{pmatrix} + (\mathbf{M}^{\kappa})'(\mathbf{W}_i^{\kappa})^{-1}\mathbf{M}^{\kappa}\right]^{-1} \\ &= \left[\begin{pmatrix} \tau_d + h_{Pi} + \omega_{Ai}^{-1} + \omega_q^{-1} & \sqrt{\kappa}(\omega_{Ai}^{-1} + \psi\omega_q^{-1}) \\ \sqrt{\kappa}(\omega_{Ai}^{-1} + \psi\omega_q^{-1}) & \tau_b + \kappa\omega_{Ai}^{-1} + \kappa\psi^2\omega_q^{-1} \end{pmatrix}\right]^{-1}, \end{aligned}$$

$$\begin{pmatrix} \widehat{d}_i^{\kappa} \\ \widehat{b}_i^{c,\kappa} \end{pmatrix} = \boldsymbol{\Sigma}_{i,\text{post}}^{\kappa} \begin{pmatrix} h_{Pi} s_i^P + s_i^A/\omega_{Ai} + \widetilde{q}/\omega_q \\ \sqrt{\kappa}(s_i^A/\omega_{Ai} + \psi\widetilde{q}/\omega_q) \end{pmatrix}.$$

The first diagonal entry of the covariance matrix is V_{di}^{κ}. Inverting the 2×2 precision matrix and collecting terms gives

$$\begin{aligned}(V_{di}^{\kappa})^{-1} &= \tau_d + h_{Pi} + \frac{1}{\omega_{Ai}} + \frac{1}{\omega_q} - \frac{\kappa(\omega_{Ai}^{-1} + \psi\omega_q^{-1})^2}{\tau_b + \kappa/\omega_{Ai} + \kappa\psi^2/\omega_q} \\ &= \tau_d + h_{Pi} + \frac{\tau_b(\omega_{Ai} + \omega_q) + \kappa(1-\psi)^2}{\tau_b\omega_{Ai}\omega_q + \kappa\omega_q + \kappa\psi^2\omega_{Ai}}.\end{aligned}$$

The definitions in the main text imply

$$\begin{aligned} &v_{Ai} = \omega_{Ai} + \kappa/\tau_b, && v_q^{\kappa} = \omega_q + \kappa\psi^2/\tau_b, \qquad c_{Aq}^{\kappa} = \kappa\psi/\tau_b, \\ &D_i^{\kappa} = \omega_{Ai}\omega_q + \frac{\kappa}{\tau_b}(\omega_q + \psi^2\omega_{Ai}), && v_{Ai} + v_q^{\kappa} - 2c_{Aq}^{\kappa} = \omega_{Ai} + \omega_q + \frac{\kappa}{\tau_b}(1-\psi)^2. \end{aligned}$$

Thus the last fraction is $(v_{Ai} + v_q^{\kappa} - 2c_{Aq}^{\kappa})/D_i^{\kappa}$, which gives (4.4).

The first row of the inverse precision matrix also gives

$$\begin{aligned}\widehat{d}_i^{\kappa} = V_{di}^{\kappa}\Bigg[&h_{Pi}s_i^P + \frac{s_i^A}{\omega_{Ai}} + \frac{\widetilde{q}}{\omega_q} \\ &- \frac{\kappa(\omega_{Ai}^{-1} + \psi\omega_q^{-1})}{\tau_b + \kappa/\omega_{Ai} + \kappa\psi^2/\omega_q}\left(\frac{s_i^A}{\omega_{Ai}} + \frac{\psi\widetilde{q}}{\omega_q}\right)\Bigg].\end{aligned}$$

Using $\tau_b\omega_{Ai}\omega_q + \kappa\omega_q + \kappa\psi^2\omega_{Ai} = \tau_b D_i^{\kappa}$, the coefficients on the AI and price signals simplify, respectively, to

$$\begin{aligned}\frac{1}{\omega_{Ai}} - \frac{\kappa(\omega_{Ai}^{-1} + \psi\omega_q^{-1})}{\omega_{Ai}(\tau_b + \kappa/\omega_{Ai} + \kappa\psi^2/\omega_q)} &= \frac{\tau_b\omega_q + \kappa\psi^2 - \kappa\psi}{\tau_b D_i^{\kappa}} = \frac{v_q^{\kappa} - c_{Aq}^{\kappa}}{D_i^{\kappa}}, \\ \frac{1}{\omega_q} - \frac{\kappa\psi(\omega_{Ai}^{-1} + \psi\omega_q^{-1})}{\omega_q(\tau_b + \kappa/\omega_{Ai} + \kappa\psi^2/\omega_q)} &= \frac{\tau_b\omega_{Ai} + \kappa - \kappa\psi}{\tau_b D_i^{\kappa}} = \frac{v_{Ai} - c_{Aq}^{\kappa}}{D_i^{\kappa}}.\end{aligned}$$

Consequently,

$$\widehat{d}_i^{\kappa} = V_{di}^{\kappa}\left[h_{Pi}s_i^P + \frac{v_q^{\kappa} - c_{Aq}^{\kappa}}{D_i^{\kappa}}s_i^A + \frac{v_{Ai} - c_{Aq}^{\kappa}}{D_i^{\kappa}}\widetilde{q}\right],$$

which establishes (4.3) and (4.5). The joint update accounts for learning about b^c; the lemma reports the marginal posterior of d needed for portfolio choice.

B.2 Proof of Lemma 2

Let $\mu_i^\kappa \equiv \widehat{v}_i^\kappa - q$. The first-order condition is $\mu_i^\kappa - \rho m_i \Omega_i^\kappa = 0$, which gives (4.8). Substitution yields

$$m_i^* \mu_i^\kappa - \frac{\rho}{2}(m_i^*)^2 \Omega_i^\kappa = \frac{(\mu_i^\kappa)^2}{2\rho\Omega_i^\kappa}.$$

The square appears because favorable and unfavorable signals generate profitable long and short positions, respectively.

B.3 Proof of Proposition 1

Using the conjectured investment rule, the manager's conditional expected payoff is

$$\widehat{v}_i^\kappa = \delta y_N + \chi z + \mu_L + \chi\omega_s s + \delta\left(\ell_{Pi}^\kappa s_i^P + \ell_{Ai}^\kappa s_i^A + \ell_{qi}^\kappa \tilde{q}\right) + g(k_0 + k_s s + k_q \tilde{q}).$$

Subtracting $q = \beta_0 + \beta_s s + \beta_q \tilde{q}$ gives

$$\begin{aligned}\widehat{v}_i^\kappa - q = {} & [\delta y_N + \chi z + \mu_L + g k_0 - \beta_0] \\ & + [\chi\omega_s + g k_s - \beta_s] s + \delta \ell_{Pi}^\kappa s_i^P + \delta \ell_{Ai}^\kappa s_i^A \\ & + [\delta \ell_{qi}^\kappa + g k_q - \beta_q]\tilde{q}.\end{aligned}$$

The equilibrium price coefficients derived below satisfy

$$\beta_0 = \delta y_N + \chi z + \mu_L + g k_0, \quad \beta_s = \chi\omega_s + g k_s, \quad \beta_q = \delta \bar{\ell}_T^\kappa + g k_q.$$

Consequently,

$$\widehat{v}_i^\kappa - q = \gamma_{Pi} s_i^P + \gamma_{Ai} s_i^A + \gamma_{qi}\tilde{q}.$$

Under the manager's subjective model, the expected excess gain satisfies:

$$\widehat{v}_i^\kappa - q = (\gamma_{Pi} + \gamma_{Ai} + \gamma_{qi})d + \sqrt{\kappa}(\gamma_{Ai} + \psi\gamma_{qi})b^c + \sqrt{1-\kappa}\gamma_{Ai} b_i^I + \gamma_{Pi} v_i^P + \gamma_{Ai}\epsilon_i^A + \zeta\gamma_{qi} n.$$

Here $v_i^P = u_i + \epsilon_i^P$ has variance h_{Pi}^{-1}. The primitive shocks in this expression are subjectively independent and mean zero, so its variance is (4.10). Moreover, $\mathbb{E}_{0,i}^\kappa[\widehat{v}_i^\kappa - q] = 0$. Because Gaussian posterior variances depend on chosen precisions but not on signal realizations, Ω_i^κ is deterministic at date 0. Taking the date-0 expectation of (4.9) therefore gives (4.12).

B.4 Proof of Proposition 2

At a symmetric equilibrium, all managers have the same residual payoff variance Ω^κ. Integrating the holding rule and imposing market clearing gives

$$0 = \int_0^1 m_i \, di + n = \frac{\int_0^1 \widehat{v}_i^\kappa \, di - q}{\rho \Omega^\kappa} + n, \qquad q = \int_0^1 \widehat{v}_i^\kappa \, di + \rho \Omega^\kappa n.$$

Aggregation uses the true signal process. Investor-specific errors wash out, so $\int_0^1 s_i^P \, di = d$ and $\int_0^1 s_i^A \, di = d + b$. Consequently,

$$\int_0^1 \widehat{d}_i^\kappa \, di = (\ell_P^\kappa + \ell_A^\kappa) d + \ell_A^\kappa b + \ell_q^\kappa \widetilde{q}.$$

Substitute this expression and the investment rule into the clearing price:

$$\begin{aligned} q = {} & \delta y_N + \chi z + \mu_L + g k_0 + (\chi \omega_s + g k_s) s \\ & + \delta(\ell_P^\kappa + \ell_A^\kappa) d + \delta \ell_A^\kappa b + (\delta \ell_q^\kappa + g k_q) \widetilde{q} + \rho \Omega^\kappa n. \end{aligned}$$

Now use $\widetilde{q} = d + \psi b + \zeta n$ and compare with $q = \beta_0 + \beta_s s + \beta_q (d + \psi b + \zeta n)$. The three stochastic coefficients must satisfy

$$\begin{aligned} d: \quad & \beta_q = \delta(\ell_P^\kappa + \ell_A^\kappa + \ell_q^\kappa) + g k_q, \\ b: \quad & \beta_q \psi = \delta \ell_A^\kappa + (\delta \ell_q^\kappa + g k_q) \psi, \\ n: \quad & \beta_q \zeta = (\delta \ell_q^\kappa + g k_q) \zeta + \rho \Omega^\kappa. \end{aligned}$$

The first equation gives (4.16). Subtracting ψ times the first equation from the second gives $\delta(\ell_P^\kappa + \ell_A^\kappa)\psi = \delta \ell_A^\kappa$. Subtracting ζ times the first equation from the third gives $\delta(\ell_P^\kappa + \ell_A^\kappa)\zeta = \rho \Omega^\kappa$. Thus

$$\psi = \frac{\ell_A^\kappa}{\ell_P^\kappa + \ell_A^\kappa}, \qquad \zeta = \frac{\rho \Omega^\kappa}{\delta(\ell_P^\kappa + \ell_A^\kappa)}.$$

Matching the constant and public-signal terms gives $\beta_0 = \delta y_N + \chi z + \mu_L + g k_0$ and $\beta_s = \chi \omega_s + g k_s$. Finally, each manager chooses information while holding this aggregate price rule fixed. The two individual first-order conditions, evaluated at symmetric choices, supply the last two equations in (4.19).

B.5 Proof of Proposition 3

Treating aggregate output as known, as maintained in the large-economy equilibrium, expected profit conditional on $(s_j, \widetilde{q}_j)$ is

$$Y_N^{\delta} Z^{\chi} M_j K_j^{g} - K_j.$$

Strict concavity follows from $0 < g < 1$. The first-order condition is

$$g Y_N^{\delta} Z^{\chi} M_j K_j^{g-1} = 1,$$

Rearranging gives $K_j^{1-g} = g Y_N^{\delta} Z^{\chi} M_j$, which proves (4.28). Taking logs and substituting (4.25) yields

$$\begin{aligned} k_j &= \Theta[\log g + \delta y_N + \chi z + \log M_j] \\ &= \Theta\left[\log g + \delta y_N + \chi z + \mu_L + \frac{1}{2}\Sigma^F\right] + \Theta\chi\omega_s s_j + \Theta\delta\omega_q^T \widetilde{q}_j. \end{aligned}$$

The constant, s_j, and $\widetilde{q}_j$ coefficients give (4.29)–(4.31).

B.6 Finite-Sample Allocation

For a finite cross section, define

$$K_N \equiv \frac{1}{N}\sum_{j=1}^{N} K_j, \qquad D_N \equiv \frac{1}{N}\sum_{j=1}^{N} M_j^{\Theta}, \qquad B_N \equiv \frac{1}{N}\sum_{j=1}^{N} X_j M_j^{\Theta-1}. \tag{B.1}$$

Proposition B.1 (Finite-sample allocation index) *The investment rule implies*

$$K_j = K_N \frac{M_j^{\Theta}}{D_N}. \tag{B.2}$$

Aggregate output admits the accounting representation

$$Y_N = Z \mathcal{A}_N^R K_N^{\eta}, \tag{B.3}$$

where realized finite-sample allocative TFP is

$$\mathcal{A}_N^R = \frac{B_N^{1/\chi}}{D_N^{\eta}} = \frac{\left[N^{-1}\sum_{j=1}^{N} X_j M_j^{\Theta-1}\right]^{1/\chi}}{\left[N^{-1}\sum_{j=1}^{N} M_j^{\Theta}\right]^{\eta}}. \tag{B.4}$$

Equation (4.28) has the form $K_j = C_N M_j^{\Theta}$. Averaging gives $K_N = C_N D_N$ and hence (B.2). Equations (3.2) and (3.4) imply

$$Y_N = Z\left[\frac{1}{N}\sum_{j=1}^{N} X_j K_j^g\right]^{1/\chi}.$$

Since $g\Theta = \Theta - 1$,

$$K_j^g = K_N^g \frac{M_j^{\Theta-1}}{D_N^g}.$$

The common factor K_N^g / D_N^g can be taken outside the cross-sectional sum. Hence

$$\begin{aligned} Y_N &= Z\left[\frac{K_N^g}{D_N^g}\frac{1}{N}\sum_j X_j M_j^{\Theta-1}\right]^{1/\chi} \\ &= Z K_N^{g/\chi}\frac{B_N^{1/\chi}}{D_N^{g/\chi}} = Z K_N^{\eta}\frac{B_N^{1/\chi}}{D_N^{\eta}}, \end{aligned}$$

where the last equality uses $g/\chi = \eta$. This proves (B.3) and (B.4).

For finite N, Proposition B.1 is an exact accounting result conditional on the limiting atomistic investment rule. The equilibrium interpretation applies in the large-N limit, where aggregate output is known conditional on Z.

B.7 Proof of Proposition 4

The law of large numbers gives $D_N \to D \equiv \mathbb{E}[M^{\Theta}]$ and

$$B_N \to \mathbb{E}[X M^{\Theta-1}] = \mathbb{E}\Big[\mathbb{E}[X \mid s, \widetilde{q}] M^{\Theta-1}\Big] = \mathbb{E}[M^{\Theta}] = D.$$

The law of iterated expectations applies because $M^{\Theta-1}$ is measurable with respect to $(s, \widetilde{q})$ and that $M = \mathbb{E}[X \mid s, \widetilde{q}]$. Since $M > 0$, $D > 0$, and the continuous mapping theorem gives

$$\mathcal{A}_N^R \longrightarrow \frac{D^{1/\chi}}{D^{\eta}} = D^{p_A},$$

where $p_A = \frac{1}{\chi} - \eta > 0$. This establishes (4.33). Next, let $\widehat{L} \equiv \mathbb{E}[\log X \mid s, \widetilde{q}]$. It is Gaussian with mean μ_L and variance $\Sigma_0 - \Sigma^F$, while

$$M = \exp(\widehat{L} + \tfrac{1}{2}\Sigma^F).$$

Hence

$$\begin{aligned}\log \mathbb{E}[M^{\Theta}] &= \Theta\mu_L + \frac{\Theta}{2}\Sigma^F + \frac{\Theta^2}{2}(\Sigma_0 - \Sigma^F) \\ &= \Theta\mu_L + \frac{\Theta^2}{2}\Sigma_0 - \frac{\Theta(\Theta-1)}{2}\Sigma^F.\end{aligned}$$

Multiplying by p_A proves (4.35).

For the aggregate capital and output expressions in the main text, multiply the firm's first-order condition by K_j and take the cross-sectional limit:

$$K = gY^{\delta}Z^{\chi}\mathbb{E}[MK_j^g].$$

Investment is measurable with respect to the firm's information, so $\mathbb{E}[MK_j^g] = \mathbb{E}[XK_j^g]$. The final-good aggregator gives $\mathbb{E}[XK_j^g] = (Y/Z)^{\chi}$. Therefore,

$$K = gY^{\delta}Z^{\chi}(Y/Z)^{\chi} = gY,$$

where $\delta + \chi = 1$. Combining $K = gY$ with $Y = Z\mathcal{A}K^{\eta}$ gives $K^{1-\eta} = gZ\mathcal{A}$ and hence (4.36).

B.8 Proof of Lemma 3

We specialize the equilibrium system already obtained in (4.19). In terms of (h_i, α_i), its information conditions are those in (4.13). Thus the system to simplify is

$$\begin{aligned}\psi &= \frac{\ell_A^{\kappa}}{\ell_P^{\kappa} + \ell_A^{\kappa}}, \\ \zeta &= \frac{\rho\Omega^{\kappa}}{\delta(\ell_P^{\kappa} + \ell_A^{\kappa})}, \\ \left.\frac{\partial \mathcal{G}_i^{\kappa}}{\partial h_i}\right|_{(h_i,\alpha_i)=(h,\alpha)} &= \mathcal{C}_h^I(h,\alpha), \\ \left.\frac{\partial \mathcal{G}_i^{\kappa}}{\partial \alpha_i}\right|_{(h_i,\alpha_i)=(h,\alpha)} &= \mathcal{C}_{\alpha}^I(h,\alpha).\end{aligned}$$

The information derivatives hold the aggregate price rule and aggregate posterior loading fixed, as in the general equilibrium system.

Posterior beliefs. Substituting $\kappa = 0$ into (4.1) gives $c_{Aq}^0 = 0$ and $v_q^0 = \zeta^2/\tau_n = 1/r$. The

covariance matrix is therefore diagonal. The posterior formulas in Lemma 1 reduce to

$$\begin{aligned} V_{di}^0 &= \frac{1}{\tau_d + h_i + r}, \\ \widehat{d}_i^0 &= V_{di}^0\left[(1-\alpha_i)h_i s_i^P + \alpha_i h_i s_i^A + r\widetilde{q}\right], \\ \ell_{Pi}^0 &= V_{di}^0(1-\alpha_i)h_i, \qquad \ell_{Ai}^0 = V_{di}^0\alpha_i h_i, \qquad \ell_{qi}^0 = V_{di}^0 r. \end{aligned} \tag{B.5}$$

The residual payoff variance in (4.7) becomes

$$\Omega_i^0 = \Omega_a + \delta^2 V_{di}^0. \tag{B.6}$$

Price equilibrium. At a symmetric information choice, $\ell_P^0 + \ell_A^0 = V_d^0 h$ and $\ell_A^0 = V_d^0 \alpha h$. Substituting these expressions into the first two equilibrium conditions gives

$$\psi = \alpha, \qquad \zeta = \frac{\rho(\Omega_a + \delta^2 V_d^0)}{\delta V_d^0 h} = \frac{\rho[\Omega_a(\tau_d + h + r) + \delta^2]}{\delta h}.$$

Using $r = \tau_n/\zeta^2$ gives (5.8). Hence, for a given total precision, the price equations determine ψ from α and r from h.

Information quantity and composition. We now simplify the remaining two equilibrium conditions. Write $V_i = V_{di}^0$ and $\bar{\ell}_T = \bar{\ell}_T^0$ within this proof. In (4.10), setting $\kappa = 0$ combines the AI error terms using $1/\tau_b + 1/\pi_{Ai} = 1/(\alpha_i h_i)$. Together with the private signal error, they contribute

$$\frac{\gamma_{Pi}^2}{(1-\alpha_i)h_i} + \frac{\gamma_{Ai}^2}{\alpha_i h_i} = \delta^2 V_i^2 h_i.$$

Substituting the other posterior loadings into that same variance formula then yields

$$H_i^0 = \delta^2\left\{\frac{[V_i(h_i + r) - \bar{\ell}_T]^2}{\tau_d} + V_i^2 h_i + \frac{(V_i r - \bar{\ell}_T)^2}{r}\right\}. \tag{B.7}$$

For a fixed aggregate price rule, both H_i^0 and Ω_i^0 depend on the manager's information choice only through h_i. Since $\mathcal{G}_i^0 = H_i^0/(2\rho\Omega_i^0)$, it follows that $\partial\mathcal{G}_i^0/\partial\alpha_i = 0$. The composition condition therefore reduces to

$$\mathcal{C}_\alpha^I(h, \alpha) = 0.$$

This statement concerns the manager's subjective ex ante value. The realized posterior mean still depends on how the manager weights the two signals.

For the quantity condition, differentiate the individual objective with respect to h_i before

imposing symmetry. Throughout this derivative, $(r, \bar{\ell}_T)$ remain fixed. At symmetry, write

$$V = \frac{1}{\tau_d + h + r}, \qquad \bar{\ell}_T = V(h + r), \qquad \Omega = \Omega_a + \delta^2 V.$$

Equation (B.7) then implies

$$H^0 = \delta^2 V^2 h \left(1 + \frac{h}{r}\right). \tag{B.8}$$

Because $\partial V_i / \partial h_i = -V_i^2$, the derivatives of the three terms in braces in (B.7), evaluated at symmetry, give

$$\left.\frac{\partial H_i^0}{\partial h_i}\right|_{(h,\alpha)} = \delta^2 \left[0 + (V^2 - 2hV^3) + 2hV^3\right] = \delta^2 V^2, \qquad \left.\frac{\partial \Omega_i^0}{\partial h_i}\right|_{(h,\alpha)} = -\delta^2 V^2.$$

The quotient rule therefore gives the marginal gross return to quantity:

$$\left.\frac{\partial \mathcal{G}_i^0}{\partial h_i}\right|_{(h,\alpha)} = \frac{\delta^2 V^2}{2\rho\Omega}\left(1 + \frac{H^0}{\Omega}\right). \tag{B.9}$$

Only after taking this individual derivative do we substitute the equilibrium price precision $r(h)$. The resulting marginal return is $m(h)$, so the quantity condition is $\mathcal{C}_h^I(h, \alpha) = m(h)$.

For completeness, under (3.13), the cost derivatives can be written using the marginal costs of effective precision:

$$\mathrm{MC}_P(h_P) \equiv c_P \frac{\tau_u^{\nu+1} h_P^{\nu-1}}{(\tau_u - h_P)^{\nu+1}}, \tag{B.10}$$

$$\mathrm{MC}_A(h_A; c_A, \tau_b) \equiv c_A \frac{\tau_b^{\nu+1} h_A^{\nu-1}}{(\tau_b - h_A)^{\nu+1}}. \tag{B.11}$$

With $h_P = (1 - \alpha)h$ and $h_A = \alpha h$, the chain rule yields

$$\mathcal{C}_h^I = (1 - \alpha)\,\mathrm{MC}_P + \alpha\,\mathrm{MC}_A, \qquad \mathcal{C}_\alpha^I = h(\mathrm{MC}_A - \mathrm{MC}_P).$$

Thus the conditions in (h, α) also imply $m(h) = \mathrm{MC}_P = \mathrm{MC}_A$. Since the two marginal costs are strictly increasing in their respective effective precisions, $\mathcal{C}_{\alpha\alpha}^I = h^2(\mathrm{MC}_P' + \mathrm{MC}_A') > 0$. The interior composition condition therefore minimizes information cost at a given total precision.

Properties of the reduced system. For every $h > 0$, the right-hand side of (5.8) is positive at $r = 0$ and strictly decreasing in r, whereas the left-hand side increases from zero without bound. The positive solution exists and is unique. To establish the comparative statics, use the

following abbreviations only in this proof:

$$T \equiv \tau_d + h + r(h), \qquad D \equiv \Omega_a T + \delta^2. \tag{B.12}$$

Implicit differentiation of the price equation gives

$$r'(h) = \frac{2r(D/h - \Omega_a)}{D + 2\Omega_a r} > 0, \tag{B.13}$$

where $D/h - \Omega_a = [\Omega_a(\tau_d + r) + \delta^2]/h > 0$. Substituting $V = 1/T$, $\Omega = D/T$, and $r = \tau_n \delta^2 h^2/(\rho^2 D^2)$ into (B.9) yields

$$m(h) = \frac{\delta^2}{2\rho}\left(\frac{1}{TD} + \frac{\delta^2 h}{T^2 D^2}\right) + \frac{\delta^2 \rho}{2\tau_n T^2} > 0. \tag{B.14}$$

Finally, let $U = TD$. Since $T' = 1 + r' > 1$, $U' = T'(2\Omega_a T + \delta^2) > \delta^2$. Hence

$$\frac{d}{dh}\left(\frac{1}{U} + \frac{\delta^2 h}{U^2}\right) = \frac{\delta^2 - U'\left(1 + 2\delta^2 h/U\right)}{U^2} < 0.$$

The final term in (B.14) also decreases with h, which establishes $m'(h) < 0$. This last derivative compares marginal returns across symmetric price equilibria; the manager's first-order condition itself holds prices fixed.

B.9 Proof of Proposition 5

Fix $\nu > 1$ and $\kappa \in [0, 1]$. All derivatives below compare symmetric equilibria along the specified technology path. The individual information derivative is taken holding the price rule fixed before imposing symmetry. We begin with the equilibrium system in (4.19).

Since $v_{Ai} = 1/h_{Ai}$, the determinant in the posterior formulas satisfies

$$D_i^\kappa = \frac{v_q^\kappa - (c_{Aq}^\kappa)^2 h_{Ai}}{h_{Ai}}.$$

Multiplying the numerators and denominators of the posterior weights by h_{Ai} gives

$$\begin{aligned} \ell_{Pi}^\kappa &= V_{di}^\kappa h_{Pi}, \\ \ell_{Ai}^\kappa &= V_{di}^\kappa h_{Ai} \frac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_{Ai}}, \\ \ell_{qi}^\kappa &= V_{di}^\kappa \frac{1 - c_{Aq}^\kappa h_{Ai}}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_{Ai}}. \end{aligned} \tag{B.15}$$

The posterior precision is equivalently

$$(V_{di}^{\kappa})^{-1} = \tau_d + h_{Pi} + \frac{1}{v_q^{\kappa}} + h_{Ai}\frac{(1 - c_{Aq}^{\kappa}/v_q^{\kappa})^2}{1 - (c_{Aq}^{\kappa})^2 h_{Ai}/v_q^{\kappa}}. \tag{B.16}$$

Differentiating the last expression yields

$$\frac{\partial (V_{di}^{\kappa})^{-1}}{\partial h_{Pi}} = 1, \qquad \frac{\partial (V_{di}^{\kappa})^{-1}}{\partial h_{Ai}} = \left(\frac{v_q^{\kappa} - c_{Aq}^{\kappa}}{v_q^{\kappa} - (c_{Aq}^{\kappa})^2 h_{Ai}}\right)^2.$$

To obtain the marginal information value, the posterior projection identities give

$$\mathrm{Var}_i^{\kappa}(\widehat{d}_i^{\kappa}) = \tau_d^{-1} - V_{di}^{\kappa}, \qquad \mathrm{Cov}_i^{\kappa}(\widehat{d}_i^{\kappa}, \widetilde{q}) = \tau_d^{-1}.$$

Using the price coefficients in Proposition 2, expected excess returns equal $\delta(\widehat{d}_i^{\kappa} - \bar{\ell}_T^{\kappa}\widetilde{q})$. Their perceived variance can therefore be written as

$$H_i^{\kappa} = \delta^2 \left[\frac{(1 - \bar{\ell}_T^{\kappa})^2}{\tau_d} + (\bar{\ell}_T^{\kappa})^2 v_q^{\kappa} - V_{di}^{\kappa}\right].$$

This is an equivalent expression for (4.10). Differentiating $H_i^{\kappa}/(2\rho\Omega_i^{\kappa})$, with $\Omega_i^{\kappa} = \Omega_a + \delta^2 V_{di}^{\kappa}$ and the aggregate price rule fixed, gives

$$\frac{\partial \mathcal{G}_i^{\kappa}}{\partial h_{Pi}} = \frac{\delta^2 (V_{di}^{\kappa})^2}{2\rho(\Omega_i^{\kappa})^2}\left\{\Omega_a + \delta^2\left[\frac{(1 - \bar{\ell}_T^{\kappa})^2}{\tau_d} + (\bar{\ell}_T^{\kappa})^2 v_q^{\kappa}\right]\right\}.$$

At symmetry, $\bar{\ell}_T^{\kappa} = 1 - \tau_d V_d^{\kappa}$. Hence

$$\mathcal{G}_{h_P}^{\kappa} = \frac{\delta^2}{2\rho[\Omega_a (V_d^{\kappa})^{-1} + \delta^2]^2}\left\{\Omega_a + \delta^2\left[\tau_d (V_d^{\kappa})^2 + (1 - \tau_d V_d^{\kappa})^2 v_q^{\kappa}\right]\right\}. \tag{B.17}$$

The AI marginal value equals this expression multiplied by the squared posterior correction in

the derivative of (B.16). Substitution into the four equilibrium conditions gives

$$\begin{aligned}
\psi &= \frac{h_A \dfrac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A}}{h_P + h_A \dfrac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A}}, \\
\frac{\zeta^2}{\tau_n} &= \frac{\rho^2[\Omega_a (V_d^\kappa)^{-1} + \delta^2]^2}{\tau_n \delta^2 \left[h_P + h_A \dfrac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A}\right]^2}, \\
\mathrm{MC}_P(h_P) &= \mathcal{G}_{h_P}^\kappa, \\
\mathrm{MC}_A(h_A; c_A, \tau_b) &= \mathcal{G}_{h_P}^\kappa \left(\frac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A}\right)^2.
\end{aligned} \tag{B.18}$$

Here $v_q^\kappa = \zeta^2/\tau_n + \kappa\psi^2/\tau_b$ and $c_{Aq}^\kappa = \kappa\psi/\tau_b$ adjust with equilibrium prices.

The first price equation implies that its denominator is $h_P/(1-\psi)$. The second price equation consequently becomes

$$\frac{\zeta^2}{\tau_n} = \frac{\rho^2[\Omega_a (V_d^\kappa)^{-1} + \delta^2]^2 (1-\psi)^2}{\tau_n \delta^2 h_P^2}. \tag{B.19}$$

The posterior weights also satisfy

$$\frac{\ell_A^\kappa}{V_d^\kappa} = \frac{\psi h_P}{1-\psi}, \qquad c_{Aq}^\kappa \ell_A^\kappa + v_q^\kappa \ell_q^\kappa = V_d^\kappa.$$

The first identity uses the price equation; the second follows directly by adding the two weight expressions in (B.15). Since $\tau_d V_d^\kappa + \ell_P^\kappa + \ell_A^\kappa + \ell_q^\kappa = 1$,

$$\begin{aligned}
(V_d^\kappa)^{-1} &= \tau_d + h_P + \frac{\ell_A^\kappa + \ell_q^\kappa}{V_d^\kappa} \\
&= \tau_d + \frac{1}{v_q^\kappa} + \frac{h_P}{1-\psi}\left(1 - \frac{c_{Aq}^\kappa \psi}{v_q^\kappa}\right).
\end{aligned}$$

Using $v_q^\kappa - c_{Aq}^\kappa \psi = \zeta^2/\tau_n$ gives the exact equilibrium identity

$$(V_d^\kappa)^{-1} = \tau_d + \frac{1 + (\zeta^2/\tau_n) h_P/(1-\psi)}{v_q^\kappa}. \tag{B.20}$$

At the no-AI equilibrium, $h_P = h_0$, $h_A = \psi = 0$, and

$$v_q^\kappa = \frac{\zeta^2}{\tau_n} = S_0 = \frac{1}{r(h_0)}, \qquad (V_d^\kappa)^{-1} = \tau_d + h_0 + r(h_0).$$

These values are independent of κ. The functions established in Appendix B.8 imply

$$\begin{aligned} r'(h_0) &= \frac{2r(h_0)\{\Omega_a[\tau_d + r(h_0)] + \delta^2\}}{h_0\{\Omega_a[\tau_d + h_0 + 3r(h_0)] + \delta^2\}} > 0, \\ s'(h_0) &= -\frac{r'(h_0)}{r(h_0)^2} < 0, \\ \mathrm{MC}_P'(h_0) &= m_0\left[\frac{\nu - 1}{h_0} + \frac{\nu + 1}{\tau_u - h_0}\right] > 0. \end{aligned} \tag{B.21}$$

We next differentiate the symmetric marginal value in (B.17). This is a derivative across symmetric configurations, rather than a second derivative of the individual's fixed-price objective. Holding posterior precision fixed gives

$$\frac{\partial \mathcal{G}_{h_P}^\kappa}{\partial v_q^\kappa} = \frac{\delta^4(1 - \tau_d V_d^\kappa)^2}{2\rho[\Omega_a(V_d^\kappa)^{-1} + \delta^2]^2}.$$

Evaluating at the no-AI point and using $r(h_0) = \tau_n\delta^2 h_0^2/\{\rho^2[\Omega_a(\tau_d + h_0 + r(h_0)) + \delta^2]^2\}$ yields

$$\left.\frac{\partial \mathcal{G}_{h_P}^\kappa}{\partial v_q^\kappa}\right|_0 = \frac{\delta^2\rho r(h_0)}{2\tau_n}\left(\frac{h_0 + r(h_0)}{h_0[\tau_d + h_0 + r(h_0)]}\right)^2.$$

Along the no-AI price relation at a given private precision h, posterior precision is $\tau_d + h + r(h)$ and price-error variance is $s(h)$. The chain rule therefore gives

$$m'(h_0) = [1 + r'(h_0)]\left.\frac{\partial \mathcal{G}_{h_P}^\kappa}{\partial[(V_d^\kappa)^{-1}]}\right|_0 + s'(h_0)\left.\frac{\partial \mathcal{G}_{h_P}^\kappa}{\partial v_q^\kappa}\right|_0.$$

Solving for the first partial derivative and substituting into the total differential gives, at the no-AI point,

$$\begin{aligned} d\mathcal{G}_{h_P}^\kappa = {} & \frac{m'(h_0)}{1 + r'(h_0)}\, d[(V_d^\kappa)^{-1}] \\ & + \frac{\delta^2\rho r(h_0)}{2\tau_n}\left(\frac{h_0 + r(h_0)}{h_0[\tau_d + h_0 + r(h_0)]}\right)^2\left\{dv_q^\kappa - \frac{s'(h_0)}{1 + r'(h_0)} d[(V_d^\kappa)^{-1}]\right\}. \end{aligned} \tag{B.22}$$

This identity does not impose $\mathcal{G}_{h_P}^\kappa = m(h)$ away from the no-AI point. The term in braces re-

tains departures from the no-AI relation between posterior precision and price-error variance.

Fix $\tau_b > 0$ and parameterize the local price and private-information equilibrium by h_A. The AI first-order condition then determines c_A. The first equation in (B.18) can be written as

$$\psi h_P = (1-\psi) h_A \frac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A}.$$

At $h_A = 0$, the ratio on the right equals one because $c_{Aq}^\kappa = \kappa\psi/\tau_b = 0$. Differentiating the product identity and evaluating at this boundary leaves $h_0 (d\psi/dh_A)|_0 = 1$. Moreover, the exact variance derivative is

$$\frac{dv_q^\kappa}{dh_A} = \frac{d(\zeta^2/\tau_n)}{dh_A} + \frac{2\kappa\psi}{\tau_b}\frac{d\psi}{dh_A}.$$

Only after taking the fixed-accuracy boundary limit does its last term vanish. Thus

$$\left.\frac{d\psi}{dh_A}\right|_0 = \frac{1}{h_0}, \qquad \left.\frac{dv_q^\kappa}{dh_A}\right|_0 = \left.\frac{d(\zeta^2/\tau_n)}{dh_A}\right|_0.$$

The quotient rule applied to (B.20) gives at this boundary

$$\begin{aligned}\left.\frac{d[(V_d^\kappa)^{-1}]}{dh_A}\right|_0 = h_0 r(h_0) \left.\frac{d(\zeta^2/\tau_n)}{dh_A}\right|_0 + \left.\frac{dh_P}{dh_A}\right|_0 + 1 \\ - [r(h_0)^2 + h_0 r(h_0)] \left.\frac{d(\zeta^2/\tau_n)}{dh_A}\right|_0.\end{aligned}$$

The two terms with $h_0 r(h_0)$ cancel. Logarithmic differentiation of (B.19) supplies the second equation below:

$$\begin{aligned}\left.\frac{d[(V_d^\kappa)^{-1}]}{dh_A}\right|_0 &= \left.\frac{dh_P}{dh_A}\right|_0 + 1 - r(h_0)^2 \left.\frac{d(\zeta^2/\tau_n)}{dh_A}\right|_0, \\ \left.\frac{d(\zeta^2/\tau_n)}{dh_A}\right|_0 &= \frac{2}{r(h_0)}\left[\frac{\Omega_a}{\Omega_a[\tau_d + h_0 + r(h_0)] + \delta^2} \left.\frac{d[(V_d^\kappa)^{-1}]}{dh_A}\right|_0 - \frac{1 + dh_P/dh_A|_0}{h_0}\right].\end{aligned} \tag{B.23}$$

Substitute the first line into the second. The coefficient of the noise-variance derivative is $1 + 2\Omega_a r(h_0)/\{\Omega_a[\tau_d + h_0 + r(h_0)] + \delta^2\} > 0$. Dividing by this coefficient and using (B.21) gives

$$\begin{aligned}\left.\frac{d(\zeta^2/\tau_n)}{dh_A}\right|_0 &= s'(h_0)\left(1 + \left.\frac{dh_P}{dh_A}\right|_0\right), \\ \left.\frac{d[(V_d^\kappa)^{-1}]}{dh_A}\right|_0 &= [1 + r'(h_0)]\left(1 + \left.\frac{dh_P}{dh_A}\right|_0\right).\end{aligned}$$

The correction term in (B.22) is therefore zero. Differentiating the private-information condition yields

$$\mathrm{MC}_P'(h_0)\left.\frac{dh_P}{dh_A}\right|_0 = m'(h_0)\left(1+\left.\frac{dh_P}{dh_A}\right|_0\right).$$

Consequently,

$$\left.\frac{dh_P}{dh_A}\right|_0 = \lambda_0 - 1, \qquad \left.\frac{dh}{dh_A}\right|_0 = \lambda_0, \qquad \left.\frac{d(\zeta^2/\tau_n)}{dh_A}\right|_0 = s'(h_0)\lambda_0 < 0. \tag{B.24}$$

The positive price coefficient above and $\mathrm{MC}_P'(h_0) - m'(h_0) > 0$ establish the nonsingularity of this local subsystem, justifying its parameterization by h_A.

The AI condition can now be inverted along this branch:

$$c_A = \mathcal{G}^{\kappa}_{h_A} h_A^{1-\nu}(1 - h_A/\tau_b)^{\nu+1}.$$

Since $\mathcal{G}^{\kappa}_{h_A} \to m_0 > 0$, this expression tends to infinity as $h_A \downarrow 0$. Its exact logarithmic derivative is

$$\frac{1}{c_A}\frac{dc_A}{dh_A} = \frac{1}{\mathcal{G}^{\kappa}_{h_A}}\frac{d\mathcal{G}^{\kappa}_{h_A}}{dh_A} - \frac{\nu-1}{h_A} - \frac{\nu+1}{\tau_b - h_A}. \tag{B.25}$$

The first and last terms remain bounded at fixed positive τ_b. The middle term tends to minus infinity, so the inverse is locally decreasing and

$$h_A = \left(\frac{m_0}{c_A}\right)^{1/(\nu-1)}[1+o(1)], \qquad \frac{dh_A}{dc_A} = -\frac{h_A}{(\nu-1)c_A}[1+o(1)].$$

The true common-error term has derivative $2\psi(d\psi/dh_A)/\tau_b \to 0$. Hence

$$\left.\frac{dS_q^T}{dh_A}\right|_0 = s'(h_0)\lambda_0 < 0, \qquad \lim_{c_A\to\infty}\frac{c_A}{h_A}\frac{d\log\mathcal{A}}{dc_A} = \frac{\Gamma_0 s'(h_0)\lambda_0}{\nu-1} < 0.$$

This proves part (i), including the information expansion in (5.4).

Now fix $c_A > 0$ and let $\tau_b \downarrow 0$. Feasibility implies $0 < h_A < \tau_b$. On the positive-loading branch, the posterior correction in the first equation of (B.18) is positive. It is at most one because $c^{\kappa}_{Aq}h_A = \kappa\psi h_A/\tau_b < 1$ and the denominator minus the numerator equals $c^{\kappa}_{Aq}(1 - c^{\kappa}_{Aq}h_A) \geq 0$. It follows that $\psi \leq h_A/h_P = O(\tau_b)$ and $v_q^{\kappa} \to 1/r(h_0)$.

Divide the product form of the first price equation by τ_b and use $c^{\kappa}_{Aq} = \kappa\psi/\tau_b$. Since the relevant ratios are bounded,

$$h_0\frac{\psi}{\tau_b} = \frac{h_A}{\tau_b}\left(1 - \kappa r(h_0)\frac{\psi}{\tau_b}\right) + o(1).$$

Solving for the price-loading ratio gives

$$\begin{aligned} \frac{\psi}{\tau_b} &= \frac{h_A/\tau_b}{h_0 + \kappa r(h_0) h_A/\tau_b} + o(1), \\ \frac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A} &= \frac{h_0}{h_0 + \kappa r(h_0) h_A/\tau_b} + o(1). \end{aligned} \tag{B.26}$$

The AI first-order condition becomes

$$c_A \tau_b^{\nu-1} \frac{(h_A/\tau_b)^{\nu-1}}{(1 - h_A/\tau_b)^{\nu+1}} = \mathcal{G}_{h_P}^\kappa \left(\frac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A} \right)^2. \tag{B.27}$$

If h_A/τ_b converged along a subsequence to a value below one, the left side would vanish because $\nu > 1$. The right side would have a strictly positive limit, since $\mathcal{G}_{h_P}^\kappa \to m_0$ and the correction in (B.26) stays bounded away from zero. Thus $h_A/\tau_b \to 1$. The other two limits in (5.5) follow from (B.26) and $\alpha = h_A/h$. Substitution back into the AI condition gives

$$1 - \frac{h_A}{\tau_b} = \left(\frac{c_A [h_0 + \kappa r(h_0)]^2}{m_0 h_0^2} \right)^{1/(\nu+1)} \tau_b^{(\nu-1)/(\nu+1)} [1 + o(1)]. \tag{B.28}$$

In raw precision units, this implies

$$\pi_A = \frac{h_A}{1 - h_A/\tau_b} = \left(\frac{m_0 h_0^2}{c_A [h_0 + \kappa r(h_0)]^2} \right)^{1/(\nu+1)} \tau_b^{2/(\nu+1)} [1 + o(1)].$$

Hence $\pi_A \to 0$ while $\pi_A/\tau_b \to \infty$: acquisition noise becomes small relative to model error even as acquisition itself vanishes.

To justify taking limits of derivatives, first regard $(\tau_b, h_A/\tau_b)$ as inputs to the two price equations and the private-information condition, and solve for $(h_P, \zeta^2/\tau_n, \psi/\tau_b)$. At $\tau_b = 0$, the first two values are (h_0, S_0), independent of h_A/τ_b, and the third is given by (B.26). The no-AI price coefficient is positive, the private-information coefficient is $\mathrm{MC}_P'(h_0) - m'(h_0) > 0$, and the loading equation has coefficient $h_0 + \kappa r(h_0) h_A/\tau_b > 0$. Thus the local Jacobian is nonsingular and this subsystem has a smooth extension near $(\tau_b, h_A/\tau_b) = (0, 1)$.

Rewrite the AI equation exactly as

$$1 - \frac{h_A}{\tau_b} = \tau_b^{(\nu-1)/(\nu+1)} \left\{ \frac{c_A (h_A/\tau_b)^{\nu-1}}{\mathcal{G}_{h_P}^\kappa \left[\dfrac{v_q^\kappa - c_{Aq}^\kappa}{v_q^\kappa - (c_{Aq}^\kappa)^2 h_A} \right]^2} \right\}^{1/(\nu+1)}.$$

The expression in braces is positive and smooth at the boundary. Treating τ_b and $\tau_b^{(\nu-1)/(\nu+1)}$ temporarily as separate inputs, the derivative of the left minus right side with respect to h_A/τ_b is -1 when both inputs vanish. The implicit function theorem therefore gives a smooth solution in these two inputs. Returning to the actual path yields

$$\frac{d(h_A/\tau_b)}{d\tau_b} = O(\tau_b^{-2/(\nu+1)}), \qquad \tau_b \frac{d(h_A/\tau_b)}{d\tau_b} \to 0.$$

Since h_P and ζ^2/τ_n do not depend on this ratio at $\tau_b = 0$, their partial derivatives with respect to it are $O(\tau_b)$. The induced terms in their total derivatives vanish. The price-loading ratio is smooth and bounded as well. Consequently, the required derivatives have finite limits and

$$\frac{dh_A}{d\tau_b} \to 1, \qquad \frac{d\psi}{d\tau_b} \to \frac{1}{h_0 + \kappa r(h_0)}, \qquad \frac{d\alpha}{d\tau_b} \to \frac{1}{h_0}. \tag{B.29}$$

These regularity properties justify the derivative limits used below.

The exact variance derivative along the accuracy path is

$$\frac{dv_q^\kappa}{d\tau_b} = \frac{d(\zeta^2/\tau_n)}{d\tau_b} + \kappa \left(\frac{2\psi}{\tau_b} \frac{d\psi}{d\tau_b} - \frac{\psi^2}{\tau_b^2} \right).$$

Unlike the fixed-accuracy path, the common-error term has a nonzero limit:

$$\left.\frac{dv_q^\kappa}{d\tau_b}\right|_{0^+} = \left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \frac{\kappa}{[h_0 + \kappa r(h_0)]^2}. \tag{B.30}$$

In the remainder of the proof, every derivative marked 0^+ is the limit along this accuracy path. Differentiating (B.20), the numerator and denominator contributions give

$$\begin{aligned}\left.\frac{d[(V_d^\kappa)^{-1}]}{d\tau_b}\right|_{0^+} &= h_0 r(h_0) \left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \left.\frac{dh_P}{d\tau_b}\right|_{0^+} + \frac{h_0}{h_0 + \kappa r(h_0)} \\ &\quad - r(h_0)[h_0 + r(h_0)] \left[\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \frac{\kappa}{[h_0 + \kappa r(h_0)]^2} \right].\end{aligned}$$

Combining terms gives the first equation of the derivative system:

$$\left.\frac{d[(V_d^\kappa)^{-1}]}{d\tau_b}\right|_{0^+} = \left.\frac{dh_P}{d\tau_b}\right|_{0^+} - r(h_0)^2 \left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \frac{h_0^2 - \kappa r(h_0)^2}{[h_0 + \kappa r(h_0)]^2}. \tag{B.31}$$

Logarithmic differentiation of (B.19) gives the second equation:

$$\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} = \frac{2}{r(h_0)}\left[\frac{\Omega_a}{\Omega_a[\tau_d + h_0 + r(h_0)] + \delta^2}\left.\frac{d[(V_d^\kappa)^{-1}]}{d\tau_b}\right|_{0^+} - \frac{1}{h_0}\left.\frac{dh_P}{d\tau_b}\right|_{0^+} - \frac{1}{h_0 + \kappa r(h_0)}\right]. \tag{B.32}$$

Finally, differentiate the private-information condition and use (B.22) and (B.30):

$$\begin{aligned}\mathrm{MC}_P'(h_0)\left.\frac{dh_P}{d\tau_b}\right|_{0^+} = {} & \frac{m'(h_0)}{1 + r'(h_0)}\left.\frac{d[(V_d^\kappa)^{-1}]}{d\tau_b}\right|_{0^+} \\ & + \frac{\delta^2 \rho r(h_0)}{2\tau_n}\left(\frac{h_0 + r(h_0)}{h_0[\tau_d + h_0 + r(h_0)]}\right)^2 \\ & \times \left\{\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \frac{\kappa}{[h_0 + \kappa r(h_0)]^2} - \frac{s'(h_0)}{1 + r'(h_0)}\left.\frac{d[(V_d^\kappa)^{-1}]}{d\tau_b}\right|_{0^+}\right\}.\end{aligned} \tag{B.33}$$

Substitute (B.31) into (B.32). Collecting the noise-variance derivative on the left and using (B.21) gives

$$\begin{aligned}\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} = {} & s'(h_0)\left[\left.\frac{dh_P}{d\tau_b}\right|_{0^+} + \frac{h_0}{h_0 + \kappa r(h_0)}\right] \\ & - \frac{2\kappa\Omega_a[h_0 + r(h_0)]}{\{\Omega_a[\tau_d + h_0 + 3r(h_0)] + \delta^2\}[h_0 + \kappa r(h_0)]^2}.\end{aligned} \tag{B.34}$$

Solving this equation for $dh_P/d\tau_b|_{0^+}$ and substituting back into (B.31) gives a relation between the two remaining derivatives. To simplify its coefficients, (B.21) implies

$$\frac{\Omega_a[\tau_d + h_0 + r(h_0)] + \delta^2}{\{\Omega_a[\tau_d + h_0 + 3r(h_0)] + \delta^2\}[1 + r'(h_0)]} = \frac{h_0}{h_0 + 2r(h_0)}.$$

It follows that

$$\frac{1}{1 + r'(h_0)}\left.\frac{d[(V_d^\kappa)^{-1}]}{d\tau_b}\right|_{0^+} = \frac{1}{s'(h_0)}\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \frac{\kappa}{s'(h_0)[h_0 + \kappa r(h_0)]^2}\left(1 + \frac{h_0}{h_0 + 2r(h_0)}\right). \tag{B.35}$$

In particular, the braces in (B.33) reduce to

$$\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \frac{\kappa}{[h_0 + \kappa r(h_0)]^2} - \frac{s'(h_0)}{1 + r'(h_0)}\left.\frac{d[(V_d^\kappa)^{-1}]}{d\tau_b}\right|_{0^+} = -\frac{\kappa h_0}{[h_0 + 2r(h_0)][h_0 + \kappa r(h_0)]^2}.$$

The private-information derivative equation therefore becomes

$$\mathrm{MC}'_P(h_0)\left.\frac{dh_P}{d\tau_b}\right|_{0^+} = \frac{m'(h_0)}{s'(h_0)}\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} + \frac{\kappa m'(h_0)}{s'(h_0)[h_0+\kappa r(h_0)]^2}\left(1+\frac{h_0}{h_0+2r(h_0)}\right) - \frac{\kappa h_0}{[h_0+2r(h_0)][h_0+\kappa r(h_0)]^2}\frac{\delta^2\rho r(h_0)}{2\tau_n}\left(\frac{h_0+r(h_0)}{h_0[\tau_d+h_0+r(h_0)]}\right)^2.$$

Now solve (B.34) explicitly for the private-information derivative:

$$\left.\frac{dh_P}{d\tau_b}\right|_{0^+} = \frac{1}{s'(h_0)}\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} - \frac{h_0}{h_0+\kappa r(h_0)} + \frac{2\kappa\Omega_a[h_0+r(h_0)]}{s'(h_0)\{\Omega_a[\tau_d+h_0+3r(h_0)]+\delta^2\}[h_0+\kappa r(h_0)]^2}. \tag{B.36}$$

Substitute it into the preceding private-information equation and multiply by $s'(h_0)[h_0+\kappa r(h_0)]^2$. Moving the noise-variance terms to the left yields

$$\begin{aligned}&[\mathrm{MC}'_P(h_0)-m'(h_0)][h_0+\kappa r(h_0)]^2\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+}\\ &\quad= \mathrm{MC}'_P(h_0)s'(h_0)h_0[h_0+\kappa r(h_0)] - \frac{2\kappa\Omega_a[h_0+r(h_0)]\,\mathrm{MC}'_P(h_0)}{\Omega_a[\tau_d+h_0+3r(h_0)]+\delta^2}\\ &\qquad+\kappa m'(h_0)\left(1+\frac{h_0}{h_0+2r(h_0)}\right) - \frac{\kappa h_0 s'(h_0)}{h_0+2r(h_0)}\frac{\delta^2\rho r(h_0)}{2\tau_n}\left(\frac{h_0+r(h_0)}{h_0[\tau_d+h_0+r(h_0)]}\right)^2.\end{aligned}$$

Use $s'(h_0) = -r'(h_0)/r(h_0)^2$ and the identity

$$\frac{2\Omega_a[h_0+r(h_0)]}{\Omega_a[\tau_d+h_0+3r(h_0)]+\delta^2} - s'(h_0)h_0r(h_0) = 1+[1+r'(h_0)]\frac{h_0}{h_0+2r(h_0)}.$$

The right side of the preceding derivative equation then becomes

$$\begin{aligned}&\mathrm{MC}'_P(h_0)h_0^2s'(h_0) - \kappa[\mathrm{MC}'_P(h_0)-m'(h_0)]\\ &\quad-\frac{\kappa h_0}{h_0+2r(h_0)}\left\{\mathrm{MC}'_P(h_0)-m'(h_0)+r'(h_0)\,\mathrm{MC}'_P(h_0) - \frac{\delta^2\rho[h_0+r(h_0)]^2r'(h_0)}{2\tau_n h_0^2 r(h_0)[\tau_d+h_0+r(h_0)]^2}\right\}.\end{aligned}$$

Dividing by the positive coefficient on the left and using λ_0 gives

$$\begin{aligned}\left.\frac{d(\zeta^2/\tau_n)}{d\tau_b}\right|_{0^+} &= \frac{h_0^2 s'(h_0)\lambda_0 - \kappa}{[h_0+\kappa r(h_0)]^2} \\ &\quad - \frac{\kappa h_0}{[h_0+2r(h_0)][h_0+\kappa r(h_0)]^2} \\ &\quad \times \left\{1 + r'(h_0)\lambda_0\left[1 - \frac{\delta^2\rho[h_0+r(h_0)]^2}{2\tau_n h_0^2 r(h_0)[\tau_d+h_0+r(h_0)]^2\,\mathrm{MC}_P'(h_0)}\right]\right\}.\end{aligned} \tag{B.37}$$

Equation (B.36) now determines the private-information response; total information satisfies $dh/d\tau_b|_{0^+} = 1 + dh_P/d\tau_b|_{0^+}$. For general κ, this response need not equal λ_0.

By (B.29),

$$\left.\frac{d(\psi^2/\tau_b)}{d\tau_b}\right|_{0^+} = \lim_{\tau_b\downarrow 0}\left(\frac{2\psi}{\tau_b}\frac{d\psi}{d\tau_b} - \frac{\psi^2}{\tau_b^2}\right) = \frac{1}{[h_0+\kappa r(h_0)]^2}.$$

Adding this to (B.37) gives $dS_q^T/d\tau_b|_{0^+}$. Since

$$\left.\frac{d\log\mathcal{A}}{d\tau_b}\right|_{0^+} = -\Gamma_0\left.\frac{dS_q^T}{d\tau_b}\right|_{0^+},$$

the full expression can be collected as

$$\left.\frac{d\log\mathcal{A}}{d\tau_b}\right|_{0^+} = \frac{\Gamma_0}{[h_0+\kappa r(h_0)]^2}\left[-h_0^2 s'(h_0)\lambda_0 - 1 + \kappa B_0\right], \tag{B.38}$$

where we define

$$B_0 \equiv 1 + \frac{h_0}{h_0+2r(h_0)}\left\{1 + r'(h_0)\lambda_0\left[1 - \frac{\delta^2\rho[h_0+r(h_0)]^2}{2\tau_n h_0^2 r(h_0)[\tau_d+h_0+r(h_0)]^2\,\mathrm{MC}_P'(h_0)}\right]\right\}. \tag{B.39}$$

Every term in B_0 is evaluated at the no-AI equilibrium, so B_0 is independent of κ. Writing $r_0 = r(h_0)$ and $s_0' = s'(h_0)$ and factoring h_0^2 out of the bracket gives (5.6). Thus B_0 only abbreviates the coefficient on κ in the full expression. The fundamental variance $\Sigma_0 = \chi^2/\tau_a + \delta^2/\tau_d$ remains fixed along this comparison; model-error variance affects firms' posterior uncertainty through S_q^T.

The sign of B_0 requires a separate argument. Under (5.17), (B.21) implies $\mathrm{MC}_P'(h_0) \geq m_0/h_0$.

The no-AI marginal value in (B.17) and the price equation give

$$\frac{m_0}{r(h_0)} - \frac{\delta^2\rho[h_0+r(h_0)]^2}{2\tau_n h_0^2 r(h_0)[\tau_d+h_0+r(h_0)]^2} = \frac{\delta^2\left\{\Omega_a+\delta^2\tau_d/[\tau_d+h_0+r(h_0)]^2\right\}}{2\rho r(h_0)\{\Omega_a[\tau_d+h_0+r(h_0)]+\delta^2\}^2} > 0.$$

Dividing the resulting bound by $\mathrm{MC}'_P(h_0)$ yields

$$\frac{\delta^2\rho[h_0+r(h_0)]^2}{2\tau_n h_0^2 r(h_0)[\tau_d+h_0+r(h_0)]^2\,\mathrm{MC}'_P(h_0)} < \frac{m_0}{r(h_0)\,\mathrm{MC}'_P(h_0)} \le \frac{h_0}{r(h_0)}.$$

Also, (B.21) gives $0 < r'(h_0) < 2r(h_0)/h_0$, and $0 < \lambda_0 < 1$. If $r(h_0) \ge h_0$, the square bracket in (B.39) is positive, so $B_0 > 1$. If $r(h_0) < h_0$, these bounds instead imply

$$\begin{aligned} B_0 &> 1 + \frac{h_0}{h_0+2r(h_0)}\left[1 + r'(h_0)\lambda_0\left(1-\frac{h_0}{r(h_0)}\right)\right] \\ &> 1 + \frac{h_0}{h_0+2r(h_0)}\left(\frac{2r(h_0)}{h_0}-1\right) = \frac{4r(h_0)}{h_0+2r(h_0)} > 0. \end{aligned}$$

Hence the stated curvature condition ensures $B_0 > 0$. For general $\nu > 1$, the relabelling itself does not impose this sign.

Full neglect. Setting $\kappa = 0$ in (B.37) gives $d(\zeta^2/\tau_n)/d\tau_b|_{0^+} = s'(h_0)\lambda_0$. Equation (B.36) then gives $dh_P/d\tau_b|_{0^+} = \lambda_0 - 1$, which proves the complete-neglect derivative in (5.15). For the level expansion used in the text, at $\kappa = 0$ the private condition is exactly $m(h) = \mathrm{MC}_P(h - h_A)$. Its derivative gives

$$\frac{dh}{dh_A} = \frac{\mathrm{MC}'_P(h_P)}{\mathrm{MC}'_P(h_P) - m'(h)} \longrightarrow \lambda_0.$$

Thus (5.13) holds along either boundary. Since $S_q^T = s(h) + h_A^2/(\tau_b h^2)$,

$$S_q^T - S_0 = s'(h_0)\lambda_0 h_A + o(h_A) + \frac{h_A^2}{\tau_b h_0^2} + o(h_A).$$

The last remainder follows from $h - h_0 = O(h_A)$ and $h_A/\tau_b < 1$. This proves (5.14).

For the threshold statement, multiply (5.6) by $[h_0+\kappa r(h_0)]^2/\Gamma_0 > 0$. The resulting expression is $-h_0^2 s'(h_0)\lambda_0 - 1 + \kappa B_0$, which is affine in κ, with coefficients evaluated at the common no-AI economy. If it is negative at zero and positive at one, its slope is strictly positive and it has exactly one zero in $(0,1)$. Its sign gives the sign of the initial accuracy effect on either side. Strict negativity at zero also persists for sufficiently small positive κ without any additional

endpoint condition.

Correct recognition. Substitute (B.39) into (B.38) and set $\kappa = 1$. The term -1 cancels with the leading constant in B_0. Substituting $s'(h_0) = -r'(h_0)/r(h_0)^2$ and collecting the remaining terms gives

$$\left.\frac{d\log\mathcal{A}}{d\tau_b}\right|_{0^+,\kappa=1} = \frac{\Gamma_0 h_0}{[h_0+2r(h_0)][h_0+r(h_0)]^2} \times\left\{1+\frac{r'(h_0)\lambda_0[h_0+r(h_0)]^2}{r(h_0)^2}\left[1-\frac{\delta^2\rho r(h_0)}{2\tau_n h_0^2[\tau_d+h_0+r(h_0)]^2\,\mathrm{MC}_P'(h_0)}\right]\right\}. \tag{B.40}$$

Using the definition of λ_0, this derivative is strictly positive if and only if

$$\mathrm{MC}_P'(h_0)\left[1+\frac{r'(h_0)[h_0+r(h_0)]^2}{r(h_0)^2}\right] > m'(h_0)+\frac{\delta^2\rho[h_0+r(h_0)]^2 r'(h_0)}{2\tau_n h_0^2 r(h_0)[\tau_d+h_0+r(h_0)]^2}. \tag{B.41}$$

A sufficient condition is

$$\mathrm{MC}_P'(h_0) \geq \frac{\delta^2\rho r(h_0)}{2\tau_n h_0^2[\tau_d+h_0+r(h_0)]^2},$$

because the square bracket in (B.40) is then nonnegative and the braces remain strictly positive.

To verify the stated condition on ν, (B.21) gives

$$\frac{h_0\,\mathrm{MC}_P'(h_0)}{m_0} = \nu-1+(\nu+1)\frac{h_0}{\tau_u-h_0} \geq 1 \quad\Longleftrightarrow\quad \nu \geq 2-\frac{3h_0}{\tau_u}.$$

At the no-AI point, (B.17) and the price equation imply

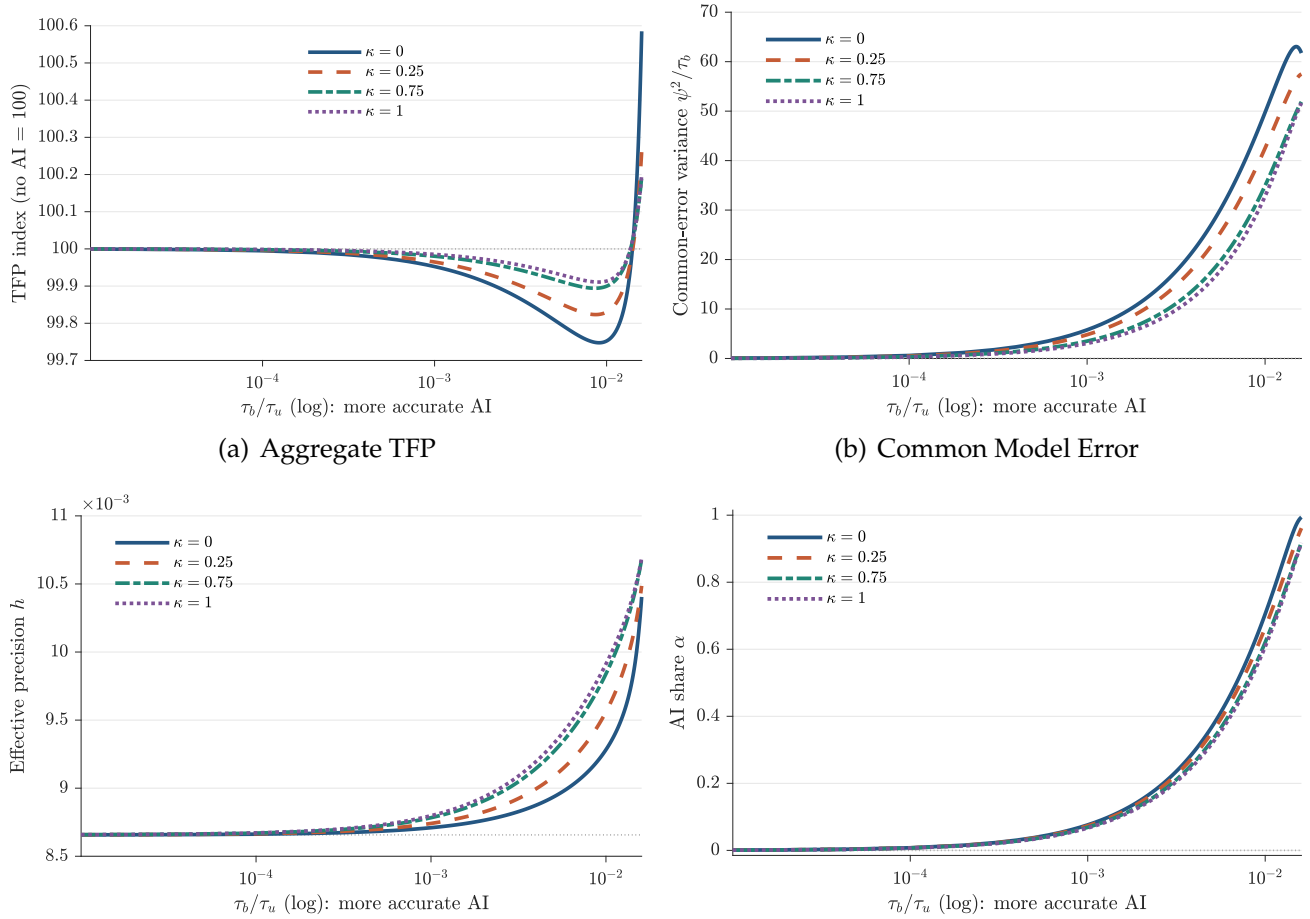
$$\frac{m_0}{\delta^2\rho r(h_0)/\{2\tau_n h_0^2[\tau_d+h_0+r(h_0)]^2\}} = \frac{\Omega_a[\tau_d+h_0+r(h_0)]^2}{\delta^2}+\tau_d+\frac{[h_0+r(h_0)]^2}{r(h_0)} > h_0.$$

Consequently, under (5.17),

$$\mathrm{MC}_P'(h_0) \geq \frac{m_0}{h_0} > \frac{\delta^2\rho r(h_0)}{2\tau_n h_0^2[\tau_d+h_0+r(h_0)]^2}.$$

The initial accuracy effect is therefore strictly positive. This is a sufficient condition, while (B.41) provides the exact criterion for general $\nu > 1$.

The sufficient condition in (5.17) need not hold when acquisition costs are close to linear. Figure B.1 illustrates an initial TFP decline with nearly linear costs, $\nu = 1.01$, even when investors correctly recognize the common model error, $\kappa = 1$.



(a) Aggregate TFP

(b) Common Model Error

(c) Total Information

(d) AI Information Share

Figure B.1: Equilibrium Effect of AI Accuracy Changes

Notes: This figure shows the equilibrium effect as AI accuracy τ_b changes, where we choose parameters such that even with $\kappa = 1$, a marginal increase in τ_b could initially decrease the equilibrium aggregate TFP. Parameters are $c_A = 0.2$, $c_P = 1.7$, $\tau_n = 100$, $\tau_u = 1$, $\nu = 1.01$.

B.10 Model Training from Market Outcomes

We allow market information to affect subsequent model training by repeating the baseline economy. Only model precision carries over between periods; conditional on $\tau_{b,t}$, fundamentals and signal errors retain the baseline distributions and independence assumptions. After trading and investment, the population precision of the price signal, $\tau_{q,t}^T$, determines the quality of the next model update:

$$\tau_{b,t+1} = \bar{\tau}_b \mathcal{T}(\tau_{q,t}^T), \qquad \bar{\tau}_b > 0. \tag{B.42}$$

Here $\bar{\tau}_b$ measures exogenous training efficiency. The function $\mathcal{T}$ is positive and continuously differentiable, with $\mathcal{T}' \geq 0$ and both $\mathcal{T}$ and $\mathcal{T}'$ finite near r_0. This reduced-form technology assumes that more informative market observations improve model accuracy, holding other training inputs fixed.

Conditional on model precision, the baseline equilibrium equations are unchanged. On a regular symmetric branch, a stationary equilibrium satisfies $\tau_b = \bar{\tau}_b \mathcal{T}(\tau_q^T(c_A, \tau_b; \kappa))$. Assume local stability, $|\bar{\tau}_b \mathcal{T}'(\tau_q^T)\, \partial \tau_q^T / \partial \tau_b| < 1$. Since information affects TFP through price precision, differentiating the stationary condition gives

$$\begin{aligned} \frac{d \log \mathcal{A}}{d c_A} &= \frac{\partial \log \mathcal{A} / \partial c_A}{1 - \bar{\tau}_b \mathcal{T}'(\tau_q^T) \partial \tau_q^T / \partial \tau_b}, \\ \frac{d \log \mathcal{A}}{d \bar{\tau}_b} &= \frac{\mathcal{T}(\tau_q^T)\, \partial \log \mathcal{A} / \partial \tau_b}{1 - \bar{\tau}_b \mathcal{T}'(\tau_q^T) \partial \tau_q^T / \partial \tau_b}. \end{aligned} \tag{B.43}$$

Partial derivatives are those of the baseline equilibrium at the stationary (c_A, τ_b), holding the other technology parameter fixed. Total derivatives also account for the adjustment in trained precision.

The reciprocal of the denominator is the training feedback multiplier. It exceeds one when better AI improves price informativeness: better market data then support further gains in model accuracy. When better AI reduces price informativeness, poorer training data dampen the initial change in accuracy, and the multiplier lies below one. Local stability keeps the denominator positive, so feedback changes the magnitude of the baseline productivity effects without reversing their signs.

At both boundaries in Proposition 5, this multiplier tends to one. At the high-cost boundary, trained precision approaches $\bar{\tau}_b \mathcal{T}(r_0) > 0$, while $\partial \tau_q^T / \partial \tau_b \to 0$ because prices no longer depend on unused AI information. At the low-accuracy boundary, $\bar{\tau}_b \to 0$ and $\partial \tau_q^T / \partial \tau_b$ remains finite, so the feedback term again vanishes. Hence

$$\lim_{c_A \to \infty} \frac{c_A}{h_A} \frac{d \log \mathcal{A}}{d c_A} = \frac{\Gamma_0 s_0' \lambda_0}{\nu - 1} < 0, \tag{B.44}$$

$$\left.\frac{d\log\mathcal{A}}{d\bar{\tau}_b}\right|_{0^+} = \mathcal{T}(r_0)\left.\frac{\partial\log\mathcal{A}}{\partial\tau_b}\right|_{0^+}. \tag{B.45}$$

Cheaper access therefore raises TFP near the high-cost boundary. The initial accuracy effect differs from the baseline only by the positive factor $\mathcal{T}(r_0)$, leaving its sign criterion and any interior correlation-belief threshold unchanged. These results compare stationary equilibria; transition dynamics require a separate analysis.

C General Information Costs

C.1 Proof of Proposition 6

All comparative statics are taken along the symmetric equilibrium branch. When obtaining an individual's information first-order condition, the aggregate price rule is held fixed. The functions $m(h)$ and $s(h)$ subsequently incorporate the equilibrium adjustment of that price rule. We keep the private-information cost and all parameters other than the specified AI parameter fixed.

We first express acquisition costs in effective-precision coordinates. The raw and effective AI precisions satisfy

$$h_A = \frac{\tau_b\pi_A}{\tau_b+\pi_A}, \qquad \pi_A = \frac{\tau_b h_A}{\tau_b - h_A}, \qquad 0 < h_A < \tau_b. \tag{C.1}$$

Holding τ_b fixed gives

$$\frac{\partial h_A}{\partial \pi_A} = \left(\frac{\tau_b}{\tau_b+\pi_A}\right)^2, \qquad \frac{\partial \pi_A}{\partial h_A} = \left(1-\frac{h_A}{\tau_b}\right)^{-2}.$$

The marginal cost of effective AI precision is therefore

$$\mathrm{MC}_A(h_A; c_A, \tau_b) = c_A f'(\pi_A)\left(1-\frac{h_A}{\tau_b}\right)^{-2}. \tag{C.2}$$

At $\kappa = 0$, information quantity and composition satisfy $\mathcal{C}_h^I = m(h)$ and $\mathcal{C}_\alpha^I = 0$ at an interior choice. Since $h_P = (1-\alpha)h$ and $h_A = \alpha h$, the chain rule gives

$$\mathcal{C}_h^I = (1-\alpha)\,\mathrm{MC}_P + \alpha\,\mathrm{MC}_A, \qquad \mathcal{C}_\alpha^I = h(\mathrm{MC}_A - \mathrm{MC}_P).$$

The composition condition equates the two marginal costs; the quantity condition equates their

common value to $m(h)$. Thus the nonlinear system is

$$m(h) = \mathrm{MC}_P(h - h_A), \qquad m(h) = \mathrm{MC}_A(h_A; c_A, \tau_b). \tag{C.3}$$

Equivalently, the AI condition in raw precision is

$$m(h) \left(\frac{\tau_b}{\tau_b + \pi_A} \right)^2 = c_A f'(\pi_A). \tag{C.4}$$

Using (C.1) in this equality proves (6.6).

The private-information condition determines how private and total precision adjust to AI acquisition. Recall the no-AI equilibrium and its marginal quantities:

$$\begin{aligned} m(h_0) &= \mathrm{MC}_P(h_0) = m_0 > 0, \qquad & S_0 &= s(h_0), \\ \lambda_0 &= \frac{\mathrm{MC}_P'(h_0)}{\mathrm{MC}_P'(h_0) - m'(h_0)} \in (0,1), \quad & \Gamma_0 &= -\left. \frac{\partial \log \mathcal{A}}{\partial S_q^T} \right|_{S_q^T = S_0} > 0. \end{aligned} \tag{C.5}$$

The derivative with respect to h of the first equation in (C.3) is $m'(h) - \mathrm{MC}_P'(h - h_A) < 0$. The implicit function theorem therefore determines h as a differentiable function of h_A near $(h_0, 0)$. Differentiating that equation gives

$$m'(h) \frac{dh}{dh_A} = \mathrm{MC}_P'(h_P) \left(\frac{dh}{dh_A} - 1 \right).$$

Solving for the derivative, and writing λ for this derivative within the proof, yields

$$\lambda \equiv \frac{dh}{dh_A} = \frac{\mathrm{MC}_P'(h_P)}{\mathrm{MC}_P'(h_P) - m'(h)} \in (0,1), \qquad \frac{dh_P}{dh_A} = \lambda - 1 < 0. \tag{C.6}$$

Differentiating $\alpha = h_A / h$ then gives

$$\frac{d\alpha}{dh_A} = \frac{1 - \alpha\lambda}{h} > 0. \tag{C.7}$$

Consequently, as $h_A \to 0$ along this branch,

$$\begin{aligned} h - h_0 &= \lambda_0 h_A + o(h_A), \qquad & h_P - h_0 &= -(1 - \lambda_0) h_A + o(h_A), \\ \alpha &= \frac{h_A}{h_0} + o(h_A), \qquad & m(h) &\to m_0, \qquad \lambda \to \lambda_0. \end{aligned} \tag{C.8}$$

In particular, positive AI information raises h above h_0 on the branch, so $m(h) < m_0$.

These conditions also locate the AI-adoption boundary. At $\pi_A = 0$ and any $\tau_b > 0$, the

marginal conversion from raw to effective precision is one. The marginal gross return to entering AI research is therefore m_0, while the marginal cost is $c_A f'_+(0)$. Positive AI acquisition requires $c_A f'_+(0) < m_0$. To see why, convexity and (C.2) imply $\mathrm{MC}_A \geq c_A f'_+(0)$, whereas the private-information response derived above gives $m(h) < m_0$ when $h_A > 0$.

For later use, differentiate (C.2) at fixed c_A and τ_b:

$$\frac{\partial \mathrm{MC}_A}{\partial h_A} = c_A f''(\pi_A)\left(1 - \frac{h_A}{\tau_b}\right)^{-4} + \frac{2c_A f'(\pi_A)}{\tau_b}\left(1 - \frac{h_A}{\tau_b}\right)^{-3} > 0. \tag{C.9}$$

The first term is nonnegative and the second is strictly positive. Hence, along the private-information equation, $\mathrm{MC}_A - m(h)$ is strictly increasing in h_A, with derivative $\partial \mathrm{MC}_A / \partial h_A - m'(h)\lambda > 0$. Any root on this branch is unique.

First fix $\tau_b > 0$ and suppose $f'_+(0) = 0$. At an interior equilibrium, (C.2) implies

$$0 < f'(\pi_A) \leq \frac{m(h)}{c_A} \leq \frac{m_0}{c_A}.$$

As $c_A \to \infty$, the right-hand side vanishes. Since f' is nondecreasing and $f'(\pi) > 0$ for every $\pi > 0$, this forces $\pi_A \to 0$. It follows from (C.1) that $h_A \to 0$ and $h_A/\tau_b \to 0$. To verify that the branch exists near this boundary, fix any sufficiently small positive h_A in the neighborhood where the private-information condition determines h. At that value, $\mathrm{MC}_A - m(h) \to \infty$ as $c_A \to \infty$, while its value at $h_A = 0$ is $-m_0 < 0$. Continuity and strict monotonicity give a unique root between these two values for sufficiently large c_A.

Next suppose $f'_+(0) > 0$, keeping τ_b fixed. At an interior equilibrium,

$$m_0 \geq m(h) \geq c_A f'_+(0)\left(1 - \frac{h_A}{\tau_b}\right)^{-2}.$$

Rearranging gives the bound

$$0 < \frac{h_A}{\tau_b} \leq 1 - \sqrt{\frac{c_A f'_+(0)}{m_0}}. \tag{C.10}$$

The upper bound vanishes as $c_A \uparrow m_0/f'_+(0)$, so $h_A \to 0$ and $h_A/\tau_b \to 0$ at this finite boundary. Existence just below the threshold again follows by continuity: $\mathrm{MC}_A - m(h) < 0$ at zero, while at the threshold itself it is strictly positive at any sufficiently small positive h_A. Its derivative is strictly positive, so the nearby root is unique.

If $c_A f'_+(0) \geq m_0$, the no-AI choice satisfies the entry condition with the opposite weak inequality and there is no positive root on the branch. This is the no-adoption region described after the proposition; the proposition itself restricts attention to the strict interior side.

We now hold model accuracy fixed and vary the acquisition cost. At fixed h_A and τ_b, lin-

earity in c_A gives $\partial \mathrm{MC}_A / \partial c_A = \mathrm{MC}_A / c_A = m(h)/c_A$ at equilibrium. Differentiating the AI equation in (C.3), using $dh = \lambda\, dh_A$, gives

$$m'(h)\lambda \frac{dh_A}{dc_A} = \frac{\partial \mathrm{MC}_A}{\partial h_A}\frac{dh_A}{dc_A} + \frac{m(h)}{c_A}.$$

Solving,

$$\frac{dh_A}{dc_A} = -\frac{m(h)/c_A}{\partial \mathrm{MC}_A / \partial h_A - m'(h)\lambda} < 0. \tag{C.11}$$

Together with (C.6)–(C.7), this also gives $dh/dc_A < 0$, $d\alpha/dc_A < 0$, and $dh_P/dc_A > 0$.

The true price-error variance is

$$S_q^T = s(h) + \frac{\alpha^2}{\tau_b} = s(h) + \frac{h_A^2}{\tau_b h^2}. \tag{C.12}$$

Its exact derivative at fixed τ_b is therefore

$$\frac{dS_q^T}{dc_A} = \left[s'(h)\lambda + \frac{2\alpha(1-\alpha\lambda)}{\tau_b h} \right] \frac{dh_A}{dc_A}. \tag{C.13}$$

At either of the cost boundaries identified above, the bracket tends to $s'(h_0)\lambda_0 < 0$. Since $dh_A/dc_A < 0$ at each interior point, $dS_q^T/dc_A > 0$ sufficiently close to the boundary.

Holding the production parameters fixed, the aggregate TFP formula implies the exact differential

$$d\log \mathcal{A} = -\Gamma_0 \left(\frac{1+\tau_d S_0}{1+\tau_d S_q^T} \right)^2 dS_q^T. \tag{C.14}$$

The multiplying factor is strictly positive, proving (6.4). For comparison, the level expansion in this case is

$$S_q^T - S_0 = s'(h_0)\lambda_0 h_A + o(h_A), \qquad \frac{\alpha^2}{\tau_b} = O(h_A^2).$$

The marginal conclusion follows from (C.11)–(C.13), not from differentiating this remainder.

For the accuracy comparison, we instead hold the acquisition cost fixed. Fix $c_A > 0$ satisfying $c_A f'_+(0) < m_0$ and let $\tau_b \downarrow 0$. Since $0 < h_A < \tau_b$, we have $h_A \to 0$. The private-information condition then gives $h \to h_0$ and $m(h) \to m_0$. The local interior branch exists for sufficiently small τ_b: $\mathrm{MC}_A - m(h)$ is negative at zero, strictly increasing in h_A, and diverges to infinity as $h_A \uparrow \tau_b$. For small τ_b, the entire interval $0 \le h_A < \tau_b$ lies in the private-information neighborhood established by the implicit function theorem above.

We next show that $\pi_A \to 0$. Otherwise, there would be a sequence with $\pi_A \ge \epsilon$ for some

fixed $\epsilon > 0$. Along that sequence, (C.4) would imply

$$0 < c_A f'(\epsilon) \leq c_A f'(\pi_A) \leq m_0 \left(\frac{\tau_b}{\tau_b + \epsilon} \right)^2 \longrightarrow 0,$$

a contradiction. Continuity of the right derivative therefore gives $f'(\pi_A) \to f'_+(0)$. Taking limits in (6.6), with $0 < 1 - h_A/\tau_b < 1$, proves (6.7).

For the remaining proof only, abbreviate this limiting fraction by

$$\omega_0 \equiv 1 - \sqrt{\frac{c_A f'_+(0)}{m_0}} \in (0, 1]. \tag{C.15}$$

Thus $h_A = \omega_0 \tau_b + o(\tau_b)$, and (C.8) gives

$$h - h_0 = \lambda_0 \omega_0 \tau_b + o(\tau_b), \quad h_P - h_0 = -(1 - \lambda_0)\omega_0 \tau_b + o(\tau_b), \quad \alpha = \frac{\omega_0}{h_0} \tau_b + o(\tau_b). \tag{C.16}$$

If $f'_+(0) > 0$, then $0 < \omega_0 < 1$ and

$$\frac{\pi_A}{\tau_b} = \frac{h_A/\tau_b}{1 - h_A/\tau_b} \longrightarrow \frac{\omega_0}{1 - \omega_0}. \tag{C.17}$$

If $f'_+(0) = 0$, then $\omega_0 = 1$ and the same identity gives $\pi_A/\tau_b \to \infty$, even though $\pi_A \to 0$.

Substitution into (C.12) yields

$$S_q^T - S_0 = \left[s'(h_0)\lambda_0 \omega_0 + \frac{\omega_0^2}{h_0^2} \right] \tau_b + o(\tau_b). \tag{C.18}$$

This establishes the right derivative of the continuous extension at $\tau_b = 0$. We now establish the limit of the derivatives at positive τ_b directly from the nonlinear system.

To obtain the marginal accuracy effect directly, differentiate the nonlinear equilibrium system. When h_A is held fixed, the derivatives needed in (C.2) are

$$\frac{\partial \pi_A}{\partial \tau_b} = -\frac{h_A^2}{(\tau_b - h_A)^2}, \qquad \frac{\partial}{\partial \tau_b} \left(1 - \frac{h_A}{\tau_b} \right) = \frac{h_A}{\tau_b^2}.$$

Applying the product and chain rules gives

$$-\frac{\partial \mathrm{MC}_A}{\partial \tau_b} = c_A f''(\pi_A) \left(1 - \frac{h_A}{\tau_b} \right)^{-4} \left(\frac{h_A}{\tau_b} \right)^2 + \frac{2 c_A f'(\pi_A)}{\tau_b} \left(1 - \frac{h_A}{\tau_b} \right)^{-3} \frac{h_A}{\tau_b} > 0. \tag{C.19}$$

Now differentiate the equilibrium AI equation at fixed c_A:

$$m'(h)\lambda\frac{dh_A}{d\tau_b} = \frac{\partial \mathrm{MC}_A}{\partial h_A}\frac{dh_A}{d\tau_b} + \frac{\partial \mathrm{MC}_A}{\partial \tau_b}.$$

Hence the exact derivative is

$$\frac{dh_A}{d\tau_b} = \frac{-\partial \mathrm{MC}_A/\partial \tau_b}{\partial \mathrm{MC}_A/\partial h_A - m'(h)\lambda} > 0. \tag{C.20}$$

Divide the numerator and denominator by $c_A f'(\pi_A)/[\tau_b(1-h_A/\tau_b)^3]$.

Using $m(h) = c_A f'(\pi_A)/(1-h_A/\tau_b)^2$ and the identity $\tau_b/(1-h_A/\tau_b) = \tau_b + \pi_A$ gives the equivalent formula

$$\frac{dh_A}{d\tau_b} = \frac{h_A}{\tau_b}\frac{2 + \pi_A f''(\pi_A)/f'(\pi_A)}{2 + (\tau_b+\pi_A)f''(\pi_A)/f'(\pi_A) - \dfrac{m'(h)}{m(h)}\lambda\tau_b\left(1-\dfrac{h_A}{\tau_b}\right)}. \tag{C.21}$$

The derivative limit depends on whether the initial marginal acquisition cost is positive or zero. First suppose $f'_+(0) > 0$. The acquisition limits derived above imply $f'(\pi_A) \to f'_+(0) > 0$ and $\tau_b/\pi_A \to (1-\omega_0)/\omega_0 < \infty$. The additional curvature condition gives

$$\frac{\pi_A f''(\pi_A)}{f'(\pi_A)} \to 0, \qquad \frac{(\tau_b+\pi_A)f''(\pi_A)}{f'(\pi_A)} = \left(1+\frac{\tau_b}{\pi_A}\right)\frac{\pi_A f''(\pi_A)}{f'(\pi_A)} \to 0.$$

Also, $m'(h)/m(h)$ and λ remain bounded, while $\tau_b(1-h_A/\tau_b) \to 0$. Every term in (C.21) now has a limit, yielding $dh_A/d\tau_b \to \omega_0$.

Next suppose $f'_+(0) = 0$. We cannot assume that $\pi_A f''(\pi_A)/f'(\pi_A)$ has a finite limit. Instead, compare the two nonnegative terms in (C.9) with their counterparts in (C.19). Because $0 < h_A/\tau_b < 1$, these formulas imply

$$\left(\frac{h_A}{\tau_b}\right)^2 \le \frac{-\partial \mathrm{MC}_A/\partial \tau_b}{\partial \mathrm{MC}_A/\partial h_A} \le \frac{h_A}{\tau_b}. \tag{C.22}$$

Both bounds tend to one. Moreover, the second term in (C.9), together with the equilibrium condition, gives

$$\frac{\partial \mathrm{MC}_A}{\partial h_A} \ge \frac{2m(h)}{\tau_b(1-h_A/\tau_b)} \longrightarrow \infty.$$

Since $m'(h)\lambda$ is bounded,

$$\frac{\partial \mathrm{MC}_A/\partial h_A}{\partial \mathrm{MC}_A/\partial h_A - m'(h)\lambda} \longrightarrow 1.$$

Multiplying this limit by the middle ratio in (C.22) and using (C.20) yields $dh_A/d\tau_b \to 1 = \omega_0$.

Thus in both cases the exact nonlinear system gives

$$\begin{aligned} \frac{dh_A}{d\tau_b} &\to \omega_0, & \frac{dh}{d\tau_b} &\to \lambda_0\omega_0, \\ \frac{dh_P}{d\tau_b} &\to -(1-\lambda_0)\omega_0, & \frac{d\alpha}{d\tau_b} &\to \frac{\omega_0}{h_0}. \end{aligned} \tag{C.23}$$

The last three limits use (C.6)–(C.7). They are not inferred by differentiating the remainders in (C.16).

These information responses determine the change in price informativeness and TFP. Differentiating (C.12) at fixed c_A gives

$$\frac{dS_q^T}{d\tau_b} = s'(h)\frac{dh}{d\tau_b} + \frac{2\alpha}{\tau_b}\frac{d\alpha}{d\tau_b} - \frac{\alpha^2}{\tau_b^2}. \tag{C.24}$$

The three terms represent the change in noise-trader contamination, the change in the common-error loading, and the direct reduction in model-error variance, respectively. Using the acquisition limits and (C.23), their limits are

$$s'(h_0)\lambda_0\omega_0, \qquad \frac{2\omega_0^2}{h_0^2}, \qquad -\frac{\omega_0^2}{h_0^2}.$$

Consequently,

$$\lim_{\tau_b\downarrow 0}\frac{dS_q^T}{d\tau_b} = s'(h_0)\lambda_0\omega_0 + \frac{\omega_0^2}{h_0^2}. \tag{C.25}$$

This agrees with the right derivative obtained from (C.18). Since $S_q^T \to S_0$, the factor multiplying dS_q^T in (C.14) tends to $-\Gamma_0$. It follows that

$$\left.\frac{d\log\mathcal{A}}{d\tau_b}\right|_{0^+} = \lim_{\tau_b\downarrow 0}\frac{d\log\mathcal{A}}{d\tau_b} = \Gamma_0\omega_0\left[[-s'(h_0)]\lambda_0 - \frac{\omega_0}{h_0^2}\right]. \tag{C.26}$$

Substituting (C.15) proves (6.5). Both Γ_0 and ω_0 are positive, so the expression is negative precisely under (6.8), and positive under the reverse strict inequality. The existence of this derivative limit gives the same strict sign at every sufficiently small positive τ_b. If the bracket is zero, these first-order results do not determine the sign.

C.2 Cost Derivatives and Equilibrium Responses

We begin by expressing the individual optimality conditions in the two information coordinates. We suppress c_P, which is held fixed, from the cost notation. For the proof, distinguish

the two representations of the same acquisition cost by writing

$$\mathcal{C}^I(a,p;c_A,t)=\mathcal{C}\left(\frac{1}{2}\log\left(1+\frac{a+p}{\tau_d}\right),\frac{a}{a+p};c_A,c_P,t\right),\quad a=h_A,\quad p=h_P,\quad t=\tau_b. \tag{C.27}$$

The inverse transformation is

$$a=\alpha H(\mathcal{K}),\qquad p=(1-\alpha)H(\mathcal{K}),\qquad H'(\mathcal{K})=2(\tau_d+H(\mathcal{K})). \tag{C.28}$$

Thus the $\mathcal{C}_{h_Ah_P}$ used in the main text is $\mathcal{C}^I_{ap}$; it is distinct from $\mathcal{C}_{\mathcal{K}\alpha}$.

In a unilateral information deviation, manager i holds the aggregate price rule fixed. Under $\kappa=0$, write gross information value as $\mathcal{V}_i(h_i)$. The objective is $\mathcal{V}_i(H(\mathcal{K}_i))-\mathcal{C}(\mathcal{K}_i,\alpha_i;c_A,c_P,t)$. At a symmetric equilibrium $\mathcal{V}_i'(h)=m(h)$, so differentiation with respect to quantity and composition gives

$$H'(\mathcal{K})m(h)-\mathcal{C}_{\mathcal{K}}=0,\qquad \mathcal{C}_\alpha=0. \tag{C.29}$$

A change in composition at fixed $\mathcal{K}_i$ leaves gross value unchanged. These are the main-text information conditions.

Applying the chain rule to (C.27) also gives

$$\begin{aligned}\mathcal{C}^I_a&=\frac{\mathcal{C}_{\mathcal{K}}}{H'(\mathcal{K})}+\frac{1-\alpha}{h}\mathcal{C}_\alpha,\\ \mathcal{C}^I_p&=\frac{\mathcal{C}_{\mathcal{K}}}{H'(\mathcal{K})}-\frac{\alpha}{h}\mathcal{C}_\alpha.\end{aligned} \tag{C.30}$$

Hence the equilibrium conditions are equivalently

$$\mathcal{C}^I_a(a,p;c_A,t)=\mathcal{C}^I_p(a,p;c_A,t)=m(a+p),\qquad \mathcal{C}_\alpha=h(\mathcal{C}^I_a-\mathcal{C}^I_p). \tag{C.31}$$

The no-AI cost identity implies $\mathcal{C}^I(0,p;c_A,t)=G(p)$ and $\mathcal{C}^I_p(0,p;c_A,t)=G'(p)$.

We next differentiate the conditions across symmetric equilibria. Across symmetric equilibria, the marginal value changes with aggregate information, including its effect on prices. Its derivative is $m'(h)$; it need not equal the curvature of an individual's value at fixed prices. Let

$$B(\mathcal{K})=H'(\mathcal{K})m(H(\mathcal{K})),\qquad B'(\mathcal{K})=H''(\mathcal{K})m(h)+[H'(\mathcal{K})]^2m'(h). \tag{C.32}$$

At $\kappa=0$, B has no direct dependence on c_A or t. For either parameter ϑ, total differentiation of $\mathcal{C}_{\mathcal{K}}-B=0$ and $\mathcal{C}_\alpha=0$ yields

$$\begin{aligned}[\mathcal{C}_{\mathcal{K}\mathcal{K}}-B']\mathcal{K}_\vartheta+\mathcal{C}_{\mathcal{K}\alpha}\alpha_\vartheta&=-\mathcal{C}_{\mathcal{K}\vartheta},\\ \mathcal{C}_{\mathcal{K}\alpha}\mathcal{K}_\vartheta+\mathcal{C}_{\alpha\alpha}\alpha_\vartheta&=-\mathcal{C}_{\alpha\vartheta}.\end{aligned} \tag{C.33}$$

The coefficient matrix is

$$J = \begin{pmatrix} \mathcal{C}_{\mathcal{K}\mathcal{K}} - B' & \mathcal{C}_{\mathcal{K}\alpha} \\ \mathcal{C}_{\mathcal{K}\alpha} & \mathcal{C}_{\alpha\alpha} \end{pmatrix}. \tag{C.34}$$

The main-text regularity conditions are equivalent to positive definiteness of J, which implies $\mathcal{C}_{\alpha\alpha} > 0$, $\det J > 0$, and

$$\mathcal{C}_{\mathcal{K}\mathcal{K}} - B' - \frac{\mathcal{C}_{\mathcal{K}\alpha}^2}{\mathcal{C}_{\alpha\alpha}} > 0. \tag{C.35}$$

Solving the second line of (C.33) for α_ϑ gives (6.12). Substitution into the first line gives the quantity formula in the main-text footnote. Equivalently,

$$\begin{pmatrix} \mathcal{K}_\vartheta \\ \alpha_\vartheta \end{pmatrix} = -J^{-1} \begin{pmatrix} \mathcal{C}_{\mathcal{K}\vartheta} \\ \mathcal{C}_{\alpha\vartheta} \end{pmatrix}. \tag{C.36}$$

For changes in AI acquisition costs, the differentiated system determines the relative responses of information quantity and composition. Write $\Delta = \det J > 0$. Expanding the inverse in (C.36), and using the definitions of U_c, V_c in the main text, gives

$$\mathcal{K}_{c_A} = -\frac{U_c}{\Delta}, \qquad \alpha_{c_A} = -\frac{V_c}{\Delta}, \qquad \frac{h_{c_A}}{\alpha_{c_A}} = \frac{H'(\mathcal{K})U_c}{V_c}. \tag{C.37}$$

The last equality holds whenever $V_c \neq 0$. Under $V_c > 0$ and (6.13), $\alpha_{c_A} < 0$ and the last ratio has a strictly positive lower limit. Thus $h_{c_A} < 0$ sufficiently close to the boundary. The cost conditions are restrictions on derivatives evaluated along the local branch; positive definiteness of J alone does not imply these signs. They do not require the entry cost to be infinite.

For changes in model accuracy, the same system also captures the direct improvement in the AI signal. The same inverse gives

$$\begin{aligned} \mathcal{K}_t &= \frac{\mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\alpha t} - \mathcal{C}_{\alpha\alpha}\mathcal{C}_{\mathcal{K}t}}{\Delta}, \\ \alpha_t &= \frac{\mathcal{C}_{\mathcal{K}\alpha}\mathcal{C}_{\mathcal{K}t} - [\mathcal{C}_{\mathcal{K}\mathcal{K}} - B']\mathcal{C}_{\alpha t}}{\Delta}. \end{aligned} \tag{C.38}$$

These are interior derivatives and do not by themselves establish their limits as $t \downarrow 0$.

Differentiating $S_q^T = s(H(\mathcal{K})) + \alpha^2/t$ and combining the result with (C.36) gives an exact expression in cost derivatives at every regular interior equilibrium:

$$\begin{aligned} \frac{d\log\mathcal{A}}{d\vartheta} = \left(-\frac{\partial\log\mathcal{A}}{\partial S_q^T}\right) \Bigg[\begin{pmatrix} s'(h)H'(\mathcal{K}) & 2\alpha/t \end{pmatrix} J^{-1} \begin{pmatrix} \mathcal{C}_{\mathcal{K}\vartheta} \\ \mathcal{C}_{\alpha\vartheta} \end{pmatrix} \\ + \mathbf{1}_{\{\vartheta=t\}} \frac{\alpha^2}{t^2} \Bigg]. \end{aligned} \tag{C.39}$$

The prefactor is positive. The final term is the direct variance reduction from greater accuracy, holding information choices fixed. For cost changes, the expression simplifies to

$$\frac{d\log\mathcal{A}}{dc_A} = \frac{-\partial\log\mathcal{A}/\partial S_q^T}{\Delta}\left[s'(h)H'(\mathcal{K})U_c + \frac{2\alpha}{t}V_c\right]. \tag{C.40}$$

Thus the exact condition for a cost reduction to raise TFP is $[-s'(h)]H'(\mathcal{K})U_c > (2\alpha/t)V_c$.

C.3 Proof of Lemma 4

We first establish the limiting fraction of available AI precision acquired. Let $h(t)$ and $\alpha(t)$ denote the maintained equilibrium branch. Feasibility gives $0 < a(t)/t < 1$. Fix any $q \in (0, q_0)$. Since $h(t) \to h_0$, the joint-limit condition (6.14) implies

$$\mathcal{C}_\alpha\left(\mathcal{K}(t), \frac{qt}{h(t)}; c_A, c_P, t\right) < 0$$

for all sufficiently small t. At the same total information, the equilibrium satisfies

$$\mathcal{C}_\alpha(\mathcal{K}(t), \alpha(t); c_A, c_P, t) = 0.$$

Strict convexity in composition therefore gives $\alpha(t) > qt/h(t)$, or $a(t)/t > q$. The comparison configuration is feasible for small t: its AI precision is below t, and its private precision converges to $h_0 \in (0, \tau_u)$.

If $q_0 < 1$, fix any $q \in (q_0, 1)$. The positive sign in (6.14) similarly implies $\alpha(t) < qt/h(t)$, so $a(t)/t < q$ for small t. Taking fixed comparison fractions arbitrarily close to q_0 on each side proves $a(t)/t \to q_0$. If $q_0 = 1$, the lower comparison holds for every $q < 1$, while $a(t)/t < 1$ by feasibility. It again follows that

$$\frac{a(t)}{t} \longrightarrow q_0, \qquad a(t) = q_0 t + o(t), \qquad \frac{\alpha(t)}{t} \longrightarrow \frac{q_0}{h_0}. \tag{C.41}$$

This argument uses cost signs at fixed comparison fractions. It does not assume a limit for the equilibrium ratio.

To determine how private research adjusts, expand its marginal acquisition cost. Put $p(t) = h(t) - a(t)$. Then $p(t) \to h_0$. At each $t > 0$, continuity of $\mathcal{C}_p^I$ at $a = 0$ and the mean value theorem imply that, for some $\widetilde{a}(t) \in (0, a(t))$,

$$\begin{aligned}\mathcal{C}_p^I(a(t), p(t); c_A, t) - G'(p(t)) &= \mathcal{C}_{ap}^I(\widetilde{a}(t), p(t); c_A, t)\, a(t) \\ &= \big[\gamma_0 + o(1)\big] a(t).\end{aligned} \tag{C.42}$$

The last equality follows from (6.15), since $0 < \widetilde{a}(t) < a(t) < t$ and $p(t) \to h_0$. The joint limit controls the entire interval used by the mean value theorem; convergence of the cross partial only at the equilibrium endpoint would not suffice.

This expansion determines the order of the change in total information. By the private-information condition in (C.31),

$$m(h(t)) = G'(h(t) - a(t)) + [\gamma_0 + o(1)]a(t). \tag{C.43}$$

Because G'' is continuous near h_0, the mean value theorem gives $G'(h - a) - G'(h) = -[G''(h_0) + o(1)]a$. Also $m(h_0) - G'(h_0) = 0$, so the same theorem applied to $m - G'$ gives

$$[m'(h_0) - G''(h_0) + o(1)](h - h_0) = [\gamma_0 - G''(h_0) + o(1)]a. \tag{C.44}$$

The coefficient on the left is bounded away from zero for small t. Dividing by this coefficient proves, rather than assumes, that $h - h_0 = O(a)$ and

$$\frac{h - h_0}{a} \longrightarrow \frac{G''(h_0) - \gamma_0}{G''(h_0) - m'(h_0)}. \tag{C.45}$$

This conclusion also holds when the limiting numerator is zero.

Combining the utilization limit with the information response gives the boundary expansion. Multiplying (C.45) by $a/t \to q_0$ yields

$$\lim_{t \downarrow 0} \frac{h - h_0}{t} = q_0 \frac{G''(h_0) - \gamma_0}{G''(h_0) - m'(h_0)} = L_0. \tag{C.46}$$

Thus $h = h_0 + L_0 t + o(t)$, proving the lemma. For completeness, $p = h_0 + (L_0 - q_0)t + o(t)$ and $\alpha = (q_0/h_0)t + o(t)$. These are boundary expansions. Their proof does not differentiate an $o(t)$ remainder or assert convergence of the interior derivatives.

C.4 Proof of Proposition 7

Consider first a cost reduction near the no-AI boundary. Fix $t > 0$. By $V_c > 0$, (C.37), and (6.13), $\alpha_{c_A} < 0$ and h_{c_A}/α_{c_A} has a strictly positive lower limit. Hence the cost branch can locally be parameterized by α. Along it,

$$\frac{dS_q^T}{d\alpha} = s'(h)\frac{dh}{d\alpha} + \frac{2\alpha}{t}, \qquad \frac{dh}{d\alpha} = \frac{h_{c_A}}{\alpha_{c_A}} = \frac{H'(\mathcal{K})U_c}{V_c}. \tag{C.47}$$

There is a constant $R > 0$ such that $dh/d\alpha \geq R$ sufficiently close to the boundary. Continuity and $s'(h_0) < 0$ give a constant $\varepsilon > 0$ such that $s'(h) \leq -\varepsilon$ there. Consequently,

$$\frac{dS_q^T}{d\alpha} \leq -\varepsilon R + \frac{2\alpha}{t} < 0$$

for sufficiently small α. Multiplication by $\alpha_{c_A} < 0$ gives $dS_q^T/dc_A > 0$. Since $\partial \log \mathcal{A}/\partial S_q^T < 0$, it follows that $d \log \mathcal{A}/dc_A < 0$. This argument concerns the interior side of the boundary and is unchanged if the entry threshold is finite. If an interval of costs instead has zero AI acquisition, the maintained no-AI cost identity makes information and TFP independent of c_A on that interval.

For the accuracy comparison, the lemma determines the order of both components of true price error. Fix $c_A > 0$ and let $t \downarrow 0$. The lemma gives $a = q_0 t + o(t)$ and $h = h_0 + L_0 t + o(t)$. Because $h_0 > 0$,

$$\alpha = \frac{a}{h} = \frac{q_0}{h_0} t + o(t), \qquad \frac{\alpha^2}{t} = \frac{q_0^2}{h_0^2} t + o(t). \tag{C.48}$$

Differentiability of s gives $s(h) - s(h_0) = s'(h_0) L_0 t + o(t)$. Adding the two components yields

$$S_q^T - s(h_0) = \left[s'(h_0) L_0 + \frac{q_0^2}{h_0^2} \right] t + o(t). \tag{C.49}$$

This is the true price-error variance, including the full common model error despite managers' correlation neglect.

The aggregate TFP formula translates this price-error expansion into the productivity effect. Let $\mathcal{A}_0$ denote TFP in the no-AI economy. The aggregate TFP mapping is differentiable at $S_q^T = s(h_0)$, where $\partial \log \mathcal{A}/\partial S_q^T = -\Gamma_0 < 0$. Since (C.49) is $O(t)$,

$$\log \mathcal{A} - \log \mathcal{A}_0 = -\Gamma_0 \left[s'(h_0) L_0 + \frac{q_0^2}{h_0^2} \right] t + o(t). \tag{C.50}$$

Dividing by t and taking the limit proves (6.18). The derivative at 0^+ in the proposition is this right boundary difference quotient. Substituting the lemma's formula for L_0 into (6.18) determines the sign of the boundary effect. It is negative when the common-error term exceeds the information gain, positive under the reverse strict inequality, and zero at equality.

Recovery of the separable-cost results. For the cost in Proposition 6, write

$$\mathcal{C}^I(a, p; c_A, t) = c_A f\left(\frac{ta}{t-a} \right) + G(p), \qquad x_A = \frac{ta}{t-a}. \tag{C.51}$$

Its cross derivative is identically zero, so (6.15) holds with $\gamma_0 = 0$. For every fixed $q \in (0,1)$, set $a = qt$ and $p = h - qt$. Then $x_A = tq/(1-q) \to 0$, and

$$\frac{\mathcal{C}_\alpha}{h} = \frac{c_A f'(tq/(1-q))}{(1-q)^2} - G'(h-qt) \longrightarrow \frac{c_A f'_+(0)}{(1-q)^2} - m_0 \tag{C.52}$$

as $t \downarrow 0$ and $h \to h_0$. If $f'_+(0) > 0$ and $c_A f'_+(0) < m_0$, the limiting expression has the unique zero $q_0 = 1 - \sqrt{c_A f'_+(0)/m_0} \in (0,1)$, with the signs required by (6.14). If $f'_+(0) = 0$, it equals $-m_0 < 0$ at every $q < 1$, giving $q_0 = 1$. The lemma consequently implies

$$L_0 = \frac{G''(h_0)}{G''(h_0) - m'(h_0)} q_0 = \lambda_0 q_0. \tag{C.53}$$

This recovers the generalized initial-marginal-cost result, including the special case $q_0 = 1$.

To verify inclusion of the cost path as well, let $F(a;t) = f(ta/(t-a))$. The equilibrium residuals in effective coordinates are $c_A F_a - m(a+p)$ and $G'(p) - m(a+p)$. Their Jacobian is

$$M = \begin{pmatrix} c_A F_{aa} - m' & -m' \\ -m' & G'' - m' \end{pmatrix}.$$

Near the no-AI equilibrium of Proposition 6, $F_{aa} > 0$, $G'' > 0$, and $m' < 0$. Thus, for every nonzero vector v, $v'Mv = c_A F_{aa} v_a^2 + G'' v_p^2 - m'(v_a + v_p)^2 > 0$. The coordinate Jacobian

$$T = \frac{\partial(a,p)}{\partial(\mathcal{K},\alpha)} = \begin{pmatrix} \alpha H' & H \\ (1-\alpha)H' & -H \end{pmatrix}$$

has determinant $-HH' \neq 0$. At a zero of the residuals the response matrix satisfies $J = T'MT$, and is therefore positive definite.

Differentiating the private condition $m(h) = G'(h-a)$ along the branch gives

$$\lambda = \frac{dh}{da} = \frac{G''(p)}{G''(p) - m'(h)} \in (0,1), \qquad \frac{da}{dc_A} = -\frac{F_a}{c_A F_{aa} - m'(h)\lambda} < 0.$$

Since $d\alpha/da = (1-\alpha\lambda)/h > 0$, $\alpha_{c_A} < 0$ and

$$\frac{h_{c_A}}{\alpha_{c_A}} = \frac{h\lambda}{1-\alpha\lambda} \longrightarrow h_0\lambda_0 > 0.$$

Equation (C.37) then verifies both restrictions, $V_c > 0$ and (6.13). When $f'_+(0) > 0$, this verification applies as $c_A \uparrow m_0/f'_+(0)$ from the active side. When $f'_+(0) = 0$, it applies as $c_A \to \infty$.

D Additional Empirical Results

This appendix reports descriptive statistics, common selling and downside outcomes, and robustness checks for the portfolio-convergence results. The exercises address persistence in portfolio similarity, the construction of the AI-reliance measure, the definition of portfolio adjustment, the threshold used to define joint GPT alignment, manager-specific time-varying shocks, and residual dependence across manager pairs.

Appendix Table D.1 reports descriptive statistics for the variables used in the main regressions. The AI-reliance row is drawn from the manager-quarter sample rather than the manager-pair panel, so each estimated manager-quarter value enters once.

Table D.1: Summary Statistics for Main Regression Variables

Variable	N	Mean	SD	Median
Panel A: AI reliance and portfolio similarity				
AI reliance	7,581	0.0691	0.0917	0.0360
Common AI reliance	890,394	0.0043	0.0110	0.0014
Average AI reliance	890,394	0.0653	0.0618	0.0406
Joint GPT alignment	890,394	0.0146	0.1199	0.0000
Portfolio overlap	890,394	0.3044	0.1741	0.2848
Cosine similarity	890,394	0.4204	0.2553	0.3940
Panel B: Validation variables				
Pair-weighted earnings-call length	859,357	0.0713	0.1784	0.0819
Pair-weighted analyst coverage	859,357	0.9911	0.3608	0.9928
Panel C: Similarity and pair-level outcomes				
Portfolio overlap	1,233,217	0.2456	0.1827	0.2105
Cosine similarity	1,233,217	0.3437	0.2641	0.2951
Future common selling, minimum denominator	1,233,217	0.1513	0.1451	0.1129
Future common selling, Jaccard denominator	1,234,800	0.0301	0.0331	0.0201
Next-quarter market downturn	1,233,217	0.2865	0.4521	0.0000
Total downside comovement $\times$10,000	1,234,801	35.5393	106.9504	0.0000
Joint downside indicator	1,234,801	0.2567	0.4368	0.0000
Conditional downside comovement $\times$10,000	298,963	137.5984	173.6902	33.6524
Panel D: Pair-level controls				
Average log portfolio value	1,233,217	24.4325	0.7752	24.3109
Absolute log portfolio value difference	1,233,217	1.1224	0.9923	0.8483
Average number of holdings	1,233,217	1504.3964	972.3907	1370.0000
Absolute difference in number of holdings	1,233,217	1486.3973	1252.5271	1161.0000
Average portfolio HHI	1,233,217	0.0279	0.0518	0.0149
Absolute portfolio HHI difference	1,233,217	0.0342	0.0975	0.0095

Notes: The AI-reliance row uses all valid top-300 manager-quarter estimates from 2018Q1 through 2024Q4. The remaining Panel A variables use the common estimation sample for the eight baseline regressions of AI reliance and portfolio similarity. Each Panel B moderator uses the sample for its corresponding validation regression. Panel C variables use the sample for the corresponding pair-outcome regression. Similarity and Panel D controls use the minimum-denominator common-selling sample. Pair-outcome samples require at least one common stock holding in quarter t. Interaction terms, fixed effects, absorbed same-type indicators, and sample-selection indicators are not reported. Means of binary variables are sample proportions.

D.1 Portfolio Similarity and Pair-Level Outcomes

We next ask whether current portfolio similarity provides an ex ante indicator of common selling and joint downside exposure, and whether this relationship strengthens during market downturns. Prior work shows that portfolio similarity predicts common trading and liquidation behavior (Pool et al., 2015; Girardi et al., 2021). We therefore relate quarter-t similarity to selling and downside outcomes in quarter $t+1$, examining whether the portfolio convergence documented in Section 2 is followed by common adjustments. The analysis is restricted to manager pairs with at least one common stock holding in quarter t, which removes pairs whose current common exposure is mechanically zero while preserving zero realizations of the future outcome variables. Because common holdings create the opportunity for common sales, the regressions map portfolio exposure into subsequent liquidation behavior.

Common selling and downside comovement. We first consider whether higher portfolio similarity predicts more common selling. Our primary outcome follows the standard definition of Pool et al. (2015). Let

$$I^{-}_{i,s,t+1} = \mathbf{1}\{Shares_{i,s,t+1} < Shares_{i,s,t}\}, \tag{D.1}$$

indicate whether manager i reduces its split-adjusted holdings of stock s between quarters t and $t+1$, or whether manager i *sells* asset s in period $t+1$. We define the common-selling fraction as (Pool et al., 2015):

$$Common_Selling_{i,k,t+1} = \frac{\sum_s \min\{I^{-}_{i,s,t+1}, I^{-}_{k,s,t+1}\}}{\min\left\{\sum_s I^{-}_{i,s,t+1}, \sum_s I^{-}_{k,s,t+1}\right\}}. \tag{D.2}$$

The numerator counts stocks sold by both managers, while the denominator is the smaller number of stocks sold by either manager. The measure is defined when both managers sell at least one stock and ranges from zero to one. Because it treats each sold stock as one observation, it captures overlap in the direction of portfolio adjustment without allowing a few large dollar trades to dominate the outcome.

We complement the baseline measure with two value-weighted definitions. For stock s, manager i's selling value between t and $t+1$, $S_{i,s,t+1}$, is the positive decline in split-adjusted shares multiplied by the quarter-$t+1$ stock price. If the share holding does not decline, $S_{i,s,t+1}$ is set to zero. Let $S_{i,t+1} = \sum_s S_{i,s,t+1}$ denote manager i's total selling, and define the pair-level common selling value as:

$$C_{i,k,t+1} = \sum_s \min\{S_{i,s,t+1}, S_{k,s,t+1}\}, \tag{D.3}$$

which measures dollar selling that can be matched stock by stock across the pair. Following Pool et al. (2015) and Girardi et al. (2021), we construct the value-based common-selling overlap

measure as:[16]

$$Common_Selling^{Val}_{i,k,t+1} = \frac{C_{i,k,t+1}}{\min\{S_{i,t+1}, S_{k,t+1}\}}. \tag{D.4}$$

$Common_Selling^{Val}_{i,k,t+1}$ measures the value of stocks that are commonly sold by the two asset managers as a fraction of the smaller aggregate selling within the manager pair. We also consider an alternative Jaccard-type common-selling measure:

$$Common_Selling^{Jac}_{i,k,t+1} = \frac{C_{i,k,t+1}}{\sum_s \max\{S_{i,s,t+1}, S_{k,s,t+1}\}}. \tag{D.5}$$

Its denominator is algebraically equal to $S_{i,t+1} + S_{k,t+1} - C_{i,k,t+1}$, the union of the pair's selling values. It is more conservative because unmatched sales by either manager enlarge the denominator. Together, the three measures distinguish overlap in sold stocks from overlap in the dollar magnitude of liquidation and show whether the results depend on the normalization of selling intensity.

We next examine whether portfolio similarity is associated with joint downside performance. Managers with similar portfolios are exposed to many of the same asset-level shocks, so their returns may experience common downside pressure. We first define the following unconditional measure of pair-level downside comovement:

$$Downside_{i,k,t+1} = \min\{R_{i,t+1}, 0\} \times \min\{R_{k,t+1}, 0\}, \tag{D.6}$$

which equals zero whenever at least one manager has a nonnegative return. We multiply this variable by 10,000 for readability. Because the unconditional downside product has a large mass at zero, its variation reflects both the incidence of negative returns for the two managers and the severity of losses conditional on both returns being negative.[17] We therefore also estimate a two-part model that separates the occurrence of joint negative returns from the severity of losses conditional on that event. The first part is a linear probability model for $\mathbf{1}\{R_{i,t+1} < 0, R_{k,t+1} < 0\}$. The second part uses $Downside_{i,k,t+1}$ conditional on both returns being negative. This decomposition shows whether similarity primarily predicts the probability of entering a joint downside state or the severity of losses within that state.

To investigate the relationship between portfolio similarity and the pair-level outcomes, we estimate the following equation for each outcome $Y_{i,k,t+1}$:

$$Y_{i,k,t+1} = \beta \times Similarity_{i,k,t} + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}. \tag{D.7}$$

[16] Girardi et al. (2021) measures common selling using the inner product of two asset managers' selling vectors. We use the sum of the smaller selling value for each stock and normalize it following Pool et al. (2015).

[17] In our sample, 92.5% of the observations in which both asset managers experience negative returns are concentrated in 7 quarters.

The control vector contains the pair average and absolute difference in log portfolio value, number of holdings, and portfolio HHI. Concentration is included here because it can directly affect portfolio return volatility and liquidation behavior. Pair fixed effects absorb persistent differences in manager matching, mandates, and investment style; quarter fixed effects absorb market-wide conditions common to all pairs. Standard errors are clustered by manager pair.

Table D.2 reports six complementary outcomes. Panels A–C show that current portfolio similarity predicts greater overlap in next-quarter sales under all three definitions. Under the baseline measure, a one-standard-deviation increase in portfolio overlap predicts 3.6 percentage points more overlap in sold stocks, while a one-standard-deviation increase in cosine similarity predicts 1.0 percentage point more. The corresponding estimates are 8.4 and 3.2 percentage points for the value-based normalized common-selling measure and 2.1 and 0.9 percentage points under the weighted-Jaccard normalization. The similar pattern across a stock-name measure and two value-weighted measures indicates that the association is not confined to a few large sales or to the scale of the smaller seller. More similar portfolios are followed by greater overlap in the assets subsequently sold.

Panels D–F show that similarity also predicts joint downside exposure. The unconditional downside coefficient is positive and significant for both overlap and cosine similarity. The two-part estimates locate the source of this association. A one-standard-deviation increase in overlap predicts a 0.62-percentage-point increase in the probability that both managers have negative returns, while a one-standard-deviation increase in cosine similarity predicts an increase of approximately 1.39 percentage points. Conditional on both returns being negative, the overlap coefficient remains positive but is not statistically distinguishable from zero; the cosine coefficient is positive and highly significant. Thus, current portfolio similarity is an ex ante indicator of joint losses. The evidence on conditional loss severity is stronger for cosine similarity than for overlap. These results link portfolio convergence to subsequent joint downside exposure.

Market downturns. Asset managers are more likely to sell during aggregate market downturns (Goldstein et al., 2017). When their portfolios are similar, those sales are more likely to involve the same assets, which can make common selling especially relevant for market fragility (Manconi et al., 2012). We ask whether the predictive relationship between similarity and common selling is stronger in such periods. We define $Market_Down_{t+1}$ as an indicator equal to one when the value-weighted market return is in the bottom 20% of sample quarters, and estimate:

$$\begin{aligned} Common_Selling^{d}_{i,k,t+1} = \beta \times Similarity_{i,k,t} + \lambda \times Similarity_{i,k,t} \times Market_Down_{t+1} \\ + \theta' Z_{i,k,t} + \varphi_{i,k} + \psi_t + \epsilon_{i,k,t}, \end{aligned} \tag{D.8}$$

Table D.2: Portfolio Similarity and Pair-Level Outcomes

Similarity measure	(1) Portfolio overlap	(2) Cosine similarity
Panel A: Future common selling		
Similarity	0.1981***	0.0372***
	(41.53)	(20.84)
Observations	1,233,217	1,233,217
Panel B: Value-based future common selling		
Similarity	0.4573***	0.1225***
	(100.97)	(67.82)
Observations	1,233,217	1,233,217
Panel C: Jaccard-weighted future common selling		
Similarity	0.1126***	0.0345***
	(84.34)	(66.27)
Observations	1,234,800	1,234,800
Panel D: Total future downside comovement		
Similarity	4.1582***	5.5694***
	(4.25)	(11.54)
Observations	1,234,801	1,234,801
Panel E: Both future returns are negative		
Similarity	0.0338***	0.0526***
	(8.02)	(30.48)
Observations	1,234,801	1,234,801
Panel F: Conditional downside comovement		
Similarity	5.8675	14.9124***
	(1.55)	(7.76)
Observations	298,963	298,963
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Notes: Similarity is measured in quarter t and all outcomes in quarter $t+1$. Panel A uses the Pool SaleOverlap measure based on the fraction of sold stock names common to both managers. Panel B scales matched selling value by the smaller of the two managers' total selling values. Panel C uses the union of the pair's selling values. Panel D reports downside comovement in the full outcome sample. Panel E is a linear probability model for both managers having negative returns. Panel F conditions on both returns being negative. Downside comovement in Panels D and F is multiplied by 10,000. Robust t-statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

where $Common_Selling^{d}_{i,k,t+1}$ denotes each of the three common-selling measures defined above, and λ captures the difference in the similarity–common-selling association between normal and downturn quarters. A positive λ means that similarity is a stronger predictor of common selling during market downturns.

Table D.3 reports positive and statistically significant estimates of λ under all three common-selling definitions. For the baseline measure, a one-standard-deviation increase in overlap predicts an additional 0.28 percentage points of common selling during market downturns, and a one-standard-deviation increase in cosine similarity predicts an additional 0.16 percentage points. For the value-based common-selling measure, the corresponding increments are 0.82 and 0.58 percentage points; under the weighted-Jaccard normalization, they are 0.18 and 0.12 percentage points. Portfolio similarity is therefore a stronger predictor of overlapping stock-level sales in market downturns, when common liquidation is more consequential.

D.2 Results without Controlling for Current Similarity

Table D.4 removes current portfolio similarity from the baseline specifications. The coefficients on common AI reliance are positive and statistically significant for both overlap and cosine similarity. They are also larger than the corresponding baseline estimates. This difference is consistent with the strong persistence in pair-level portfolio composition: current similarity absorbs much of the variation that would otherwise be attributed to common reliance. The main specifications retain current similarity because they provide a more demanding test of whether common reliance predicts incremental portfolio convergence in the next quarter.

Table D.3: Portfolio Similarity, Market Downturns, and Common Selling

Similarity measure	(1) Portfolio overlap	(2) Cosine similarity
Panel A: Future common selling		
Similarity	0.1941***	0.0355***
	(40.52)	(19.51)
Similarity × market downturn	0.0153***	0.0062***
	(9.93)	(6.10)
Observations	1,233,217	1,233,217
Panel B: Value-based future common selling		
Similarity	0.4457***	0.1165***
	(97.77)	(63.48)
Similarity × market downturn	0.0450***	0.0221***
	(29.44)	(22.50)
Observations	1,233,217	1,233,217
Panel C: Jaccard-weighted future common selling		
Similarity	0.1100***	0.0333***
	(82.56)	(63.94)
Similarity × market downturn	0.0099***	0.0046***
	(24.54)	(18.34)
Observations	1,234,800	1,234,800
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Notes: The market-downturn indicator equals one for the bottom 20% of quarterly value-weighted market returns and is absorbed by quarter fixed effects. Robust t-statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Future portfolio similarity	(1) Overlap	(2) Cosine
Panel A: Controlling for average AI reliance		
Common AI reliance	0.0833***	0.1339***
	(10.45)	(7.81)
Average AI reliance	-0.0367***	-0.0070*
	(-22.26)	(-1.74)
Panel B: Joint GPT alignment		
Joint GPT alignment	0.0044***	0.0107***
	(13.61)	(12.07)
Average AI reliance	-0.0290***	0.0028
	(-21.60)	(0.86)
Panel C: Controlling for individual AI reliance		
Common AI reliance	0.0838***	0.1343***
	(10.51)	(7.84)
AI reliance, manager i	-0.0225***	-0.0070***
	(-20.16)	(-2.58)
AI reliance, manager k	-0.0148***	-0.0005
	(-15.05)	(-0.22)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair
Observations	890,394	890,394

Table D.4: Common AI Reliance and Future Similarity without Current Similarity

D.3 Alternative Fit-Based Measures of AI Reliance

The baseline reliance measure is the increase in in-sample R^2 when the fourteen GPT scores are added to the public-information regression. Since the full specification nests the baseline specification, this increase is mechanically nonnegative. We therefore consider three alternative measures that treat model fit and the additional regressors differently.

The first measure is the fraction of residual variation from the baseline regression that is removed by the GPT scores:

$$Partial\ R^2_{i,t} = \frac{RSS_{base,i,t} - RSS_{full,i,t}}{RSS_{base,i,t}}. \tag{D.9}$$

This normalization measures the explanatory contribution of the GPT scores relative to the variation left unexplained by the baseline public-information variables. Table D.5 reports positive and statistically significant common-reliance coefficients.

The next two measures penalize the full model for the additional GPT regressors. Let $\bar{R}^2_{base,i,t}$ and $\bar{R}^2_{full,i,t}$ denote adjusted R^2 from the two specifications. We define

$$Adjusted\ R^2\ gain_{i,t} = \max\left\{\bar{R}^2_{full,i,t} - \bar{R}^2_{base,i,t}, 0\right\}, \tag{D.10}$$

$$Adjusted\ partial\ R^2\ gain_{i,t} = \max\left\{\frac{\bar{R}^2_{full,i,t} - \bar{R}^2_{base,i,t}}{1 - \bar{R}^2_{base,i,t}}, 0\right\}. \tag{D.11}$$

The truncation at zero preserves the interpretation of reliance as nonnegative exposure to information extracted by the GPT scores. Unlike the raw R^2 gain, these measures assign a positive value only when the improvement in fit is large enough to offset the degrees-of-freedom penalty.

Table D.6 shows that the estimated coefficients remain positive, but the overlap estimates become less precise. Common adjusted-R^2 gain predicts future overlap at the 10 percent level, while common adjusted partial-R^2 gain has a similar coefficient but falls just outside conventional significance levels. The cosine estimates remain statistically significant under both adjustments. These results indicate that the baseline association is not solely a consequence of the mechanical increase in raw in-sample R^2, although the precision of the overlap result depends on how strongly the additional GPT regressors are penalized.

Future portfolio similarity	(1) Overlap	(2) Cosine
Panel A: Controlling for average partial R^2		
Common partial R^2	0.0171***	0.0390***
	(5.25)	(5.75)
Average partial R^2	-0.0063***	0.0078***
	(-8.73)	(3.97)
Current similarity	0.7957***	0.6760***
	(371.16)	(223.62)
Panel B: Joint GPT alignment		
Joint GPT alignment	0.0019***	0.0104***
	(9.35)	(13.37)
Average partial R^2	-0.0050***	0.0071***
	(-8.45)	(4.46)
Current similarity	0.7957***	0.6760***
	(371.19)	(223.83)
Panel C: Controlling for individual partial R^2		
Common partial R^2	0.0172***	0.0392***
	(5.28)	(5.77)
Partial R^2, manager i	-0.0046***	0.0017
	(-9.78)	(1.35)
Partial R^2, manager k	-0.0018***	0.0058***
	(-4.06)	(4.74)
Current similarity	0.7957***	0.6760***
	(371.16)	(223.62)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair
Observations	890,394	890,394

Table D.5: Partial-R^2 AI Reliance and Future Portfolio Similarity

Future portfolio similarity	(1) Overlap	(2) Cosine
Panel A: AI reliance based on adjusted-R^2 gain		
Common adjusted-R^2 gain	0.0146*	0.0453**
	(1.73)	(2.57)
Average adjusted-R^2 gain	-0.0068***	0.0109***
	(-9.22)	(5.57)
Current similarity	0.7957***	0.6761***
	(371.40)	(223.65)
Observations	890,394	890,394
Panel B: AI reliance based on adjusted partial-R^2 gain		
Common adjusted partial-R^2 gain	0.0130	0.0452***
	(1.64)	(2.71)
Average adjusted partial-R^2 gain	-0.0068***	0.0082***
	(-9.70)	(4.39)
Current similarity	0.7957***	0.6761***
	(371.40)	(223.64)
Observations	890,394	890,394
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.6: Degrees-of-Freedom-Adjusted AI Reliance and Future Portfolio Similarity

D.4 Alternative Portfolio-Adjustment Measures

The baseline manager-level regression uses changes in portfolio weights. Such changes may reflect active trading, but they can also arise mechanically from price movements. Table D.7 reconstructs AI reliance using two alternative dependent variables. Active rebalancing removes passive price drift from portfolio-weight changes, while the split-adjusted percentage change in shares measures trading quantities more directly. Under both definitions, common AI reliance positively predicts future overlap and cosine similarity. The portfolio-convergence result therefore does not depend on treating passive changes in portfolio weights as active information-based adjustments.

	(1)	(2)
Future portfolio similarity	Overlap	Cosine
Panel A: AI reliance based on active rebalancing		
Common AI reliance	0.0104**	0.0279***
	(2.47)	(3.34)
Average AI reliance	-0.0053***	0.0042**
	(-6.87)	(2.02)
Current similarity	0.7957***	0.6760***
	(371.20)	(223.63)
Observations	890,394	890,394
Panel B: AI reliance based on split-adjusted share changes		
Common AI reliance	0.0132***	0.0269***
	(4.03)	(3.87)
Average AI reliance	-0.0063***	0.0039**
	(-9.82)	(2.17)
Current similarity	0.7958***	0.6760***
	(371.58)	(223.61)
Observations	889,769	889,769
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.7: Alternative AI Reliance Measures and Future Portfolio Similarity

D.5 Alternative Joint-Alignment Thresholds

The baseline joint-alignment indicator equals one when the fourteen GPT coefficients are jointly significant at the 5 percent level for each manager in the pair. Table D.8 varies this threshold. Requiring significance at the 1 percent level produces positive and statistically significant esti-

mates for both similarity measures, as does the baseline 5 percent threshold. When the threshold is relaxed to 10 percent, the estimates become small and statistically insignificant. The attenuation under the more permissive rule suggests that the association is concentrated among managers whose portfolio adjustments display stronger alignment with the GPT-extracted signals. A 10 percent rule includes more manager-quarter estimates with weaker evidence of alignment and therefore produces a noisier indicator.

Future portfolio similarity	(1) Overlap	(2) Cosine
Panel A: Joint GPT alignment at the 1% threshold		
Joint GPT alignment	0.0028***	0.0142***
	(8.63)	(10.98)
Average AI reliance	-0.0070***	0.0089***
	(-10.15)	(4.80)
Current similarity	0.7956***	0.6760***
	(370.97)	(223.70)
Observations	890,394	890,394
Panel B: Joint GPT alignment at the 5% threshold		
Joint GPT alignment	0.0020***	0.0105***
	(9.86)	(13.47)
Average AI reliance	-0.0076***	0.0059***
	(-10.86)	(3.19)
Current similarity	0.7956***	0.6760***
	(371.06)	(223.84)
Observations	890,394	890,394
Panel C: Joint GPT alignment at the 10% threshold		
Joint GPT alignment	-0.0002	0.0008
	(-1.25)	(1.47)
Average AI reliance	-0.0059***	0.0123***
	(-8.27)	(6.47)
Current similarity	0.7957***	0.6760***
	(371.16)	(223.65)
Observations	890,394	890,394
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.8: Alternative Joint-Alignment Thresholds and Future Portfolio Similarity

D.6 Manager-by-Quarter Fixed Effects

Table D.9 adds separate fixed effects for each manager in each quarter. These fixed effects absorb additive manager-specific shocks, including changes in fund flows, mandates, or risk constraints that may affect every pair containing the same institution. They also absorb each manager's own time-varying reliance measure. The coefficient on common reliance is therefore identified from the pair-level interaction in the two managers' reliance after accounting for these additive shocks.

The continuous common-reliance coefficient remains positive and statistically significant for future overlap. For cosine similarity, however, it becomes negative and is significant only at the 10 percent level. The joint-alignment indicator remains positive and statistically significant for both outcomes. The two measures therefore behave differently under saturated manager-specific time controls: the joint-alignment result is stable, whereas the sign of the continuous cosine estimate is sensitive to these controls.

Future portfolio similarity	(1) Overlap	(2) Cosine
Panel A: Continuous common AI reliance		
Common AI reliance	0.0111***	-0.0142*
	(2.67)	(-1.67)
Current similarity	0.7609***	0.6268***
	(290.50)	(102.70)
Observations	890,338	890,338
Panel B: Joint GPT alignment		
Joint GPT alignment	0.0008***	0.0026***
	(4.68)	(4.05)
Current similarity	0.7609***	0.6268***
	(290.52)	(102.70)
Observations	890,338	890,338
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Manager $i \times$ quarter FE	Yes	Yes
Manager $k \times$ quarter FE	Yes	Yes
SE cluster	Pair	Pair

Table D.9: Common AI Reliance and Future Portfolio Similarity with Manager-by-Quarter Fixed Effects

D.7 Two-Manager Clustered Standard Errors

The baseline regressions cluster standard errors at the manager-pair level. This covariance estimator allows arbitrary serial correlation and heteroskedasticity within a pair, which is the most immediate dependence structure in a panel of persistent pair-level outcomes. Residuals may also be correlated across different pairs that contain the same institution. Manager-wide fund flows, mandate changes, risk constraints, and trading-capacity shocks are possible sources of such dependence.

To assess the sensitivity of the empirical analysis to shared-manager dependence, Tables D.10 and D.11 cluster by the manager identities assigned to positions i and k. This estimator broadens the covariance structure beyond pair clustering, although it is not a fully permutation-invariant dyadic covariance estimator.

The two clustering dimensions overlap extensively in the dense manager-pair panel. In several multiway-clustered regressions, especially the validation specifications with many interactions, the estimated covariance matrix is not positive semidefinite. Following Cameron et al. (2011), the calculations replace its negative eigenvalues with zero. This correction leaves the coefficient estimates unchanged but makes the reported standard errors dependent on the finite-sample covariance adjustment.

Table D.10 reports the baseline coefficients under two-manager clustering. The continuous common-reliance coefficients remain statistically significant at the 5 percent level for both overlap and cosine similarity. The joint-alignment coefficient remains positive, with significance at the 10 percent level for overlap and at the 1 percent level for cosine similarity. The estimated standard errors are larger than those obtained from pair clustering.

Table D.11 applies the same covariance specification to the validation regressions. The post-GPT interaction remains positive for both outcomes but is statistically significant only for cosine similarity. The triple interaction with earnings-call length also remains positive for both outcomes, but neither estimate is statistically significant. The analyst-coverage interaction is negative and insignificant under this covariance estimator. The validation results thus have the predicted directional pattern in most specifications, but their precision is not uniform once the standard errors allow for shared-manager dependence.

Future portfolio similarity	(1) Overlap	(2) Cosine
Panel A: Controlling for average AI reliance		
Common AI reliance	0.0271**	0.0627**
	(2.41)	(2.17)
Average AI reliance	-0.0097	0.0052
	(-1.62)	(0.35)
Current similarity	0.7957***	0.6760***
	(44.40)	(24.47)
Panel B: Controlling for individual AI reliance		
Common AI reliance	0.0272**	0.0628**
	(2.43)	(2.17)
AI reliance, manager *i*	-0.0057	0.0018
	(-1.49)	(0.18)
AI reliance, manager *k*	-0.0041	0.0033
	(-1.02)	(0.35)
Current similarity	0.7956***	0.6760***
	(44.40)	(24.47)
Panel C: Joint GPT alignment		
Joint GPT alignment	0.0020*	0.0105***
	(1.84)	(2.62)
Average AI reliance	-0.0076	0.0059
	(-1.43)	(0.47)
Current similarity	0.7956***	0.6760***
	(44.39)	(24.50)
Pair controls	Yes	Yes
Pair FE	Yes	Yes
Quarter FE	Yes	Yes
SE cluster	Manager *i*, manager *k*	Manager *i*, manager *k*
Observations	890,394	890,394

Table D.10: Baseline Results with Two-Manager Clustered Standard Errors

Table D.11: Validation with Two-Manager Clustered Standard Errors

	(1) Post-GPT	(2) Call length	(3) Analyst coverage
Panel A: Future portfolio overlap			
Common reliance	0.0192**	0.0148	0.0040
	(2.02)	(1.58)	(0.23)
Common reliance × Post-GPT	0.0202	0.0207	0.0574
	(0.97)	(0.99)	(0.99)
Common reliance $\times M^X \times$ Post-GPT		0.0773	-0.0135
		(0.86)	(-0.35)
Current similarity	0.7948***	0.7907***	0.7872***
	(44.39)	(43.42)	(42.29)
Panel B: Future cosine similarity			
Common reliance	0.0180	0.0121	-0.0745*
	(0.66)	(0.43)	(-1.67)
Common reliance × Post-GPT	0.1575**	0.1704***	0.4174**
	(2.57)	(2.92)	(2.02)
Common reliance $\times M^X \times$ Post-GPT		0.2386	-0.1610
		(0.92)	(-1.22)
Current similarity	0.6736***	0.6705***	0.6644***
	(24.41)	(24.39)	(23.59)
All lower-order interactions	Yes	Yes	Yes
Pair controls	Yes	Yes	Yes
Pair FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
SE cluster	Manager i, manager k	Manager i, manager k	Manager i, manager k
Observations	890,394	859,357	859,357

Notes: Each panel and column is a separate regression. The covariance estimator clusters by both manager identities. All lower-order terms and average-reliance interactions are included. Robust t-statistics are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.8 Leave-One-Quarter-Out Robustness Checks

The post-GPT sample contains eight predictor quarters. To assess whether any single quarter disproportionately influences the validation estimates, we examine the three specifications for both dependent variables after omitting one predictor quarter at a time. Figure D.1 reports the portfolio-overlap estimates, and Figure D.2 reports the cosine-similarity estimates. The post-GPT coefficients remain positive under every omission for both similarity measures. The call-length estimates also remain positive, although precision declines under several post-GPT omissions. The analyst-coverage estimates are stable for cosine similarity but weaker and more variable for portfolio overlap. Overall, no single predictor quarter accounts for the principal validation patterns. This exercise measures the influence of individual quarters; it does not expand the time-series support of the post-GPT sample.

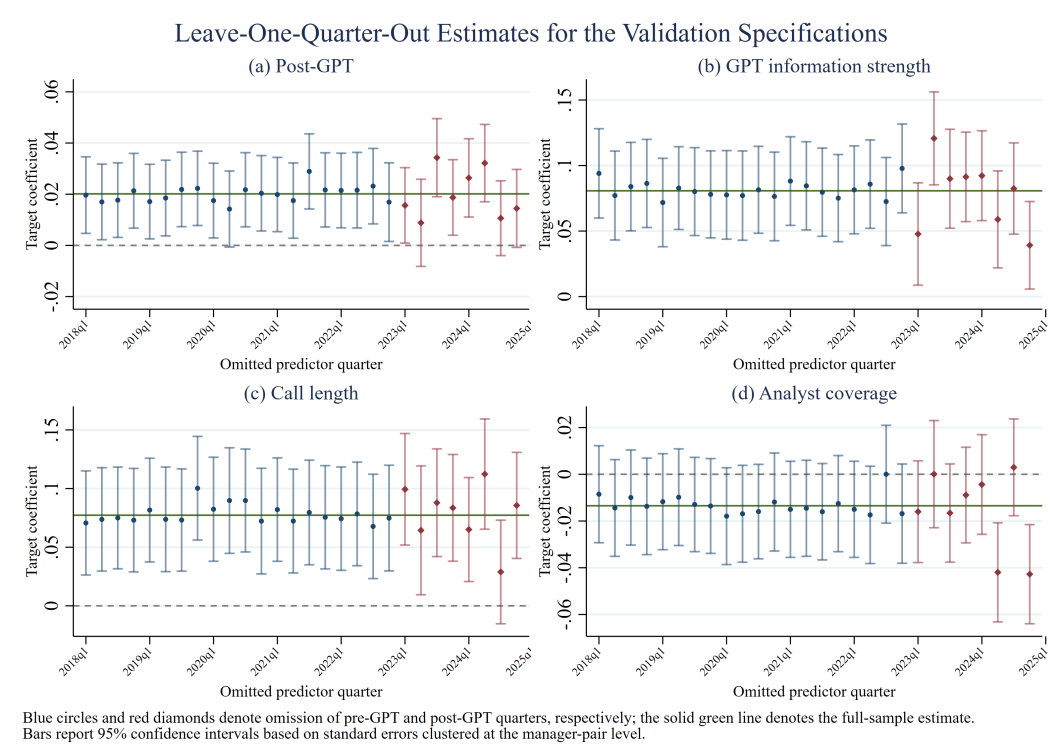


Figure D.1: Leave-One-Quarter-Out Estimates for the Validation Specifications: Portfolio Overlap

Notes: This figure shows the "leave-one-quarter-out" results for the portfolio-overlap validation regressions. We sequentially omit the predictor quarter indicated on the horizontal axis and plot the coefficients studied in the main text with 95% confidence intervals.

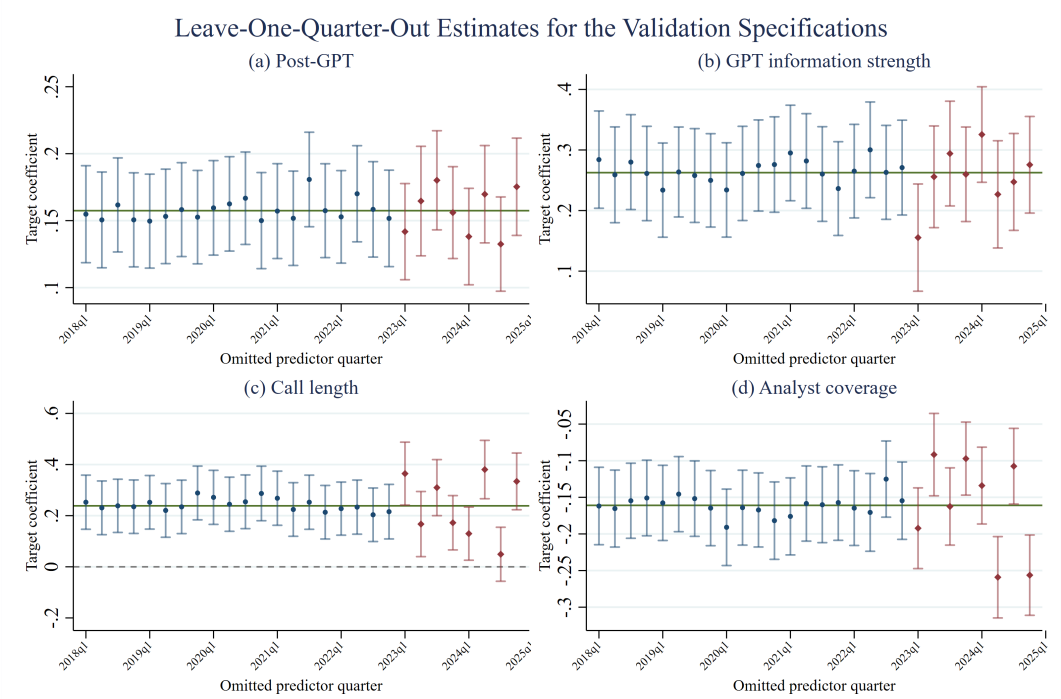


Figure D.2: Leave-One-Quarter-Out Estimates for the Validation Specifications: Cosine Similarity

Notes: This figure shows the "leave-one-quarter-out" results for the cosine-similarity validation regressions. We sequentially omit the predictor quarter indicated on the horizontal axis and plot the coefficients studied in the main text with 95% confidence intervals.

D.9 Masked-Identity Validation of the GPT Signals

To assess sensitivity to explicit identifying information, we randomly select 10% of firm-quarter observations within each signal quarter from 2018Q1 through 2024Q4 and retain all transcripts in the selected cells. The sample contains 1,438 transcripts for 1,427 firm-quarters and 519 firms. We mask company names, tickers, participant and organization names, detected product names, calendar dates, and contact details using metadata, regular expressions, and named-entity recognition. We then score both the original and masked texts with the same fixed model, `gpt-4o-mini-2024-07-18`, holding the prompt, temperature, original chunk boundaries, and aggregation rule constant. Table D.12 shows that the firm-quarter scores have a pooled Pearson correlation of 0.866, a Spearman correlation of 0.867, and an exact agreement rate of 90.7%; the mean absolute difference is 0.048 on the $[-1, 1]$ scale. Among comparisons with nonzero scores in both versions, the signs agree in 98.6% of cases. These results support the stability of the baseline signal-construction procedure to masking explicit identifiers.

Table D.12: GPT Scores with and without Explicit Identifiers

Signal	Correlation	Exact match (%)
U.S. economy	0.864	92.2
Global economy	0.832	91.3
Firm financial prospects	0.882	93.8
Industry financial prospects	0.881	92.4
Earnings	0.876	93.1
Revenue	0.887	94.0
Investment and capital expenditure	0.833	89.3
Wages and salaries	0.770	85.7
Employment	0.795	84.7
Product demand	0.915	95.8
Production quantity	0.808	87.1
Output prices	0.802	86.5
Input prices	0.698	91.5
Cost of capital	0.647	91.9
All fourteen signals pooled	0.866	90.7

Notes: Each signal uses 1,427 paired firm-quarter observations; the pooled row stacks the fourteen signals. Correlation is Pearson's correlation between scores from original and masked texts.